



Red Hat AMQ Broker 7.14

Deploying AMQ Broker on OpenShift

For Use with AMQ Broker 7.14

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Abstract

Learn how to install and deploy AMQ Broker on OpenShift Container Platform.

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CHAPTER 1. INTRODUCTION TO AMQ BROKER ON OPENSIFT CONTAINER PLATFORM

Red Hat AMQ Broker 7.14 is a high-performance messaging application that is available as a containerized image. AMQ Broker supports all OpenShift Container Platform releases that are in the full or maintenance support phases, as well as releases from version 4.14 onward that offer extended support.

Additional resources

- [OpenShift Container Platform Life Cycle Policy](#)

1.1. VERSION COMPATIBILITY AND SUPPORT

For details about OpenShift Container Platform image version compatibility, see the OpenShift Container Platform 4.x Tested Integrations article.



NOTE

All deployments of AMQ Broker on OpenShift Container Platform now use RHEL 9 based images.

Additional resources

- [OpenShift Container Platform 4.x Tested Integrations](#)

1.2. UNSUPPORTED FEATURES

Review the list of unsupported features and configurations for AMQ Broker on OpenShift.

- External clients cannot use the topology information provided by AMQ Broker
When a AMQ Core Protocol JMS client or a Red Hat build of Apache Qpid JMS client connects to a broker in an OpenShift Container Platform cluster, the broker can send the client the IP address and port information for each of the other brokers in the cluster, which serves as a failover list for clients if the connection to the current broker is lost.

The IP address provided for each broker is an internal IP address, which is not accessible to clients that are external to the OpenShift Container Platform cluster. To prevent external clients from trying to connect to a broker using an internal IP address, set the following configuration in the URI used by the client to initially connect to a broker.

Client	Configuration
AMQ Core Protocol JMS client	useTopologyForLoadBalancing=false
Red Hat build of Apache Qpid JMS client	failover.amqpOpenServerListAction=IGNORE

1.3. DOCUMENT CONVENTIONS

This document uses the following conventions for the **sudo** command, file paths, and replaceable values.

1.3.1. The **sudo** command

In this document, **sudo** is used for any command that requires root privileges. You should always exercise caution when using **sudo**, as any changes can affect the entire system.

1.3.2. About the use of file paths in this document

In this document, all file paths are valid for Linux, UNIX, and similar operating systems (for example, **/home/...**). If you are using Microsoft Windows, you should use the equivalent Microsoft Windows paths (for example, **C:\Users\...**).

1.3.3. Replaceable values

This document sometimes uses replaceable values that you must replace with values specific to your environment. Replaceable values are lowercase, enclosed by angle brackets (**< >**), and are styled using italics and **monospace** font. Multiple words are separated by underscores (**_**).

For example, in the following command, replace **<project_name>** with your own project name.

```
$ oc new-project <project_name>
```

CHAPTER 2. PLANNING A DEPLOYMENT OF AMQ BROKER ON OPENSIFT CONTAINER PLATFORM

You can use the AMQ Broker Operator to deploy and manage your brokers. The Operator can automate tasks such as installation, component health checks and updates.

Commonly, Operators are intended to provide:

- Consistent, repeatable installations
- Health checks of system components
- Over-the-air (OTA) updates
- Managed upgrades

Operators enable you to make changes while your broker instances are running, because they are always listening for changes to the Custom Resource (CR) instances that you used to configure your deployment. When you make changes to a CR, the Operator reconciles the changes with the existing broker deployment and updates the deployment to reflect the changes. In addition, the Operator provides a message migration capability, which ensures the integrity of messaging data. When a broker in a clustered deployment shuts down due to an intentional scaledown of the deployment, this capability migrates messages to a broker Pod that is still running in the same broker cluster.

2.1. OVERVIEW OF HIGH AVAILABILITY (HA)

For AMQ Broker on OpenShift, high availability means ensuring the integrity and availability of messaging data if a broker pod, node, or cluster fails. You can use the built-in HA capabilities on OpenShift or create a leader-follower deployment to provide HA.

AMQ Broker uses the HA capabilities provided in OpenShift Container Platform to mitigate pod and node failures:

- If persistent storage is enabled on AMQ Broker, each broker pod writes its data to a Persistent Volume (PV) that was claimed by using a Persistent Volume Claim (PVC). A PV remains available even after a pod is deleted. If a broker pod fails, OpenShift restarts the pod with the same name and uses the existing PV that contains the messaging data.
- You can run multiple broker pods in a cluster and distribute pods on separate nodes to recover from a node failure. Each broker pod writes its message data to its own PV which is then available to that broker pod if it is restarted on a different node.
If the mean time to repair (MTTR) for a node failure on your OpenShift cluster does not meet the availability requirements for AMQ Broker, you can create leader-follower deployments for faster recovery. You can also use leader-follower deployments to protect against a cluster or wider data center outage.

Additional resources

- [Section 4.24, “Configuring leader-follower deployments for high availability”](#)
- [Section 2.9, “Operator deployment notes”](#)
- [Section 4.17.2, “Controlling pod placement using tolerations”](#)

2.2. OVERVIEW OF THE AMQ BROKER OPERATOR CUSTOM RESOURCE DEFINITIONS

A Custom Resource Definition (CRD) is a schema of configuration items that you can modify for an OpenShift object. You can create a Custom Resource (CR) instance to configure values for items in a CRD. For example, you create a CR instance based on the main broker CRD to configure a broker.

If you are an Operator developer, what you expose through a CRD essentially becomes the API for how a deployed object is configured and used. You can directly access the CRD through regular HTTP **curl** commands, because the CRD gets exposed automatically through Kubernetes.

You can install the AMQ Broker Operator using either the OpenShift command-line interface (CLI), or the Operator Lifecycle Manager, through the OpenShift web console. In either case, the AMQ Broker Operator includes the CRDs described below.

Main broker CRD

You deploy a CR instance based on this CRD to create and configure a broker deployment. Based on how you install the Operator, this CRD is:

- The **broker_activemqartemis_crd** file in the **crds** directory of the Operator installation archive (OpenShift CLI installation method)
- The **ActiveMQArtemis** CRD in the **Custom Resource Definitions** section of the OpenShift Container Platform web console (OpenShift web console installation method)

Address CRD

You deploy a CR instance based on this CRD to create addresses and queues for a broker deployment.

Based on how you install the Operator, this CRD is:

- The **broker_activemqartemisaddress_crd** file in the **crds** directory of the Operator installation archive (OpenShift CLI installation method)
- The **ActiveMQArtemisAddresss** CRD in the **Custom Resource Definitions** section of the OpenShift Container Platform web console (OpenShift web console installation method)



NOTE

The address CRD is deprecated in 7.12. You can use the **brokerProperties** attribute in an **ActiveMQArtemis** CR instance instead of creating a CR instance based on the addresss CRD.

Security CRD

You deploy a CR instance based on this CRD to create users and associate those users with security contexts.

Based on how you install the Operator, this CRD is:

- The **broker_activemqartemissecurity_crd** file in the **crds** directory of the Operator installation archive (OpenShift CLI installation method)
- The **ActiveMQArtemisSecurity** CRD in the **Custom Resource Definitions** section of the OpenShift Container Platform web console (OpenShift web console installation method).

**NOTE**

The security CRD is deprecated in 7.12. You can use the **brokerProperties** attribute in an **ActiveMQArtemis** CR instance instead of creating a CR instance based on the security CRD.

Scaledown CRD

The **Operator** automatically creates a CR instance based on this CRD when instantiating a scaledown controller for message migration.

Based on how you install the Operator, this CRD is:

- The **broker_activemqartemiscaledown_crd** file in the **crds** directory of the Operator installation archive (OpenShift CLI installation method)
- The **ActiveMQArtemisScaledown** CRD in the **Custom Resource Definitions** section of the OpenShift Container Platform web console (OpenShift web console installation method).

**NOTE**

The scaledown CRD is deprecated in 7.12 and is not required to scale down a cluster.

Additional resources

- [Section 3.1, “Installing the Operator using the CLI”](#)
- [Section 3.2, “Installing the Operator from the software catalog by using the Openshift web console”](#)
- [Section 8.1.1, “Broker custom resource configuration reference”](#)
- [Section 8.1.2, “Address Custom Resource configuration reference”](#)

2.3. OVERVIEW OF THE AMQ BROKER OPERATOR SAMPLE CUSTOM RESOURCES

The AMQ Broker Operator archive that you download and extract during installation includes sample Custom Resource (CR) files. You can create a minimal broker and define addresses by using these sample CR files.

These sample CR files are located in the **deploy/crs** directory.

The broker Operator archive that you download and extract also includes CRs for the following example deployments in the **deploy/examples/address** and **deploy/examples/artemis** directories.

address_queue.yaml

Deploys an address and queue with different names. Deletes the queue when the CR is undeployed.

address_topic.yaml

Deploys an address with a multicast routing type. Deletes the address when the CR is undeployed.

artemis_address_settings.yaml

Deploys a broker with specific address settings.

artemis_cluster_persistence.yaml

Deploys clustered brokers with persistent storage.

artemis_enable_metrics_plugin.yaml

Enables the Prometheus metrics plugin to collect metrics.

artemis_resources.yaml

Sets CPU and memory resource limits for the broker.

artemis_single.yaml

Deploys a single broker.

2.4. CONFIGURING ITEMS NOT EXPOSED IN A CUSTOM RESOURCE DEFINITION (CRD)

You can use the **brokerProperties** attribute in an **ActiveMQArtemis** custom resource to configure any setting that is not exposed in a CRD.

Using **brokerProperties** is also a good idea if you want to configure any setting that is in the **ActiveMQArtemisAddress** or **ActiveMQArtemisSecurity** CRD as both CRDs are deprecated starting in 7.12.

Configuration settings added under a **brokerProperties** attribute are stored in a secret. This secret is mounted as a properties file on the broker pod. At startup, the properties file is applied directly to the internal java configuration bean after the XML configuration is applied.

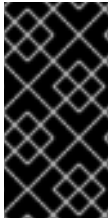
Examples

In the following example, a single property is applied to the configuration bean.

```
spec:
  ...
  brokerProperties:
    - globalMaxSize=500m
  ...
```

In the following example, multiple properties are applied to nested collections of configuration beans to create a broker connection named **target** that mirror messages with another broker.

```
spec:
  ...
  brokerProperties
    - "AMQPConnections.target.uri=tcp://<hostname>:<port>"
    - "AMQPConnections.target.connectionElements.mirror.type=MIRROR"
    - "AMQPConnections.target.connectionElements.mirror.messageAcknowledgements=true"
    - "AMQPConnections.target.connectionElements.mirror.queueCreation=true"
    - "AMQPConnections.target.connectionElements.mirror.queueRemoval=true"
  ...
```



IMPORTANT

Using the **brokerProperties** attribute provides access to many configuration items that you cannot otherwise configure for AMQ Broker on OpenShift Container Platform. If used incorrectly, some properties can have serious consequences for your deployment. Always exercise caution when configuring the broker using this method.

Status reporting for **brokerProperties**

The status of items configured in a **brokerProperties** attribute is provided in the **BrokerPropertiesApplied** status section of the **ActiveMQArtemis** CR. For example:

```
- lastTransitionTime: "2023-02-06T20:50:01Z"
  message: ""
  reason: Applied
  status: "True"
  type: BrokerPropertiesApplied
```

The **reason** field contains one of the following values to show the status of the items configured in a **brokerProperties** attribute:

Applied

OpenShift Container Platform propagated the updated secret to the properties file on each broker pod.

AppliedWithError

OpenShift Container Platform propagated the updated secret to the properties file on each broker pod. However, an error was found in the **brokerProperties** configuration. In the **status** section of the CR, check the **message** field to identify the invalid property and correct it in the CR.

OutOfSync

OpenShift Container Platform has not yet propagated the updated secret to the properties file on each broker pod. When OpenShift Container Platform propagates the updated secret to each pod, the **reason** field value changes to **Applied**.



NOTE

The broker checks periodically for configuration changes, including updates to the properties file that is mounted on the pod, and reloads the configuration if it detects any changes. However, updates to properties that are read only when the broker starts, for example, JVM settings, are not reloaded until you restart the broker.

Additional resources

- [Broker Properties](#)
- [Reloading configuration updates.](#)

2.5. WATCH OPTIONS FOR A CLUSTER OPERATOR DEPLOYMENT

When the Cluster Operator is running, you can configure it to watch for updates to AMQ Broker custom resources (CRs) in a single namespace or all namespaces. This helps you define whether the Operator manages brokers across all namespaces or only specific ones.

**NOTE**

If you installed a previous version of the AMQ Broker Operator in a namespace on your cluster, it is recommended that you do not install the AMQ Broker Operator 7.14 version to watch that namespace to avoid potential conflicts.

2.6. HOW THE OPERATOR DETERMINES THE CONFIGURATION TO USE TO DEPLOY IMAGES

You can use the **image** and **version** fields in the custom resource (CR) to control which broker container images are deployed by the Operator. Learn how the Operator selects the broker container images when both the **image** and **version** fields are present or missing in the CR.

In the **ActiveMQArtemis** CR, you can specify deploy container images:

- Specify a version number in the **spec.version** attribute and allow the Operator to choose the broker and init container images to deploy for that version number.
- Specify the registry URLs of the specific broker and init container images that you want the Operator to deploy in the **spec.deploymentPlan.image** and **spec.deploymentPlan.initImage** attributes.
- Set the value of the **spec.deploymentPlan.image** attribute to **placeholder**, which means that the Operator chooses the latest broker and init container images that are known to the Operator version.

**NOTE**

If you do not use any of these configurations to deploy container images, the Operator chooses the latest broker and init container images that are known to the Operator version.

After you save a CR, the Operator performs the following validation to determine the configuration to use.

- The Operator checks if the CR contains a **spec.version** attribute.
 - If the CR does not contain a **spec.version** attribute, the Operator checks if the CR contains a **spec.deploymentPlan.image** and a **spec.deploymentPlan.initImage** attribute.
 - If the CR contains a **spec.deploymentPlan.image** and a **spec.deploymentPlan.initImage** attribute, the Operator deploys the container images that are identified by their registry URLs.
 - If the CR does not contain a **spec.deploymentPlan.image** and a **spec.deploymentPlan.initImage** attribute, the Operator chooses the container images to deploy.
 - If the CR contains a **spec.version** attribute, the Operator verifies that the version number specified is within the valid range of versions that the Operator supports.
 - If the value of the **spec.version** attribute is not valid, the Operator stops the deployment.

- If the value of the **spec.version** attribute is valid, the Operator checks if the CR contains a **spec.deploymentPlan.image** and a **spec.deploymentPlan.initImage** attribute.
 - If the CR contains a **spec.deploymentPlan.image** and a **spec.deploymentPlan.initImage** attribute, the Operator deploys the container images that are identified by their registry URLs.
 - If the CR does not contain a **spec.deploymentPlan.image** and a **spec.deploymentPlan.initImage** attribute, the Operator chooses the container images to deploy.



NOTE

If the CR contains only one of the **spec.deploymentPlan.image** and the **spec.deploymentPlan.initImage** attributes, the Operator uses the **spec.version** number attribute to choose an image for the attribute that is not in the CR, or chooses the latest known image for that attribute if the **spec.version** attribute is not in the CR.

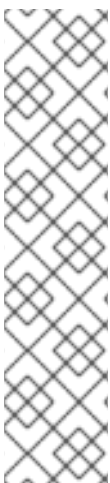
Red Hat recommends that you do not specify the **spec.deploymentPlan.image** attribute without the **spec.deploymentPlan.initImage** attribute, or vice versa, to prevent mismatched versions of broker and init container images from being deployed.

Additional resources

- [Section 2.7, “How the Operator chooses container images when a precise image is not specified in the CR”](#)

2.7. HOW THE OPERATOR CHOOSES CONTAINER IMAGES WHEN A PRECISE IMAGE IS NOT SPECIFIED IN THE CR

You can use the **image** attribute to specify the registry URLs of specific container images the Operator must deploy. Learn the sequence the Operator follows to choose the appropriate images during deployment if the **image** attribute is not configured in the CR.



NOTE

If you install the Operator using the OpenShift command-line interface, the Operator installation archive includes a sample CR file called **broker_activemqartemis_cr.yaml**. In the sample CR, the **spec.deploymentPlan.image** attribute is included and set to its default value of **placeholder**. This value indicates that the Operator does not choose a broker container image until you deploy the CR.

The **spec.deploymentPlan.initImage** attribute, which specifies the Init Container image, is **not** included in the **broker_activemqartemis_cr.yaml** sample CR file. If you do not explicitly include the **spec.deploymentPlan.initImage** attribute in your CR and specify a value, the Operator chooses a built-in Init Container image that matches the version of the Operator container image chosen.

To choose broker and Init Container images, the Operator first determines an AMQ Broker version of the images that is required. The Operator gets the version from the value of the **spec.version** attribute. If the **spec.version** attribute is not set, the Operator uses the latest version of the images that are available for AMQ Broker.

The Operator then detects your container platform. The AMQ Broker Operator can run on the following container platforms:

- OpenShift Container Platform (x86_64)
- OpenShift Container Platform on IBM Z (s390x)
- OpenShift Container Platform on IBM Power Systems (ppc64le)

Based on the version of AMQ Broker and your container platform, the Operator then references two sets of environment variables in the **operator.yaml** configuration file. These sets of environment variables specify broker and Init Container images for various versions of AMQ Broker, as described in the following section.

2.7.1. Environment variables for broker and init container images

Learn how the Operator finds the registry URLs of the container images to deploy by using environment variables.

After the Operator determines the broker version required from the custom resource configuration, and the container platform, it deploys the images related to the environment variables for that version and platform. The environment variables are specified in the Operator configuration file, **operator.yaml**, and have the following naming convention.

Table 2.1. Naming conventions for environment variables

Container platform	Naming convention
OpenShift Container Platform	RELATED_IMAGE_ActiveMQ_Artemis_Broker_Kubernetes_<AMQ_Broker_version>
OpenShift Container Platform on IBM Z	RELATED_IMAGE_ActiveMQ_Artemis_Broker_Kubernetes_<AMQ_Broker_version>_s390x
OpenShift Container Platform on IBM Power Systems	RELATED_IMAGE_ActiveMQ_Artemis_Broker_Kubernetes_<AMQ_Broker_version>_ppc64le

The following are examples of environment variable names for broker and init container images for each supported container platform.

Table 2.2. Example environment variable names

Container platform	Environment variable names
OpenShift Container Platform	RELATED_IMAGE_ActiveMQ_Artemis_Broker_Kubernetes_7140 RELATED_IMAGE_ActiveMQ_Artemis_Broker_Init_7140

Container platform	Environment variable names
OpenShift Container Platform on IBM Z	RELATED_IMAGE_ActiveMQ_Artemis_Broker_Kubernetes_7140_s390x RELATED_IMAGE_ActiveMQ_Artemis_Broker_Init_s390x_7140
OpenShift Container Platform on IBM Power Systems	RELATED_IMAGE_ActiveMQ_Artemis_Broker_Kubernetes_7140_ppc64le RELATED_IMAGE_ActiveMQ_Artemis_Broker_Init_ppc64le_7140

The value of each environment variable specifies the address of a container image that is available from Red Hat. The image name is represented by a *Secure Hash Algorithm* (SHA) value. For example:

```
- name: RELATED_IMAGE_ActiveMQ_Artemis_Broker_Kubernetes_7140
  value: registry.redhat.io/amq7/amq-broker-
        rhel9@sha256:e44742f66183e225a77f5171da5db95e51f92b6848dea9ed02895a81b1966857
```

Therefore, based on an AMQ Broker version and your container platform, the Operator determines the applicable environment variable names for the broker and init container. The Operator uses the corresponding image values when starting the broker container.

Additional resources

- [Chapter 3, Deploying AMQ Broker on OpenShift Container Platform using the AMQ Broker Operator](#)
- [Section 4.1, “How the Operator generates the broker configuration”](#)
- [Section 4.11, “Specifying a custom Init container image”](#)

2.8. VALIDATION OF IMAGE AND VERSION CONFIGURATION IN A CUSTOM RESOURCE (CR)

Learn how the Operator verifies that the image and version attributes are configured correctly in the CR and the possible status messages reported.

The Operator validation of the configuration occurs after you save the CR and covers the following checks.

Table 2.3. Operator validation of CR configuration

Validation	Purpose of validation	Status reported in CR
------------	-----------------------	-----------------------

Validation	Purpose of validation	Status reported in CR
Does the CR contain a spec.deploymentPlan.image attribute without a spec.version attribute.	A spec.deploymentPlan.image attribute without a spec.version attribute causes the Operator to restart the broker pods each time the Operator is upgraded. Pod restarts are required because the new Operator updates a label in the StatefulSet with the latest supported broker version unless a version number is explicitly set in the spec.version attribute.	The Valid condition is Unknown and the following status message is displayed: Unknown image version, set a supported broker version in spec.version when images are specified.
Does the CR contain a spec.deploymentPlan.image attribute without a spec.deploymentPlan.initImage attribute or vice versa.	With this configuration, different versions of the broker and init container images could be deployed, which might prevent your broker from starting.	The Valid condition is Unknown and the following status message is displayed: Init image and broker image must both be configured as an interdependent pair.
If the CR contain a spec.version attribute, is the version specified within the range of versions that the Operator supports.	If the value of the spec.version attribute is a broker version that is not supported by the Operator, the Operator does not proceed with the deployment of broker pods.	The Valid condition is False and the following status message is displayed: Spec.Version does not resolve to a supported broker version, reason did not find a matching broker in the supported list for <version>.
Does the version of the broker image deployed, based on the URL of a container image in the spec.deploymentPlan.image attribute, match the broker version in the spec.version attribute.	Flags a mismatch between the actual broker version deployed and the version shown in the spec.version attribute if both attributes are configured in the CR. This is for information purposes to highlight that the version shown in the spec.version attribute is not the version deployed.	The status of the BrokerVersionAligned condition is Unknown and the following message is displayed: broker version non aligned on pod <pod name>, the detected version <version> doesn't match the spec.version <version> resolved as <version>.

Additional resources

- [Viewing status information for your broker deployment](#)

2.9. OPERATOR DEPLOYMENT NOTES

Review items to consider before you use the Operator to create a broker deployment, including privilege requirements, updating CRDs in a cluster, namespace recommendations and persistent storage considerations.

The following are important considerations when deploying an Operator-based deployment:

- Deploying the Custom Resource Definitions (CRDs) that accompany the AMQ Broker Operator requires cluster administrator privileges for your OpenShift cluster. When the Operator is deployed, non-administrator users can create broker instances via corresponding Custom Resources (CRs). To enable regular users to deploy CRs, the cluster administrator must first assign roles and permissions to the CRDs.
- When you update your cluster with the CRDs for the latest Operator version, this update affects **all** projects in the cluster. Any broker pods deployed from previous versions of the Operator might become unable to update their status. When you click the Logs tab of a running broker pod in the OpenShift Container Platform web console, you see messages indicating that 'UpdatePodStatus' has failed. However, the broker pods and Operator in that project continue to work as expected. To fix this issue for an affected project, you must also upgrade that project to use the latest version of the Operator.
- While you can create more than one broker deployment in a given OpenShift project by deploying multiple Custom Resource (CR) instances, typically, you create a single broker deployment in a project, and then deploy multiple CR instances for addresses. It is recommended that you create broker deployments in separate projects.
- If you intend to deploy brokers with persistent storage and do not have container-native storage in your OpenShift cluster, you need to manually provision Persistent Volumes (PVs) and ensure that these are available to be claimed by the Operator. For example, if you want to create a cluster of two brokers with persistent storage (that is, by setting **persistenceEnabled=true** in your CR), you need to have two persistent volumes available. By default, each broker instance requires storage of 2 GiB.
If you specify **persistenceEnabled=false** in your CR, the deployed brokers uses *ephemeral* storage. Ephemeral storage means that every time you restart the broker pods, any existing data is lost.
- You must add configuration for the items listed below to the main broker CR instance **before** deploying the CR for the first time. You **cannot** add configuration for these items to a broker deployment that is already running.
 - The size and storage class of the Persistent Volume Claim (PVC) required by each broker in a deployment for persistent storage
 - Limits and requests for memory and CPU for each broker in a deployment
- If you update a parameter in your CR that the Operator is unable to dynamically update in the StatefulSet, the Operator deletes the StatefulSet and recreates it with the updated parameter value. Deleting the StatefulSet causes all pods to be deleted and recreated, which causes a temporary broker outage. An example of a CR update that the Operator cannot dynamically update in the StatefulSet is if you change **persistenceEnabled=false** to **persistenceEnabled=true**.

Additional resources

- [Understanding persistent storage](#)
- [Section 4.6.1, "Configuring storage size and storage class"](#)

- [Section 4.7.1, “Configuring broker resource limits and requests”](#)

2.10. IDENTIFYING NAMESPACES WATCHED BY EXISTING OPERATORS

Identify existing Operators that watch all namespaces. You must either uninstall them or reconfigure them to watch specific namespaces before deploying a new Operator.

If the cluster already contains installed Operators for AMQ Broker, and you want a new Operator to watch all or multiple namespaces, you must ensure that the new Operator does not watch any of the same namespaces as existing Operators. Use the following procedure to identify the namespaces watched by existing Operators.

Procedure

1. In the left pane of the OpenShift Container Platform web console, click **Workloads > Deployments**
2. In the **Project** drop-down list, select **All Projects**.
3. In the **Filter Name** box, specify a string, for example, **amq**, to display the Operators for AMQ Broker that are installed on the cluster.



NOTE

The **namespace** column displays the namespace where each operator is deployed.

4. Check the namespaces that each installed Operator for AMQ Broker is configured to **watch**.
 - a. Click the Operator name to display the Operator details and click the **YAML** tab.
 - b. Search for **WATCH_NAMESPACE** and note the namespaces that the Operator watches.
 - If the **WATCH_NAMESPACE** section has a **fieldPath** field that has a value of **metadata.namespace**, the Operator is watching the namespace where it is deployed.
 - If the **WATCH_NAMESPACE** section has a **value** field that has list of namespaces, the Operator is watching the specified namespaces. For example:

```
- name: WATCH_NAMESPACE
  value: "namespace1, namespace2"
```

- If the **WATCH_NAMESPACE** section has a **value** field that is empty or has an asterisk, the Operator is watching all the namespaces on the cluster. For example:

```
- name: WATCH_NAMESPACE
  value: ""
```

In this case, before you deploy the new Operator, you must either uninstall the existing Operator or reconfigure it to watch specific namespaces.

CHAPTER 3. DEPLOYING AMQ BROKER ON OPENSIFT CONTAINER PLATFORM USING THE AMQ BROKER OPERATOR

Before you can deploy a broker on OpenShift, you must first install the AMQ Broker Operator. You can install the Operator by using the OpenShift command-line interface (CLI) or the OpenShift web console.

3.1. INSTALLING THE OPERATOR USING THE CLI

Instead of using the OpenShift web console graphical interface, you can use the OpenShift command-line interface (CLI) to install and deploy the latest version of the Operator for AMQ Broker 7.14 on OpenShift.

3.1.1. Preparing to deploy the Operator

Prepare your local environment by downloading and extracting the necessary Operator files before starting the deployment procedure.

Procedure

1. In your web browser, navigate to the **Software Downloads** page for [AMQ Broker 7.14.0](#).
2. Ensure that the value of the **Version** drop-down list is set to **7.14.0** and the **Releases** tab is selected.
3. Next to the latest **AMQ Broker 7.14.0 Operator Installation and Example Files** click **Download**.
Download of the **amq-broker-operator-7.14.0-ocp-install-examples-rhel9.zip** compressed archive automatically begins.
4. Move the archive to your chosen directory. The following example moves the archive to a directory called **~/broker/operator**.

```
$ mkdir ~/broker/operator
$ mv amq-broker-operator-7.14.0-ocp-install-examples-rhel9.zip ~/broker/operator
```

5. In your chosen directory, extract the contents of the archive. For example:

```
$ cd ~/broker/operator
$ unzip amq-broker-operator-7.14.0-ocp-install-examples-rhel9.zip
```

6. Switch to the directory that was created when you extracted the archive. For example:

```
$ cd amq-broker-operator-7.14.0-ocp-install-examples
```

7. Log in to OpenShift Container Platform as a cluster administrator. For example:

```
$ oc login -u system:admin
```

8. Specify the project in which you want to install the Operator. You can create a new project or switch to an existing one.

- a. Create a new project:

```
$ oc new-project <project_name>
```

- b. Or, switch to an existing project:

```
$ oc project <project_name>
```

9. Specify a service account to use with the Operator.

- a. In the **deploy** directory of the Operator archive that you extracted, open the **service_account.yaml** file.
- b. Ensure that the **kind** element is set to **ServiceAccount**.
- c. If you want to change the default service account name, in the **metadata** section, replace **amq-broker-controller-manager** with a custom name.
- d. Create the service account in your project.

```
$ oc create -f deploy/service_account.yaml
```

10. Specify a role name for the Operator.

- a. Open the **role.yaml** file. This file specifies the resources that the Operator can use and modify.
- b. Ensure that the **kind** element is set to **Role**.
- c. If you want to change the default role name, in the **metadata** section, replace **amq-broker-operator-role** with a custom name.
- d. Create the role in your project.

```
$ oc create -f deploy/role.yaml
```

11. Specify a role binding for the Operator. The role binding binds the previously-created service account to the Operator role, based on the names you specified.

- a. Open the **role_binding.yaml** file.
- b. Ensure that the **name** values for **ServiceAccount** and **Role** match those specified in the **service_account.yaml** and **role.yaml** files. For example:

```
metadata:
  name: amq-broker-operator-rolebinding
subjects:
  kind: ServiceAccount
  name: amq-broker-controller-manager
roleRef:
  kind: Role
  name: amq-broker-operator-role
```

- c. Create the role binding in your project.

–

```
$ oc create -f deploy/role_binding.yaml
```

12. Specify a leader election role binding for the Operator. The role binding binds the previously-created service account to the leader election role, based on the names you specified.

- a. Create a leader election role for the Operator.

```
$ oc create -f deploy/election_role.yaml
```

- b. Create the leader election role binding in your project.

```
$ oc create -f deploy/election_role_binding.yaml
```

13. (Optional) If you want the Operator to watch multiple namespaces, complete the following steps:



NOTE

If the OpenShift Container Platform cluster already contains installed Operators for AMQ Broker, you must ensure the new Operator does not watch any of the same namespaces as existing Operators. For information on how to identify the namespaces that are watched by existing Operators, see, [Identifying namespaces watched by existing Operators](#).

- a. In the deploy directory of the Operator archive that you downloaded and extracted, open the **operator.yaml** file.
- b. If you want the Operator to watch all namespaces in the cluster, in the **WATCH_NAMESPACE** section, add a **value** attribute and set the value to an asterisk. Comment out the existing attributes in the **WATCH_NAMESPACE** section. For example:

```
- name: WATCH_NAMESPACE
  value: "*"
# valueFrom:
# fieldRef:
# fieldPath: metadata.namespace
```



NOTE

To avoid conflicts, ensure that multiple Operators do not watch the same namespace. For example, if you deploy an Operator to watch **all** namespaces on the cluster, you cannot deploy another Operator to watch individual namespaces. If Operators are already deployed on the cluster, you can specify a list of namespaces that the new Operator watches, as described in the following step.

- c. If you want the Operator to watch multiple, but not all, namespaces on the cluster, in the **WATCH_NAMESPACE** section, specify a list of namespaces. Ensure that you exclude any namespaces that are watched by existing Operators. For example:

```
- name: WATCH_NAMESPACE
  value: "namespace1, namespace2".
```

- d. In the deploy directory of the Operator archive that you downloaded and extracted, open the **cluster_role_binding.yaml** file.
- e. In the Subjects section, specify a namespace that corresponds to the OpenShift Container Platform project to which you are **deploying** the Operator. For example:

```
Subjects:
- kind: ServiceAccount
  name: amq-broker-controller-manager
  namespace: operator-project
```



NOTE

If you previously deployed brokers using an earlier version of the Operator, and you want to deploy the Operator to watch multiple namespaces, see [Before you upgrade](#).

- f. Create a cluster role in your project.

```
$ oc create -f deploy/cluster_role.yaml
```

- g. Create a cluster role binding in your project.

```
$ oc create -f deploy/cluster_role_binding.yaml
```

In the procedure that follows, you deploy the Operator in your project.

3.1.2. Deploying the Operator using the CLI

You can deploy the Operator for AMQ Broker by using the CLI after you configure access to the Red Hat Ecosystem Catalog to retrieve the Operator container image.

Prerequisites

- You must have already prepared your OpenShift project for the Operator deployment. See [Section 3.1.1, “Preparing to deploy the Operator”](#).
- Starting in AMQ Broker 7.3, you use a new version of the Red Hat Ecosystem Catalog to access container images. This new version of the registry requires you to become an authenticated user before you can access images. Before you deploy the Operator, complete the steps described in [Red Hat Container Registry Authentication](#).
- Determine if you want to use persistent or ephemeral storage for the brokers. If you want to use persistent storage, specify **persistenceEnabled=true** in your Custom Resource. The Operator then creates the required resources to deploy broker pods that have persistent volumes attached and creates the journal and other data files on the persistent volume. If you specify **persistenceEnabled=false**, the brokers use ephemeral storage. Ephemeral storage is transient local storage, which means that all data is lost each time you restart the broker pods.
- If you intend to deploy brokers with persistent storage and do not have container-native storage in your OpenShift cluster, you need to manually provision Persistent Volumes (PVs) and ensure that they are available to be claimed by the Operator. For example, if you want to create

a cluster of two brokers with persistent storage (that is, by setting **persistenceEnabled=true** in your Custom Resource), you need to have two PVs available. By default, each broker instance requires storage of 2 GiB.

For more information about provisioning persistent storage, see:

- [Understanding persistent storage](#)

Procedure

1. In the OpenShift command-line interface (CLI), log in to OpenShift as a cluster administrator. For example:

```
$ oc login -u system:admin
```

2. Switch to the project that you previously prepared for the Operator deployment. For example:

```
$ oc project <project_name>
```

3. Switch to the directory that was created when you previously extracted the Operator installation archive. For example:

```
$ cd ~/broker/operator/amq-broker-operator-7.14.0-ocp-install-examples
```

4. Deploy the CRDs that are included with the Operator. You must install the CRDs in your OpenShift cluster before deploying and starting the Operator.

- a. Deploy the main broker CRD.

```
$ oc create -f deploy/crds/broker_activemqartemis_crd.yaml
```

- b. Deploy the address CRD.

```
$ oc create -f deploy/crds/broker_activemqartemisaddress_crd.yaml
```

- c. Deploy the scaledown controller CRD.

```
$ oc create -f deploy/crds/broker_activemqartemisscaledown_crd.yaml
```

- d. Deploy the security CRD:

```
$ oc create -f deploy/crds/broker_activemqartemissecurity_crd.yaml
```

5. Link the pull secret associated with the account used for authentication in the Red Hat Ecosystem Catalog with the **default**, **deployer**, and **builder** service accounts for your OpenShift project.

```
$ oc secrets link --for=pull default <secret_name>
$ oc secrets link --for=pull deployer <secret_name>
$ oc secrets link --for=pull builder <secret_name>
```

- In the **deploy** directory of the Operator archive that you downloaded and extracted, open the **operator.yaml** file. Ensure that the value of the **spec.containers.image** property corresponds to version 7.14.0-opr-1 of the Operator, as shown below.

```
spec:
  template:
    spec:
      containers:
        image: registry.redhat.io/amq7/amq-broker-rhel9-
operator@sha256:8f4431d72e6fed9dc535f7db273ebc11b0c7809ea7ffa59ff687e74210b70144
```



NOTE

In the **operator.yaml** file, the Operator uses an image that is represented by a *Secure Hash Algorithm* (SHA) value. The comment line, which begins with a number sign (**#**) symbol, denotes that the SHA value corresponds to a specific container image tag.

- Deploy the Operator.

```
$ oc create -f deploy/operator.yaml
```

In your OpenShift project, the Operator starts in a new Pod.

In the OpenShift Container Platform web console, the information on the **Events** tab of the Operator Pod confirms that OpenShift has deployed the Operator image that you specified, has assigned a new container to a node in your OpenShift cluster, and has started the new container.

In addition, if you click the **Logs** tab within the Pod, the output should include lines resembling the following:

```
...
{"level":"info","ts":1553619035.8302743,"logger":"kubebuilder.controller","msg":"Starting Controller","controller":"activemqartemisaddress-controller"}
{"level":"info","ts":1553619035.830541,"logger":"kubebuilder.controller","msg":"Starting Controller","controller":"activemqartemis-controller"}
{"level":"info","ts":1553619035.9306898,"logger":"kubebuilder.controller","msg":"Starting workers","controller":"activemqartemisaddress-controller","worker count":1}
{"level":"info","ts":1553619035.9311671,"logger":"kubebuilder.controller","msg":"Starting workers","controller":"activemqartemis-controller","worker count":1}
```

The preceding output confirms that the newly-deployed Operator is communicating with Kubernetes, that the controllers for the broker and addressing are running, and that these controllers have started some workers.



NOTE

It is recommended that you deploy only a **single instance** of the AMQ Broker Operator in a given OpenShift project. Setting the **spec.replicas** property of your Operator deployment to a value greater than **1**, or deploying the Operator more than once in the same project is **not** recommended.

- [Section 3.2, “Installing the Operator from the software catalog by using the OpenShift web console”](#)

3.2. INSTALLING THE OPERATOR FROM THE SOFTWARE CATALOG BY USING THE OPENSIFT WEB CONSOLE

The software catalog is a user interface for discovering Operators; it works in conjunction with Operator Lifecycle Manager (OLM), which installs and manages Operators on a cluster.

3.2.1. Deploying the Operator by using the OpenShift web console

You can deploy the Operator for AMQ Broker from the software catalog by using the OpenShift web console. After you deploy the Operator, you can deploy one or more brokers.



NOTE

Using the OpenShift web console, you can install only the latest Operator version that is provided in each channel. If you want to install an earlier version of an Operator, you must install the Operator by using the CLI. For more information, see [Section 3.1, “Installing the Operator using the CLI”](#).

Prerequisites

- The **Red Hat Integration - AMQ Broker for RHEL 9 (Multiarch)** Operator must be available in the software catalog.
- You have cluster administrator privileges.

Procedure

1. Log in to the OpenShift Container Platform web console as a cluster administrator.
2. Navigate in the web console to the **Ecosystem > Software Catalog** page.
3. On the **Project** drop-down menu at the top of the page, select the project in which you want to deploy the Operator.
4. Scroll or type a keyword into the Filter by keyword box to find the **Red Hat Integration - AMQ Broker for RHEL 9 (Multiarch)** Operator.



NOTE

The software catalog contains a separate Operator, **Red Hat Integration - AMQ Broker for RHEL 8 (Multiarch)**, for 7.12.x versions of AMQ Broker. Ensure that you click the **Red Hat Integration - AMQ Broker for RHEL 9 (Multiarch)** Operator to install the Operator for 7.14. When you click this Operator, review the information pane that opens. For AMQ Broker 7.14, the latest minor version tag of this Operator is **7.14.0-opr-1**.

5. Click the **Red Hat Integration - AMQ Broker for RHEL 9 (Multiarch)** Operator. On the dialog box that is displayed, click **Install**.
6. On the **Install Operator** page:

- a. Under **Channel**, select the **7.14.x** channel to receive updates for version 7.14 only. The **7.14.x** channel is a Long Term Support (LTS) channel.
Depending on when your OpenShift Container Platform cluster was installed, you might also see channels for older versions of AMQ Broker.
- b. Under **Installation Mode**, choose which namespaces the Operator watches:
 - A specific namespace on the cluster – The Operator is installed in that namespace and only monitors that namespace for CR changes.
 - All namespaces – The Operator monitors all namespaces for CR changes.



NOTE

If you previously deployed brokers using an earlier version of the Operator, and you want deploy the Operator to watch many namespaces, see [Before you upgrade](#).

7. From the **Installed Namespace** drop-down menu, select the project in which you want to install the Operator.
8. Under **Approval Strategy**, ensure that the radio button entitled **Automatic** is selected. This option specifies that updates to the Operator do not require manual approval for installation to take place.



NOTE

The approval strategy applies only to updates between micro versions of the Operator. Automatic updates between minor Operator versions are not supported. For example, if the current Operator is version 7.13.4, an automatic update to version 7.14.x is not possible. To update between minor versions of the Operator, you must manually uninstall the current Operator and install the new Operator from the channel where Operators for that minor version are made available. For more information, see [Section 6.3, “Manually upgrading the Operator by using the OpenShift web console”](#).

9. Click **Install**.
When the Operator installation is complete, the **Installed Operators** page opens. You should see that the **Red Hat Integration - AMQ Broker for RHEL 9 (Multiarch)** Operator is installed in the project namespace that you specified.

Additional resources

- [Section 3.3.1, “Deploying a single broker instance”](#)

3.3. CREATING OPERATOR-BASED BROKER DEPLOYMENTS

You can deploy a single instance of a broker or create multiple instances that automatically form a cluster and provide message load balancing.

3.3.1. Deploying a single broker instance

You must prepare a custom resource (CR) that contains the configuration for the broker you want to deploy. You can create a CR in a YAML file or in the OpenShift web console.



NOTE

- While you can create more than one broker deployment in a given OpenShift project by deploying multiple Custom Resource (CR) instances, typically, you create a single broker deployment in a project, and then deploy multiple CR instances for addresses.
Red Hat recommends you create broker deployments in separate projects.
- In AMQ Broker 7.14, if you want to configure the following items, you must add the appropriate configuration to the main broker CR instance **before** deploying the CR for the first time.
 - [The size and storage class of the Persistent Volume Claim \(PVC\) required by each broker in a deployment for persistent storage](#)
 - [Limits and requests for memory and CPU for each broker in a deployment](#)

Prerequisites

- The AMQ Broker Operator is installed.
 - To use the OpenShift command-line interface (CLI) to install the AMQ Broker Operator, see [Section 3.1, “Installing the Operator using the CLI”](#).
 - To use the OpenShift web console to install the AMQ Broker Operator, see [Section 3.2, “Installing the Operator from the software catalog by using the Openshift web console”](#).
- You should understand how the Operator chooses a broker container image to use for your broker deployment. For more information, see [Section 2.7, “How the Operator chooses container images when a precise image is not specified in the CR”](#).
- Starting in AMQ Broker 7.3, you use a new version of the Red Hat Ecosystem Catalog to access container images. This new version of the registry requires you to become an authenticated user before you can access images. Before you deploy the broker, complete the steps described in [Red Hat Container Registry Authentication](#).

Procedure

1. Start configuring a Custom Resource (CR) instance for the broker deployment.
 - a. Using the OpenShift command-line interface:
 - i. Log in to OpenShift as a user that has privileges to deploy CRs in the project in which you are creating the deployment.

```
oc login -u <user> -p <password> --server=<host:port>
```

- ii. Open the sample CR file called **broker_activemqartemis_cr.yaml** that was included in the **deploy/crs** directory of the Operator installation archive that you downloaded and extracted.



NOTE

The **broker_activemqartemis_cr.yaml** sample CR uses a naming convention of **ex-aa0**. This naming convention denotes that the CR is an **example** resource for the AMQ Broker **Operator**. When you deploy this sample CR, the resulting StatefulSet uses the name **ex-aa0-ss**. Furthermore, broker Pods in the deployment are directly based on the StatefulSet name, for example, **ex-aa0-ss-0**, **ex-aa0-ss-1**, and so on. The application name in the CR appears in the deployment as a label on the StatefulSet. You might use this label in a Pod selector, for example.

- b. Using the OpenShift Container Platform web console:
 - i. Log in to the console as a user that has privileges to deploy CRs in the project in which you are creating the deployment.
 - ii. Start a new CR instance based on the main broker CRD. In the left pane, click **Administration > Custom Resource Definitions**
 - iii. Click the **ActiveMQArtemis** CRD.
 - iv. Click the **Instances** tab.
 - v. Click **Create ActiveMQArtemis**.
Within the console, a YAML editor opens, enabling you to configure a CR instance.
2. The **size** property specifies the number of brokers to deploy. A value of **2** or greater specifies a clustered broker deployment. However, to deploy a single broker instance, ensure that the value is set to **1**.
3. Deploy the CR instance.
 - a. Using the OpenShift command-line interface:
 - i. Save the CR file.
 - ii. Switch to the project in which you are creating the broker deployment.


```
$ oc project <project_name>
```
 - iii. Create the CR instance.


```
$ oc create -f <path/to/custom_resource_instance>.yaml
```
 - b. Using the OpenShift web console:
 - i. When you have finished configuring the CR, click **Create**.
4. In the OpenShift Container Platform web console, click **Workloads > StatefulSets**. You see a new StatefulSet called **ex-aa0-ss**.
 - a. Click the **ex-aa0-ss** StatefulSet. You see that there is one Pod, corresponding to the single broker that you defined in the CR.

- b. Within the StatefulSet, click the **Pods** tab. Click the **ex-aao-ss** Pod. On the **Events** tab of the running Pod, you see that the broker container has started. The **Logs** tab shows that the broker itself is running.
5. To test that the broker is running normally, access a shell on the broker Pod to send some test messages.
 - a. Using the OpenShift Container Platform web console:
 - i. Click **Workloads > Pods**
 - ii. Click the **ex-aao-ss** Pod.
 - iii. Click the **Terminal** tab.
 - b. Using the OpenShift command-line interface:
 - i. Get the Pod names and internal IP addresses for your project.

```
$ oc get pods -o wide
```

NAME	STATUS	IP
amq-broker-operator-54d996c	Running	10.129.2.14
ex-aao-ss-0	Running	10.129.2.15

- ii. Access the shell for the broker Pod.

```
$ oc rsh ex-aao-ss-0
```

6. From the shell, use the **artemis** command to send some test messages. Specify the internal IP address of the broker Pod in the URL. For example:

```
sh-4.2$ ./amq-broker/bin/artemis producer --url tcp://10.129.2.15:61616 --destination queue://demoQueue
```

The preceding command automatically creates a queue called **demoQueue** on the broker and sends a default quantity of 1000 messages to the queue.

You should see output that resembles the following:

```
Connection brokerURL = tcp://10.129.2.15:61616
Producer ActiveMQQueue[demoQueue], thread=0 Started to calculate elapsed time ...

Producer ActiveMQQueue[demoQueue], thread=0 Produced: 1000 messages
Producer ActiveMQQueue[demoQueue], thread=0 Elapsed time in second : 3 s
Producer ActiveMQQueue[demoQueue], thread=0 Elapsed time in milli second : 3492 milli
seconds
```

Additional resources

- [Section 8.1, "Custom resource configuration reference"](#)
- [Chapter 5, Connecting to AMQ Management Console for an Operator-based broker deployment](#)

3.3.2. Deploying clustered brokers

If you want to deploy multiple brokers that form a cluster, set the value to the **size** attribute to greater than 1 in the custom resource (CR). A clustered configuration enables brokers to connect to each other and redistribute messages as needed, for load balancing.

Prerequisites

- A broker instance is already deployed. See [Section 3.3.1, “Deploying a single broker instance”](#).

Procedure

1. Open the CR that you used to create a broker deployment.
2. For a clustered deployment, ensure that the value of **deploymentPlan.size** is **2** or greater. For example:

```
apiVersion: broker.amq.io/v1beta1
kind: ActiveMQArtemis
metadata:
  name: ex-aao
spec:
  deploymentPlan:
    size: 4
    image: placeholder
  ...
```



NOTE

In the **metadata** section, you need to include the **namespace** property and specify a value **only** if you are using the OpenShift Container Platform web console to create your CR instance. The value that you should specify is the name of the OpenShift project for your broker deployment.

3. Save the modified CR file.
4. Log in to OpenShift as a user that has privileges to deploy CRs in the project in which you previously created your broker deployment.

```
$ oc login -u <user> -p <password> --server=<host:port>
```

5. Switch to the project in which you previously created your broker deployment.

```
$ oc project <project_name>
```

6. At the command line, apply the change:

```
$ oc apply -f <path/to/custom_resource_instance>.yaml
```

In the OpenShift Container Platform web console, additional broker Pods starts in your project, according to the number specified in your CR. By default, the brokers running in the project are clustered.

7. Open the **Logs** tab of each Pod. The logs show that OpenShift has established a cluster connection bridge on each broker. Specifically, the log output includes a line like the following:

```
targetConnector=ServerLocatorImpl (identity=(Cluster-connection-
bridge::ClusterConnectionBridge@6f13fb88
```

3.3.3. Applying Custom Resource (CR) changes to running broker deployments

Understand the limitations and effects of applying CR changes to a running broker.

The following are some important things to note about applying Custom Resource (CR) changes to running broker deployments:

- You cannot dynamically update the **persistenceEnabled** attribute in your CR. To change this attribute, scale your cluster down to zero brokers. Delete the existing CR. Then, recreate and redeploy the CR with your changes, also specifying a deployment size.
- If you create a broker deployment with persistent storage (that is, by setting **persistenceEnabled=true** in your CR), you might need to provision Persistent Volumes (PVs) for the AMQ Broker Operator to claim for your broker Pods. If you scale down the size of your broker deployment, the Operator releases any PVs that it previously claimed for the broker Pods that are now shut down. However, if you *remove* your broker deployment by deleting your CR, AMQ Broker Operator **does not** release Persistent Volume Claims (PVCs) for any broker Pods that are still in the deployment when you remove it. In addition, these unreleased PVs are unavailable to any new deployment. In this case, you need to manually release the volumes. For more information, see [Release a persistent volume](#) in the OpenShift documentation.
- In AMQ Broker 7.14, if you want to configure the following items, you must add the appropriate configuration to the main CR instance **before** deploying the CR for the first time.
 - The size and storage class of the Persistent Volume Claim (PVC) required by each broker in a deployment for persistent storage.
 - Limits and requests for memory and CPU for each broker in a deployment.
- During an active scaling event, any further changes that you apply are queued by the Operator and executed only when scaling is complete. For example, suppose that you scale the size of your deployment down from four brokers to one. Then, while scaledown is taking place, you also change the values of the broker administrator user name and password. In this case, the Operator queues the user name and password changes until the deployment is running with one active broker.
- All CR changes – apart from changing the size of your deployment, or changing the value of the **expose** attribute for acceptors, connectors, or the console – cause existing brokers to be restarted. If you have multiple brokers in your deployment, only one broker restarts at a time.

Additional resources

- [Section 4.6.1, “Configuring storage size and storage class”](#)
- [Section 4.7.1, “Configuring broker resource limits and requests”](#)
- [Section 3.1.2, “Deploying the Operator using the CLI”](#)

3.4. CHANGING THE LOGGING LEVEL FOR THE OPERATOR

The default logging level for AMQ Broker Operator is **info**, which logs information and error messages. You can change the default logging level to increase or decrease the detail that is written to the Operator logs.

If you use the OpenShift Container Platform command-line interface to install the Operator, you can set the new logging level in the Operator configuration file, **operator.yaml**, either before or after you install the Operator. If you use the OpenShift web console, you can use the OpenShift Container Platform web console to set the logging level in the Operator subscription after you install the Operator.

The other available logging levels for the Operator are:

error

Writes error messages only to the log.

debug

Write all messages to the log including debugging messages.

Procedure

1. Using the OpenShift Container Platform command-line interface:

- a. Log in as a cluster administrator. For example:

```
$ oc login -u system:admin
```

- b. If the Operator is not installed, complete the following steps to change the logging level.

- i. In the **deploy** directory of the Operator archive that you downloaded and extracted, open the **operator.yaml** file.
- ii. Change the value of the **zap-log-level** attribute to **debug** or **error**. For example:

```
apiVersion: apps/v1
kind: Deployment
metadata:
  labels:
    control-plane: controller-manager
  name: amq-broker-controller-manager
spec:
  containers:
  - args:
    - --zap-log-level=error
  ...
```

- iii. Save the **operator.yaml** file.

- iv. Install the Operator.

- c. If the Operator is already installed, use the **sed** command to change the logging level in the **deploy/operator.yaml** file and redeploy the Operator. For example, the following command changes the logging level from **info** to **error** and redeploys the Operator:

```
$ sed 's/--zap-log-level=info/--zap-log-level=error/' deploy/operator.yaml | oc apply -f -
```

2. Using the OpenShift Container Platform web console:

- a. Log in to the OpenShift Container Platform as a cluster administrator.
- b. In the left pane, click **Ecosystem > Installed Operators**
- c. Click the **Red Hat Integration - AMQ Broker for RHEL 9 (Multiarch)** Operator.
- d. Click the **Subscriptions** tab.
- e. Click **Actions**.
- f. Click **Edit Subscription**.
- g. Click the **YAML** tab.
Within the console, a YAML editor opens, enabling you to edit the subscription.
- h. In the **config** element, add an environment variable called **ARGS** and specify a logging level of **info**, **debug** or **error**. In the following example, an **ARGS** environment variable that specifies a logging level of **debug** is passed to the Operator container.

```
apiVersion: operators.coreos.com/v1alpha1
kind: Subscription
spec:
  ...
  config:
    env:
      - name: ARGS
        value: "--zap-log-level=debug"
    ...
```

- i. Click Save.

3.5. CONFIGURING LEADER ELECTION SETTINGS FOR THE OPERATOR

Multiple instance of the Operator might be running when, for example, rolling out an upgrade. When multiple instances run concurrently, it is necessary to avoid contention between Operators via leader election. You can customize the settings used by the AMQ Broker Operator for leader elections.

If you use the OpenShift Container Platform command-line interface to install the operator, you can configure the leader elections settings in the operator configuration file, **operator.yaml**, either before or after installation. If you use the OpenShift web console, you can use the OpenShift Container Platform web console to configure the leader elections settings in the operator subscription after installation.

Procedure

1. Using the OpenShift Container Platform web console:
 - a. Log in to the OpenShift Container Platform as a cluster administrator.
 - b. In the left pane, click **Ecosystem > Installed Operators**
 - c. Click the **Red Hat Integration - AMQ Broker for RHEL 9 (Multiarch)** Operator.
 - d. Click the **Subscriptions** tab.

- e. Click **Actions**.
- f. Click **Edit Subscription**.
- g. Click the **YAML** tab.
Within the console, a YAML editor opens, enabling you to edit the subscription.
- h. In the **config** section, add an environment variable named **ARGS** and specify the leader election settings in the variable value. For example:

```
apiVersion: operators.coreos.com/v1alpha1
kind: Subscription
spec:
  ..
  config:
    env:
      - name: ARGS
        value: "--lease-duration=18 --renew-deadline=12 --retry-period=3"
```

- i. Click **Save**.

lease-duration

The duration, in seconds, that a non-leader operator waits before it attempts to acquire the lease that was not renewed by the previous leader. The default is 15.

renew-deadline

The duration, in seconds, the operator waits between attempts to renew the leader role before it stops leading. The default is 10.

retry-period

The duration, in seconds, that the operator waits between attempts to acquire and renew the leader role. The default is 2.

2. Using the OpenShift Container Platform command-line interface:

- a. Log in as a cluster administrator. For example:

```
$ oc login -u system:admin
```

- b. In the **deploy** directory of the operator archive that you downloaded and extracted, open the **operator.yaml** file.
- c. Set the values of the leader election settings. For example:

```
apiVersion: apps/v1
kind: Deployment
...
template
  ..
  spec:
    containers:
      - args:
        - --lease-duration=60
        - --renew-deadline=40
        - --retry-period=5
      ..
```

- d. Save the **operator.yaml** file.
- e. If the operator is already installed, apply the updated settings.

```
$ oc apply -f deploy/operator.yaml
```

- f. If the operator is not installed, install the operator.

3.6. VIEWING STATUS INFORMATION FOR YOUR BROKER DEPLOYMENT

You can view the status of a series of standard conditions reported by OpenShift Container Platform on the state of your broker deployment. You can also view additional status information provided in the Custom Resource (CR) for your deployment.

Procedure

1. Open the CR instance for the broker deployment.
 - a. Using the OpenShift command-line interface:
 - i. Log in to OpenShift Container Platform as a user that has privileges to view CRs in the project for the broker deployment.
 - ii. View the CR for your deployment.

```
$ oc get ActiveMQArtemis <CR instance name> -n <namespace> -o yaml
```

- b. Using the OpenShift Container Platform web console:
 - i. Log in to the console as a user that has privileges to deploy CRs in the project for the broker deployment.
 - ii. In the left pane, click **Ecosystem > Installed Operators**
 - iii. Click the **Red Hat Integration - AMQ Broker for RHEL 9 (Multiarch)** operator.
 - iv. Click the **ActiveMQ Artemis** tab.
 - v. Click the name of the ActiveMQ Artemis instance.

2. View the status of the OpenShift Container Platform conditions for your broker deployment.

- a. Using the OpenShift command-line interface:
 - i. Go to the **status** section of the CR and view the **conditions** details.

- b. Using the OpenShift Container Platform web console:

- i. In the **Details** tab, scroll down to the **Conditions** section.

A condition has a status and a type. It might also have a reason, a message and other details. A condition has a status value of **True** if the condition is met, **False** if the condition is not met, or **Unknown** if the status of the condition cannot be determined.

The **Valid** condition can also have a status of **Unknown** to flag an anomaly in the configuration that does not affect the broker deployment. For more information, see [Section 2.8, “Validation of image and version configuration in a custom resource \(CR\)”](#).

Status information is provided for the following conditions:

Table 3.1. Status information for a broker deployment

Condition name	Displays the status of...
Valid	The validation of the CR. If the status of the Valid condition is False , the Operator does not complete the reconciliation and update the StatefulSet until you first resolve the issue that caused the false status.
Deployed	The availability of the StatefulSet, Pods and other resources.
Ready	A top-level condition which summarizes the other more detailed conditions. The Ready condition has a status of True only if none of the other conditions have a status of False .
BrokerPropertiesApplied	The properties configured in the CR that use the brokerProperties attribute. For more information about the BrokerPropertiesApplied condition, see Section 2.4, “Configuring items not exposed in a custom resource definition (CRD)” .
JaasPropertiesApplied	The Java Authentication and Authorization Service (JAAS) login modules configured in the CR. For more information about the JaasPropertiesApplied condition, see Section 4.3.1, “Setting up user authentication by creating new JAAS login modules” .
ScalingDown	Information relating to a scaledown operation that is in progress. This condition is displayed only while a scaledown is in progress.

- View additional status information for your broker deployment in the **status** section of the CR. The following additional status information is displayed:

deploymentPlanSize

The number of broker Pods in the deployment.

podstatus

The status and name of each broker pod in the deployment.

version

The version of the broker and the registry URLs of the broker and init container images that are deployed.

upgrade

The ability of the Operator to apply major, minor, patch and security updates to the deployment, which is determined by the values of the **spec.deploymentPlan.image** and **spec.version** attributes in the CR.

- If the **spec.deploymentPlan.image** attribute specifies the registry URL of a broker container image, the status of all upgrade types is **False**, which means that the Operator cannot upgrade the existing container images.
- If the **spec.deploymentPlan.image** attribute is not in the CR or has a value of **placeholder**, the configuration of the **spec.version** attribute affects the **upgrade** status as follows:
 - The status of **securityUpdates** is **True**, irrespective of whether the **spec.version** attribute is configured or its value.
 - The status of **patchUpdates** is **True** if the value of the **spec.version** attribute has only a major and a minor version, for example, '7.12', so the Operator can upgrade to the latest patch version of the container images.
 - The status of **minorUpdates** is **True** if the value of the **spec.version attribute** has only a major version, for example, '7', so the Operator can upgrade to the latest minor and patch versions of the container images.
 - The status of **majorUpdates** is **True** if the **spec.version** attribute is not in the CR, so any available upgrades can be deployed, including an upgrade from 7.x.x to 8.x.x, if this version is available.

CHAPTER 4. CONFIGURING A BROKER

You can modify the settings for a broker by updating the custom resource (CR) that you used to create the broker.

4.1. HOW THE OPERATOR GENERATES THE BROKER CONFIGURATION

Before you use custom resource (CR) instances to configure your broker deployment, learn how the Operator uses the configuration in a CR to generate the final **broker.xml** configuration file.

When you create an Operator-based broker deployment, a Pod for each broker runs in a StatefulSet in your OpenShift project. An application container for the broker runs within each Pod.

The Operator runs a type of container called an *Init Container* when initializing each Pod. In OpenShift Container Platform, Init Containers are specialized containers that run before application containers. Init Containers can include utilities or setup scripts that are not present in the application image.

By default, the AMQ Broker Operator uses a built-in Init Container. The Init Container uses the main CR instance for your deployment to generate the configuration used by each broker application container.

If you specify address settings in the CR, the Operator generates a default configuration and then merges or replaces that configuration with the configuration specified in the CR. This process is described in the section that follows.

4.1.1. How the Operator generates the address settings configuration

A default address settings configuration is generated for the broker. If you specify a custom address settings configuration in the broker custom resource (CR) instance for your deployment, the default configuration is updated.

1. The Operator runs the Init Container before the broker application container. The Init Container generates a **default** address settings configuration. The default address settings configuration is shown below.

```
<address-settings>
  <!--
    if you define auto-create on certain queues, management has to be auto-create
  -->
  <address-setting match="activemq.management#">
    <dead-letter-address>DLQ</dead-letter-address>
    <expiry-address>ExpiryQueue</expiry-address>
    <redelivery-delay>0</redelivery-delay>
  <!--
    with -1 only the global-max-size is in use for limiting
  -->
  <max-size-bytes>-1</max-size-bytes>
  <message-counter-history-day-limit>10</message-counter-history-day-limit>
  <address-full-policy>PAGE</address-full-policy>
  <auto-create-queues>true</auto-create-queues>
  <auto-create-addresses>true</auto-create-addresses>
  <auto-create-jms-queues>true</auto-create-jms-queues>
  <auto-create-jms-topics>true</auto-create-jms-topics>
</address-setting>
```

```

<!-- default for catch all -->
<address-setting match="#">
  <dead-letter-address>DLQ</dead-letter-address>
  <expiry-address>ExpiryQueue</expiry-address>
  <redelivery-delay>0</redelivery-delay>
  <!--
    with -1 only the global-max-size is in use for limiting
  -->
  <max-size-bytes>-1</max-size-bytes>
  <message-counter-history-day-limit>10</message-counter-history-day-limit>
  <address-full-policy>PAGE</address-full-policy>
  <auto-create-queues>true</auto-create-queues>
  <auto-create-addresses>true</auto-create-addresses>
  <auto-create-jms-queues>true</auto-create-jms-queues>
  <auto-create-jms-topics>true</auto-create-jms-topics>
</address-setting>
</address-settings>

```

2. If you also specified an address settings configuration in your Custom Resource (CR) instance, the Init Container processes that configuration and converts it to XML.
3. Based on the value of the **applyRule** property in the CR, the Init Container *merges* or *replaces* the default address settings configuration shown above with the configuration that you have specified in the CR. The result of this merge or replacement is the final address settings configuration that the broker will use.
4. When the Init Container has finished generating the broker configuration (including address settings), the broker application container starts. When starting, the broker container copies its configuration from the installation directory previously used by the Init Container. You can inspect the address settings configuration in the **broker.xml** configuration file. For a running broker, this file is located in the **/home/jboss/amq-broker/etc** directory.

Additional resources

- [Section 4.2.2, "Configuring address settings"](#)

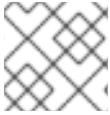
4.1.2. Directory structure of a broker Pod

Get familiar with the directory structure of a AMQ Broker Pod to help you locate configuration files, data storage directories, and logs.

When you create a broker deployment, a Pod for each broker runs in a StatefulSet in your OpenShift project. An application container for the broker runs within each Pod.

The Operator runs a type of container called an *Init Container* when initializing each Pod. In OpenShift Container Platform, Init Containers are specialized containers that run before application containers. Init Containers can include utilities or setup scripts that are not present in the application image.

When generating the configuration for a broker instance, the Init Container uses files contained in a default installation directory. This installation directory is on a volume that the Operator mounts to the broker Pod and which the Init Container and broker container share. The path that the Init Container uses to mount the shared volume is defined in an environment variable called **CONFIG_INSTANCE_DIR**. The default value of **CONFIG_INSTANCE_DIR** is **/amq/init/config**. In the documentation, this directory is referred to as **<install_dir>**.

**NOTE**

You cannot change the value of the **CONFIG_INSTANCE_DIR** environment variable.

By default, the installation directory has the following sub-directories:

Table 4.1. Pod directories

Sub-directory	Contents
<install_dir>/bin	Binaries and scripts needed to run the broker.
<install_dir>/etc	Configuration files.
<install_dir>/data	The broker journal.
<install_dir>/lib	JARs and libraries needed to run the broker.
<install_dir>/log	Broker log files.
<install_dir>/tmp	Temporary web application files.

When the Init Container has finished generating the broker configuration, the broker application container starts. When starting, the broker container copies its configuration from the installation directory previously used by the Init Container. When the broker Pod is initialized and running, the broker configuration is located in the **/home/jboss/amq-broker** directory (and subdirectories) of the broker.

Additional resources

- [Section 2.7, “How the Operator chooses container images when a precise image is not specified in the CR”](#)
- [Section 4.11, “Specifying a custom Init container image”](#)

4.2. CONFIGURING ADDRESSES AND QUEUES

By default, AMQ Broker automatically creates a queue when a client sends a message to or attempts to consume a message from a queue. You can also create queues manually.

4.2.1. Configuring addresses and queues

You can configure addresses and queues by using the **brokerProperties** attribute in the **ActiveMQArtemis** CR instance for your broker deployment. Or, you can configure addresses and queues in the **ActiveMQArtemisAddress** CR.

**NOTE**

The **ActiveMQArtemisAddress** CR is deprecated in AMQ Broker 7.12.

4.2.1.1. Configuring addresses and queues by using `brokerProperties`

You can configure addresses and queues under the **`brokerProperties`** attribute in the broker custom resource (CR) and also configure settings for each queue that you create.

Prerequisites

You created a broker deployment. For more information, see [Section 3.3.1, “Deploying a single broker instance”](#).

Procedure

1. Edit the **ActiveMQArtemis** CR instance for your broker deployment.
2. In the **`spec`** section of the CR, add a **`brokerProperties`** attribute if it is not already in the CR.

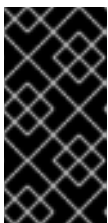
```
spec:
  ...
  brokerProperties:
  ...
```

3. Configure an address in the format:
- **`addressConfigurations.<address name>.routingTypes=<routing type>`**

For example:

```
spec:
  ...
  brokerProperties:
  - addressConfigurations.usa-news-address.routingTypes=MULTICAST
  ...
```

4. Configure a queue for the address you created in the format:
- **`addressConfigurations.<address name>.queueConfigs.<queue name>.address=<address>`**



IMPORTANT

The value of `<address>` for the **`.address`** setting must match the `<address name>` for each queue you create. If these values are different, separate addresses are created for each. In the following example, both the address name and the **`.address`** setting have the same value of **`usa-news-address`**.

```
spec:
  ...
  brokerProperties:
  - addressConfigurations.usa-news-address.queueConfigs.usa-news-queue.address=usa-news-address
  ...
```

5. Add a separate line for each setting you want to configure for a queue in the format:
 - **`addressConfigurations.<address name>.queueConfigs.<queue name>.<queue setting>`**

For example:

```
spec:
  ...
  brokerProperties:
    - addressConfigurations.usa-news-address.queueConfigs.usa-news-
      queue.routingType=ANYCAST
    - addressConfigurations.usa-news-address.queueConfigs.usa-news-
      queue.purgeOnNoConsumers=true
    - addressConfigurations.usa-news-address.queueConfigs.usa-news-
      queue.maxConsumers=20
  ...
```

6. Save the CR.
7. Check that no errors were detected in the **brokerProperties** configuration by reviewing the **status** section of the **ActiveMQArtemis** CR. For more information, see [Section 2.4](#), “Configuring items not exposed in a custom resource definition (CRD)”.

4.2.1.2. Configuring addresses and queues in the **ActiveMQArtemisAddress** CR

You can configure addresses and queues in the **ActiveMQArtemisAddress** CR. You must create a separate, uniquely named, CR instance for each address and/or queues that you want to create on the broker.

Prerequisites

- You installed the custom resource definition (CRD) required to create addresses and queues on your brokers by installing the AMQ Broker Operator. For information on two alternative ways to install the Operator, see:
 - [Section 3.1, “Installing the Operator using the CLI”](#).
 - [Section 3.2, “Installing the Operator from the software catalog by using the Openshift web console”](#).
- You are familiar with how to use a CR instance to create a broker deployment. For more information, see [Section 3.3.1, “Deploying a single broker instance”](#).

Procedure

1. Start configuring a custom resource (CR) instance to define addresses and queues for the broker deployment.
 - a. Using the OpenShift command-line interface:
 - i. Log in to OpenShift as a user that has privileges to deploy CRs in the project for the broker deployment.

```
oc login -u <user> -p <password> --server=<host:port>
```

- ii. Open the sample CR file called **broker_activemqartemisaddress_cr.yaml** that was included in the **deploy/crs** directory of the Operator installation archive that you downloaded and extracted.

- b. Using the OpenShift Container Platform web console:
 - i. Log in to the console as a user that has privileges to deploy CRs in the project for the broker deployment.
 - ii. Start a new CR instance based on the address CRD. In the left pane, click **Administration > Custom Resource Definitions**
 - iii. Click the **ActiveMQArtemisAddresss** CRD.
 - iv. Click the **Instances** tab.
 - v. Click **Create ActiveMQArtemisAddress**
Within the console, a YAML editor opens, enabling you to configure a CR instance.
2. In the **spec** section of the CR, add lines to define an address, queue, and routing type. For example:

```
apiVersion: broker.amq.io/v1beta1
kind: ActiveMQArtemisAddress
metadata:
  name: myAddressDeployment0
  namespace: myProject
spec:
  ...
  addressName: myAddress0
  queueName: myQueue0
  routingType: anycast
  ...
```

The preceding configuration defines an address named **myAddress0** with a queue named **myQueue0** and an **anycast** routing type.



NOTE

In the **metadata** section, you need to include the **namespace** property and specify a value **only** if you are using the OpenShift Container Platform web console to create your CR instance. The value that you should specify is the name of the OpenShift project for your broker deployment.

3. Deploy the CR instance.
 - a. Using the OpenShift command-line interface:
 - i. Save the CR file.
 - ii. Switch to the project for the broker deployment.

```
$ oc project <project_name>
```

- iii. Create the CR instance.

```
$ oc create -f <path/to/address_custom_resource_instance>.yaml
```

- b. Using the OpenShift web console:

- i. When you finish configuring the CR, click **Create**.

4.2.2. Configuring address settings

You configure address settings in either the **addressSettings** or the **brokerProperties** attribute in the **ActiveMQArtemis** CR instance. The precise location depends on whether you create addresses and queues in an **ActiveMQArtemisAddress** CR instance or an **ActiveMQArtemis** broker CR instance.

The following examples show how to use both methods to configure a dead letter address and queue for specific address patterns. A dead letter address and queue can be used by the broker to store messages that cannot be delivered to a client to prevent infinite delivery attempts. A system administrator can later consume undelivered messages from a dead letter queue to inspect the messages.

4.2.2.1. Configuring address settings by using **brokerProperties**

You can specify settings for individual or groups of addresses under the **brokerProperties** attribute in the broker custom resource (CR).

Procedure

1. Edit the **ActiveMQArtemis** CR instance for your broker deployment.
2. Create a dead letter address and queue to receive undelivered messages. For example:

```
spec:
  ...
  brokerProperties:
    ...
    - addressConfigurations.usDeadLetter.routingTypes=MULTICAST
    - addressConfigurations.usDeadLetter.queueConfigs.usDeadLetter-
      queue.address=usDeadLetter
```

For more information on creating addresses and queues by using **brokerProperties**, see, [Section 4.2.1.1, "Configuring addresses and queues by using **brokerProperties**"](#).

3. Add separate lines under the **brokerProperties** attribute in the format **addressSettings**. **<address name>.<address setting>** to:
 - Set the dead letter address for undelivered messages to the dead letter address you created.
 - Specify the number of delivery attempts after which a message that cannot be delivered to a matching address is sent to the dead letter address.

For example:

```
spec:
  ...
  brokerProperties:
    ...
    - addressSettings.usa-news.deadLetterAddress=usDeadLetter
    - addressSettings.usa-news.maxDeliveryAttempts=5
    ...
```

You can use an asterisk (*) or a number sign (#) character as wildcards to create address patterns. Matching of patterns is done at each delimiter boundary, which is represented by a period (.). The number sign character matches any sequence of zero or more words and can be used at the end of the address string. The asterisk character matches a single word and can be used anywhere within the address string. For example:

```
spec:
  ...
  brokerProperties:
    ...
    - addressSettings."usa-news.*".deadLetterAddress=usDeadLetter
    - addressSettings."europe-news.#".deadLetterAddress=euDeadLetter
    ...
```

In the preceding example, the following addresses are matched:

- The **usa-news.*** address pattern matches any word that follows the **usa-news.** string, such as **usa-news.domestic** and **usa-news.intl**, but not **usa-news.domestic.politics**.
- The **europe-news.#** address pattern matches any address that starts with **europe-news**, such as **europe-news**, **europe-news.politics** and **europe-news.politics.fr**.



NOTE

In **brokerProperties** entries, a period (.) is a reserved character. If you want to create an address pattern that contains a period, you must enclose the address in quotation marks. For example, "usa-news.*"

4. Save the CR.
5. Check that no errors were detected in the **brokerProperties** configuration by reviewing the **status** section of the **ActiveMQArtemis** CR. For more information, see [Section 2.4, "Configuring items not exposed in a custom resource definition \(CRD\)"](#).

4.2.2.2. Configuring addresses settings by using **addressSettings** in the **ActiveMQArtemis** custom resource (CR) instance

If you configure addresses and queues in an **ActiveMQArtemisAddress** CR, you must configure settings for those addresses under the **addressSettings** attribute in the broker CR.

The example procedure shows how to configure a setting to limit the delivery attempts for a dead letter address and queue that you configured in an **ActiveMQArtemisAddress** CR.

Prerequisites

You created an address and queue with the following details.

addressName: myDeadLetterAddress

queueName: myDeadLetterQueue

routingType: anycast

For information on creating addresses and queues, see [Section 4.2.1, "Configuring addresses and queues"](#).

Procedure

1. Edit the **ActiveMQArtemis** CR instance for your broker deployment.
2. In the **spec** section of the CR, add a new **addressSettings** section that contains a single **addressSetting** section, as shown below.

```
spec:
  ...
  addressSettings:
    addressSetting:
```

3. Add a single instance of the **match** property to the **addressSetting** block. Specify an address-matching expression. For example:

```
spec:
  ...
  addressSettings:
    addressSetting:
      - match: myAddress
```

match

Specifies the address, or set of addresses to which the broker applies the configuration that follows. In this example, the value of the **match** property corresponds to a single address called **myAddress**.

4. Add properties related to undelivered messages and specify values. For example:

```
spec:
  ...
  addressSettings:
    addressSetting:
      - match: myAddress
      deadLetterAddress: myDeadLetterAddress
      maxDeliveryAttempts: 5
```

deadLetterAddress

Address to which the broker sends undelivered messages.

maxDeliveryAttempts

Maximum number of delivery attempts that a broker makes before moving a message to the configured dead letter address.

In the preceding example, if the broker makes five unsuccessful attempts to deliver a message to an address that begins with **myAddress**, the broker moves the message to the specified dead letter address, **myDeadLetterAddress**.

5. (Optional) Apply similar configuration to another address or set of addresses. For example:

```
spec:
  ...
  addressSettings:
    addressSetting:
      - match: myAddress
```

```

deadLetterAddress: myDeadLetterAddress
maxDeliveryAttempts: 5
- match: 'myOtherAddresses#'
deadLetterAddress: myDeadLetterAddress
maxDeliveryAttempts: 3

```

In this example, the value of the second **match** property includes a hash wildcard character. The wildcard character means that the preceding configuration is applied to **any** address that begins with the string **myOtherAddresses**.



NOTE

If you use a wildcard expression as a value for the **match** property, you must enclose the value in single quotation marks, for example, **'myOtherAddresses#'**.

- At the beginning of the **addressSettings** section, add the **applyRule** property and specify a value. For example:

```

spec:
  ...
  applyRule: merge_all
  addressSetting:
    - match: myAddress
      deadLetterAddress: myDeadLetterAddress
      maxDeliveryAttempts: 5
    - match: 'myOtherAddresses#'
      deadLetterAddress: myDeadLetterAddress
      maxDeliveryAttempts: 3

```

The **applyRule** property specifies how the Operator applies the configuration that you add to the CR for each matching address or set of addresses. The values that you can specify are:

merge_all

- For address settings specified in both the CR **and** the default configuration that match the same address or set of addresses:
 - Replace any property values specified in the default configuration with those specified in the CR.
 - Keep any property values that are specified uniquely in the CR **or** the default configuration. Include each of these in the final, merged configuration.
- For address settings specified in either the CR **or** the default configuration that uniquely match a particular address or set of addresses, include these in the final, merged configuration.

merge_replace

- For address settings specified in both the CR **and** the default configuration that match the same address or set of addresses, include the settings specified in the **CR** in the final, merged configuration. **Do not** include any properties specified in the default configuration, even if these are not specified in the CR.
- For address settings specified in either the CR **or** the default configuration that uniquely

- For address settings specified in either the CR or the default configuration that uniquely match a particular address or set of addresses, include these in the final, merged configuration.

replace_all

Replace **all** address settings specified in the default configuration with those specified in the CR. The final, merged configuration corresponds exactly to that specified in the CR.



NOTE

If you do not explicitly include the **applyRule** property in your CR, the Operator uses a default value of **merge_all**.

- Save the CR instance.

4.2.2.3. Configurable address and queue settings

The address and queue settings that you can configure for a broker on OpenShift are **fully equivalent** to those of standalone broker deployed on Linux or Windows. However, the format of address and queue settings names on OpenShift is camel case in contrast to lower case for standalone brokers.

For OpenShift deployments, address and queue settings are in *camel case*, for example, **defaultQueueRoutingType**. For standalone deployments, address and queue settings are in lower case and use a dash (-) separator, for example, **default-queue-routing-type**.

The following table shows some further examples of this naming difference.

Table 4.2. Examples of differences in the names of configuration items

Configuration item for standalone broker deployment	Configuration item for OpenShift broker deployment
address-full-policy	addressFullPolicy
auto-create-queues	autoCreateQueues
default-queue-routing-type	defaultQueueRoutingType
last-value-queue	lastValueQueue

Additional resources

- [Section 4.2.2, “Configuring address settings”](#)
- [Section 8.1, “Custom resource configuration reference”](#)
- [Configuring addresses and queues](#)

4.2.3. Deleting addresses and queues

Depending on how you created addresses and queues, you can delete addresses and queues by removing **brokerProperties** entries in the **ActiveMQArtemis** CR for your broker deployment or by using the **ActiveMQArtemisAddress** CR.

4.2.3.1. Deleting addresses and queues that were created using **brokerProperties**

You can delete individual addresses and queues by removing the entries from under the **brokerProperties** attribute.

Prerequisites

- You created a broker deployment. For more information, see [Section 3.3.1, “Deploying a single broker instance”](#).

Procedure

1. Edit the **ActiveMQArtemis** CR instance for your broker deployment.
2. Add the following **brokerProperties** entries to allow the broker to delete any address, which is represented by the number sign (#), and associated queues that it no longer finds in the CR.

```
spec:
  ...
  brokerProperties:
    - addressSettings.#.configDeleteAddresses=FORCE
    - addressSettings.#.configDeleteQueues=FORCE
  ...
```

3. Under the **brokerProperties** attribute, delete all the lines that reference an address and queue that you want to remove. For example, delete all the lines that reference the **usa-news** address to remove this address and queue:

```
spec:
  ...
  brokerProperties:
    - addressConfigurations.usa-news.queueConfigs.usa-news-
      queue.routingType=MULTICAST
    - addressConfigurations.usa-news.queueConfigs.usa-news-
      queue.purgeOnNoConsumers=true
    - addressConfigurations.usa-news.queueConfigs.usa-news-queue.maxConsumers=20
  ...
```

4. Save the CR.
When the broker applies the updated configuration, it deletes addresses and queues that you removed from the CR.

4.2.3.2. Deleting addresses and queues in the **ActiveMQArtemisAddress** CR

You can delete addresses and queues in the **ActiveMQArtemisAddress** CR if you created the addresses and queues in the CR.

Procedure

1. Ensure that you have an address CR file with the details, for example, the **name**, **addressName** and **queueName**, of the address and queue you want to delete. For example:

```
apiVersion: broker.amq.io/v1beta1
kind: ActiveMQArtemisAddress
metadata:
  name: myAddressDeployment0
  namespace: myProject
spec:
  ...
  addressName: myAddress0
  queueName: myQueue0
  routingType: anycast
  ...
```

2. In the **spec** section of the address CR, add the **removeFromBrokerOnDelete** attribute and set it to a value of **true**.

```
..
spec:
  addressName: myAddress1
  queueName: myQueue1
  routingType: anycast
  removeFromBrokerOnDelete: true
```

Setting the **removeFromBrokerOnDelete** attribute to **true** causes the Operator to remove the address and any associated message for all brokers in the deployment when you delete the address CR.

3. Apply the updated address CR to set the **removeFromBrokerOnDelete** attribute for the address you want to delete.

```
$ oc apply -f <path/to/address_custom_resource_instance>.yaml
```

4. Delete the address CR to delete the address from the brokers in the deployment.

```
$ oc delete -f <path/to/address_custom_resource_instance>.yaml
```

Additional resources

- [Section 8.1, "Custom resource configuration reference"](#)
- [Configuring addresses and queues](#)
- [Using Init Containers to perform tasks before a pod is deployed](#)

4.3. CONFIGURING AUTHENTICATION AND AUTHORIZATION

You can configure security settings, including authentication using Java Authentication and Authorization Service (JAAS) login modules and authorization rules.

By default, AMQ Broker uses a Java Authentication and Authorization Service (JAAS) properties login module to authenticate and authorize users. The configuration for the default JAAS login module is stored in a **/home/jboss/amq-broker/etc/login.config** file on each broker Pod and reads user and role

information from the **artemis-users.properties** and **artemis-roles.properties** files in the same directory. You add the user and role information to the properties files in the default login module by updating the **ActiveMQArtemisSecurity** Custom Resource (CR).

An alternative to updating the **ActiveMQArtemisSecurity** CR to add user and role information to the default properties files is to configure one or more JAAS login modules in a secret. This secret is mounted as a file on each broker Pod. Configuring JAAS login modules in a secret offers the following advantages over using the **ActiveMQArtemisSecurity** CR to add user and role information.

- If you configure a properties login module in a secret, the brokers do not need to restart each time you update the property files. For example, when you add a new user to a properties file and update the secret, the changes take effect without requiring a restart of the broker.
- You can configure JAAS login modules that are not defined in the **ActiveMQArtemisSecurity** CRD to authenticate users. For example, you can configure an LDAP login module or any other JAAS login module.

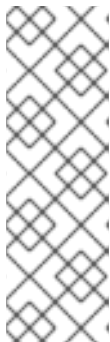
Both methods of configuring authentication and authorization for AMQ Broker are described in the following sections.

4.3.1. Setting up user authentication by creating new JAAS login modules

You can configure new JAAS login modules, which are stored in an OpenShift secret, to authenticate users with AMQ Broker. In the **ActiveMQArtemis** custom resource (CR), you must assign broker permissions to roles that are defined in the new login module.

Procedure

1. Create a text file with your new JAAS login modules configuration and save the file as **login.config**. By saving the file as **login.config**, the correct key is inserted in the secret that you create from the text file.



NOTE

You must include the credentials required by the Operator to authenticate with the broker in the **login.config** file. You can do this by adding the default properties login module, which uses the **artemis-users.properties** and **artemis.roles.properties** files, as shown in the following examples. Alternatively, you can add the contents of the **artemis-users.properties** and **artemis.roles.properties** to other user and roles properties files configured in the **login.config** file.

Example login.config files

The following example has a properties login module that is used to authenticate users and the default properties login module required by the Operator to authenticate with the broker:

```
activemq {
  org.apache.activemq.artemis.spi.core.security.jaas.PropertiesLoginModule sufficient
    reload=true
    org.apache.activemq.jaas.properties.user="new-users.properties"
    org.apache.activemq.jaas.properties.role="new-roles.properties";

  org.apache.activemq.artemis.spi.core.security.jaas.PropertiesLoginModule sufficient
```



```

reload=false
org.apache.activemq.jaas.properties.user="artemis-users.properties"
org.apache.activemq.jaas.properties.role="artemis-roles.properties"
baseDir="/home/jboss/amq-broker/etc";
};

```

The following example has an LDAP login module that is used to authenticate users and the default properties login module required by the Operator to authenticate with the broker:

```

activemq {
  org.apache.activemq.artemis.spi.core.security.jaas.LDAPLoginModule sufficient
    debug=true
    reload=true
    initialContextFactory="com.sun.jndi.ldap.LdapCtxFactory"
    connectionURL="ldap://ldap.company.com:389"
    connectionUsername="cn=read-only-admin,dc=example,dc=com"
    connectionPassword="password"
    connectionProtocol="s"
    connectionTimeout="5000"
    authentication="simple"
    userBase="dc=example,dc=com"
    userSearchMatching="(uid={0})"
    userSearchSubtree=true
    readTimeout="5000"
    roleBase="dc=example,dc=com"
    roleName="cn"
    roleSearchMatching="(uniqueMember={0})"
    roleSearchSubtree=false
  ;
  org.apache.activemq.artemis.spi.core.security.jaas.PropertiesLoginModule sufficient
    reload=false
    org.apache.activemq.jaas.properties.user="artemis-users.properties"
    org.apache.activemq.jaas.properties.role="artemis-roles.properties"
    baseDir="/home/jboss/amq-broker/etc"
  ;
};

```



NOTE

Place the properties files that contain the credentials used by the Operator to authenticate with the broker at the end of the **login.config** file. This ensures that the properties files in all login modules are loaded when the broker authenticates the Operator. Otherwise, the custom resource might have a status message showing that some property files are not visible on the broker, until all login modules are used.

2. If the **login.config** file you created includes a properties login module, ensure that the users and roles files specified in the module contain user and role information. For example:

new-users.properties

```

ruben=ruben01!
anne=anne01!
rick=rick01!
bob=bob01!

```

new-roles.properties

```
admin=ruben, rick
group1=bob
group2=anne
```

3. Use the **oc create secret** command to create a secret from the text file that you created with the new login module configuration. If the login module configuration includes a properties login module, also include the associated users and roles files in the secret. For example:

```
oc create secret generic custom-jaas-config --from-file=login.config --from-file=new-
users.properties --from-file=new-roles.properties
```

**NOTE**

The secret name must have a suffix of **-jaas-config** so the Operator can recognize that the secret contains login module configuration and propagate any updates to each broker pod.

For more information about how to create secrets, see [Secrets](#) in the Kubernetes documentation.

4. Add the secret you created to the custom resource (CR) instance for your broker deployment.
 - a. Create an **extraMounts** element and a **secrets** element and add the name of the secret. The following example adds a secret named **custom-jaas-config** to the CR.

```
deploymentPlan:
...
extraMounts:
  secrets:
    - "custom-jaas-config"
...
```

5. In the CR, grant permissions to the roles that are configured on the broker.
 - a. In the **spec** section of the CR, add a **brokerProperties** element and add the permissions. You can grant a role permissions to a single address. Or, you can specify a wildcard match using the **#** sign to grant a role permissions to all addresses. For example:

```
spec:
...
brokerProperties:
  - securityRoles.#.group2.send=true
  - securityRoles.#.group1.consume=true
  - securityRoles.#.group1.createAddress=true
  - securityRoles.#.group1.createNonDurableQueue=true
  - securityRoles.#.group1.browse=true
...
```

In the example, the group2 role is assigned **send** permissions to all addresses and the group1 role is assigned **consume**, **createAddress**, **createNonDurableQueue** and **browse** permissions to all addresses.



NOTE

In a Java properties file, a colon (:) is a reserved character that is used to separate a key and a value in a key/value pair. If you want to grant permissions to a fully qualified queue name (FQQN), which consists of an address name and a queue name separated by colons (::), you must use the backslash (\) character to escape the colon characters in the FQQN. For example:

```
spec:
  ...
  brokerProperties:
    - 'securityRoles."my-address\:\:my-queue".group2.send=true'
```

6. Save the CR.

The Operator mounts the **login.config** file in the secret in a **/amq/extra/secrets/secret name** directory on each pod and configures the broker JVM to read the mounted **login.config** file instead of the default **login.config** file. If the **login.config** file contains a properties login module, the referenced users and roles properties file are also mounted on each pod.

7. View the status information in the CR to verify that the brokers in your deployment are using the JAAS login modules in the secret for authentication.

- a. In the CR, navigate to the **status** section.
- b. Verify that a **JaasPropertiesApplied** type is present, which indicates that the broker is using the JAAS login modules configured in the secret. For example:

```
- lastTransitionTime: "2023-02-06T20:50:01Z"
  message: ""
  reason: Applied
  status: "True"
  type: JaasPropertiesApplied
```

When you update any of the files in the secret, the value of the **reason** field shows **OutOfSync** until OpenShift Container Platform propagates the latest files in the secret to each broker pod. For example, if you add a new user to the **new-users-properties** file and update the secret, you see the following status information until the updated file is propagated to each pod:

```
- lastTransitionTime: "2023-02-06T20:55:20Z"
  message: 'new-users.properties status out of sync, expected: 287641156, current: 2177044732'
  reason: OutOfSync
  status: "False"
  type: JaasPropertiesApplied
```

8. When you update user or role information in a properties file that is referenced in the secret, use the **oc set data** command to update the secret. You must add all the files to the secret again, including the **login.config** file. For example, if you add a new user to the **new-users.properties**

file that you created earlier in this procedure, use the following command to update the **custom-jaas-config** secret:

```
oc set data secret/custom-jaas-config --from-file=login.config=login.config --from-file=new-users.properties=new-users.properties --from-file=new-roles.properties=new-roles.properties
```



NOTE

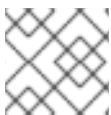
The broker JVM reads the configuration in the **login.config** file only when it starts. If you change the configuration in the **login.config** file, for example, to add a new login module, and update the secret, the broker does not use the new configuration until the broker is restarted.

Additional resources

- [Section 8.2, “Example JAAS login module configurations”](#)
- [JAAS Login Configuration File](#)

4.3.2. Setting up user authentication by using the default JAAS login module

You can use the broker’s default JAAS properties login module to authenticate users with AMQ Broker.



NOTE

The **ActiveMQArtemisSecurity** CR is deprecated starting in AMQ Broker 7.12.

4.3.2.1. Setting up user authentication by using the default JAAS login module

If you want to use the default login module to authenticate users, you must add users and roles and assign permissions to those roles in the **ActiveMQArtemisSecurity** custom resource (CR).

You can deploy the **ActiveMQArtemisSecurity** CR before or after you create a broker deployment. However, if you deploy the security CR after creating the broker deployment, the broker pod is restarted to accept the new configuration.

Prerequisites

- You installed the AMQ Broker Operator. For information on two alternative ways to install the Operator, see:
 - [Section 3.1, “Installing the Operator using the CLI”](#).
 - [Section 3.2, “Installing the Operator from the software catalog by using the Openshift web console”](#).
- You are familiar with broker security as described in [Securing brokers](#)

Procedure

1. Start configuring a Custom Resource (CR) instance to define users and associated security configuration for the broker deployment.
 - a. Using the OpenShift command-line interface:

- i. Log in to OpenShift as a user that has privileges to deploy CRs in the project for the broker deployment.

```
oc login -u <user> -p <password> --server=<host:port>
```

- ii. Edit the CR for your deployment.

```
oc edit ActiveMQArtemis <CR instance name> -n <namespace>
```

- b. Using the OpenShift Container Platform web console:

- i. Log in to the console as a user that has privileges to deploy CRs in the project for the broker deployment.
- ii. In the left pane, click **Ecosystem > Installed Operators**
- iii. Click the **Red Hat Integration - AMQ Broker for RHEL 9 (Multiarch)** operator.
- iv. Click the **AMQ Broker** tab.
- v. Click the name of the ActiveMQArtemis instance name
- vi. Click the **YAML** tab.
Within the console, a YAML editor opens, enabling you to configure a CR instance.

2. In the **spec** section of the CR, add lines to define users and roles. For example:

```
apiVersion: broker.amq.io/v1beta1
kind: ActiveMQArtemisSecurity
metadata:
  name: ex-prop
spec:
  loginModules:
    propertiesLoginModules:
      - name: "prop-module"
    users:
      - name: "sam"
        password: "samspassword"
        roles:
          - "sender"
      - name: "rob"
        password: "robspassword"
        roles:
          - "receiver"
  securityDomains:
    brokerDomain:
      name: "activemq"
      loginModules:
        - name: "prop-module"
          flag: "sufficient"
  securitySettings:
    broker:
      - match: "#"
      permissions:
        - operationType: "send"
        roles:
```

```

- "sender"
- operationType: "createAddress"
  roles:
    - "sender"
- operationType: "createDurableQueue"
  roles:
    - "sender"
- operationType: "consume"
  roles:
    - "receiver"
...

```



NOTE

Always specify values for the elements in the preceding example. For example, if you do not specify values for **securityDomains.brokerDomain** or values for roles, the resulting configuration might cause unexpected results.

The preceding configuration defines two users:

- a **propertiesLoginModule** named **prop-module** that defines a user named **sam** with a role named **sender**.
- a **propertiesLoginModule** named **prop-module** that defines a user named **rob** with a role named **receiver**.

The properties of these roles are defined in the **brokerDomain** and **broker** sections of the **securityDomains** section. For example, the **send** role is defined to allow users with that role to create a durable queue on any address. By default, the configuration applies to all deployed brokers defined by CRs in the current namespace. To limit the configuration to particular broker deployments, use the **applyToCrNames** option described in [Section 8.1.3, "Security Custom Resource configuration reference"](#).



NOTE

In the **metadata** section, you need to include the **namespace** property and specify a value **only** if you are using the OpenShift Container Platform web console to create your CR instance. The value that you should specify is the name of the OpenShift project for your broker deployment.

3. Deploy the CR instance.

- Using the OpenShift command-line interface:
 - Save the CR file.
 - Switch to the project for the broker deployment.

```
$ oc project <project_name>
```

- Create the CR instance.

```
$ oc create -f <path/to/security_custom_resource_instance>.yaml
```

- b. Using the OpenShift web console:
 - i. When you have finished configuring the CR, click **Create**.

Additional resources

- [Section 8.1.3, “Security Custom Resource configuration reference”](#)
- [Section 3.3.1, “Deploying a single broker instance”](#)

4.3.2.2. Storing user passwords in a secret

User passwords are stored in clear text in the **ActiveMQArtemisSecurity** custom resource (CR). You can adopt a more secure way of handling passwords by excluding them from the CR and storing them in a secret instead.

When you apply the CR, the Operator retrieves each user’s password from the secret and inserts it in the **artemis-users.properties** file on the broker pod.

Procedure

1. Use the **oc create secret** command to create a secret and add each user’s name and password. The secret name must follow a naming convention of **security-properties-module name**, where *module name* is the name of the login module configured in the CR. For example:

```
oc create secret generic security-properties-prop-module \
  --from-literal=sam=samspassword \
  --from-literal=rob=robspassword
```

2. In the **spec** section of the CR, add the user names that you specified in the secret along with the role information, but do not include each user’s password. For example:

```
apiVersion: broker.amq.io/v1beta1
kind: ActiveMQArtemisSecurity
metadata:
  name: ex-prop
spec:
  loginModules:
    propertiesLoginModules:
      - name: "prop-module"
        users:
          - name: "sam"
            roles:
              - "sender"
          - name: "rob"
            roles:
              - "receiver"
  securityDomains:
    brokerDomain:
      name: "activemq"
      loginModules:
        - name: "prop-module"
          flag: "sufficient"
  securitySettings:
    broker:
```

```

- match: "#"
  permissions:
    - operationType: "send"
      roles:
        - "sender"
    - operationType: "createAddress"
      roles:
        - "sender"
    - operationType: "createDurableQueue"
      roles:
        - "sender"
    - operationType: "consume"
      roles:
        - "receiver"
    ...

```

3. Deploy the CR instance.

a. Using the OpenShift command-line interface:

i. Save the CR file.

ii. Switch to the project for the broker deployment.

```
$ oc project <project_name>
```

iii. Create the CR instance.

```
$ oc create -f <path/to/address_custom_resource_instance>.yaml
```

b. Using the OpenShift web console:

i. When you finish configuring the CR, click **Create**.

Additional resources

- [Providing sensitive data to pods](#)

4.3.3. Changing the default security manager

By default, AMQ Broker uses the **ActiveMQJAASSecurityManager** security manager to handle user authentication and authorization. You can configure the broker to use a different security manager if you want more control over the implementation of broker security.

Procedure

1. Edit the custom resource (CR) instance for your broker deployment.
2. In the **spec** section of the CR, add an **env** element and add Java system properties to the value of a **JAVA_ARGS_APPEND** environment variables. For example:

```

apiVersion: broker.amq.io/v1beta1
kind: ActiveMQArtemis
metadata:
  name: ex-aao

```



```
spec:
  ...
  env:
    - name: JAVA_ARGS_APPEND
      value: -Dsecurityconfig.manager.className=<class name> -
        Dsecurityconfig.manager.properties.<key>
    ...
```

securityconfig.manager.className

The fully qualified class name of the security manager implementation.

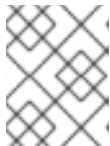
securityconfig.manager.properties.<key>

Properties to pass to the security manager. Each property key after the **properties.** prefix is passed as a configuration property to the security manager.

For example:

```
apiVersion: broker.amq.io/v1beta1
kind: ActiveMQArtemis
metadata:
  name: ex-aa0
spec:
  ...
  env:
    - name: JAVA_ARGS_APPEND
      value: -
        Dsecurityconfig.manager.className=org.apache.activemq.artemis.spi.core.security.ActiveMQBasicSecurityManager -Dsecurityconfig.manager.properties.bootstrapUser=admin -
        Dsecurityconfig.manager.properties.bootstrapPassword=admin
    ...
```

3. Save the CR.



NOTE

If you change the default security domain used to handle authentication, the broker ignores any custom security manager configured."

4.3.4. Changing the default JAAS security domain

By default, AMQ Broker uses the **activemq** security domain to handle user authentication and authorization. If you have multiple security domains configured in the **login.config** file, you can configure the broker to use a different default domain.

Prerequisites

You configured the security domain that you want the broker to use in the **login.config** file.

Procedure

1. Edit the custom resource (CR) instance for your broker deployment.
2. In the **spec** section of the CR, add an **env** element and add Java system properties to the value of a **JAVA_ARGS_APPEND** environment variables. For example:

```

apiVersion: broker.amq.io/v1beta1
kind: ActiveMQArtemis
metadata:
  name: ex-aao
spec:
  ...
  env:
    - name: JAVA_ARGS_APPEND
      value: -Dsecurityconfig.jaas.domain=MyDomain -
        Dsecurityconfig.jaas.certificateDomain=MyCertDomain
    ...

```

securityconfig.jaas.domain

The default JAAS security domain to use for client authentication.

securityconfig.jaas.certificateDomain

The default JAAS security domain for certificate-based authentication. Use this domain to authenticate clients by using their TLS/SSL certificates.

3. Save the CR.

4.4. ADDING THIRD-PARTY JAR FILES

You can add third-party JAR files, such as JDBC database drivers, to your deployment so the broker can access external resources at run time.

You must configure the Operator to make the third-party JAR file available on a mounted volume on each broker pod and add the volume path for the JAR file to the broker's Java classpath.

If a JAR file is less than 1 MB in size, you can add the JAR file to a secret or ConfigMap and configure the Operator to mount the JAR file on each broker pod. If a JAR file is larger than the 1 MB limit for secrets and ConfigMaps, you can configure the Operator to mount a shared volume on each broker pod and download the JAR file to that volume.

4.4.1. Using a secret or configMap to mount a JAR file on broker pods

If a JAR file for an external resource, such as a JDBC database, is less than 1 MB, you can use a secret or configMap to mount the third-party JAR file on each broker pod. You must also modify the broker's Java classpath to load the JAR file from the mounted location at runtime.

The size limit of 1 MB is an OpenShift limit for an individual OpenShift secret or configMap.

The following procedure assume that you are using a secret to mount the JAR file.

Procedure

1. Use the **oc create secret** command to create a secret that contains the third-party JAR file that you want to add. For example:

```
oc create secret generic log4j-template --from-file=log4j-layout-template-json-2.22.1.jar
```

For more information about how to create secrets, see [Secrets](#) in the Kubernetes documentation.

2. Edit the CR for your broker deployment and configure the Operator to mount the secret that contains the third-party JAR file on each broker pod. For example, the following configuration mounts a secret named **log4j-template**.

```
deploymentPlan:
  ...
  extraMounts:
    secrets:
      - "log4j-template"
  ...
```

The JAR file is mounted in a **/amq/extra/secrets/secret name** directory on each broker pod. For example, **/amq/extra/secrets/postgresql-driver/log4j-template.jar**.

3. Create an **ARTEMIS_EXTRA_LIBS** environment variable to extend the broker's Java classpath so the broker loads the JAR file from the mounted directory on each pod. For example:

```
spec:
  ...
  env:
    - name: ARTEMIS_EXTRA_LIBS
      value: /amq/extra/secrets/log4j-template
```

4. Save the CR.

4.4.2. Downloading the JAR file to a volume on each broker container

If a JAR file for an external resource, such as a JDBC database, is larger than 1 MB, you cannot use a secret or configMap to mount the JAR file. Instead, you can configure the Operator to download the JAR file to a persistent shared volume that the Operator mounts on each broker container.

Prerequisites

A persistent shared volume is available to mount on each broker container.

Procedure

1. Edit the **ActiveMQArtemis** CR for your broker deployment.
2. In the broker CR, use the **extraVolumes** and **extraVolumeMounts** attributes to add a persistent volume and mount the volume on each broker container. For example:

```
deploymentPlan:
  ...
  extraVolumes:
    - name: extra-volume
      persistentVolumeClaim:
        claimName: extra-jars
  extraVolumeMounts:
    - name: extra-volume
      mountPath: /opt/extra-lib
  ...
```

- Use the **resourceTemplates** attribute to customize the StatefulSet resource for the deployment. In the customization, use an init container to mount the **extra-volume** volume that you created on each container and to download the JAR file to the volume. For example:

```
spec:
  ...
  resourceTemplates:
  - selector:
      kind: StatefulSet
    patch:
      kind: StatefulSet
      spec:
        template:
          spec:
            initContainers:
            - name: mysql-jdbc-driver-init
              volumeMounts:
              - mountPath: /opt/extra-lib
                name: extra-volume
              image: curlimages/curl:8.6.0
              command:
              - /bin/sh
              args:
              - -c
              - "if ! [ -f /opt/extra-lib/mysql-connector.jar ]; then curl
https://repo1.maven.org/maven2/mysql/mysql-connector-java/8.0.23/mysql-connector-java-
8.0.23.jar --output /opt/extra-lib/mysql-connector.jar ; fi"
```

In the example, a curl image is used to download a **mysql-connector.jar** file to the mounted path of the volume, **/opt/extra-lib**, if the file is not already on the volume.

- Create an **ARTEMIS_EXTRA_LIBS** environment variable to extend the broker's Java classpath so the broker loads the JAR file from the shared volume. For example:

```
spec:
  ...
  env:
  - name: ARTEMIS_EXTRA_LIBS
    value: /opt/extra-lib
```

- Save the CR.

4.5. CONFIGURING MESSAGE PERSISTENCE

By default, AMQ Broker does not persist message data so messages do not survive a broker restart. You can configure AMQ Broker to persist messages in journal files on the filesystem or in a JDBC database of your choice.

If you enable persistence, the default method of persisting messages is to journals on the filesystem.

Additional resources

- [Red Hat AMQ 7 Supported Configurations](#)

4.5.1. Configuring journal-based persistence

When you enable persistence, messages are persisted in journal files on the filesystem by default.

Procedure

1. Edit the **ActiveMQArtemis** custom resource (CR) for your broker deployment.
2. Set the **persistenceEnabled** attribute to **true**. For example:

```
spec:
  ...
  deploymentPlan:
    persistenceEnabled: true
  ...
```



NOTE

When **persistenceEnabled** is set to **true**, the Operator creates the required resources to deploy broker pods that have persistent volumes. In addition, the Operator configures the broker to create the journal and other data files on the persistent volume.

3. Save the CR.

4.5.2. Configuring database persistence

You can configure AMQ Broker to persist messages in a database by using a Java Database Connectivity (JDBC) connection. In the **ActiveMQArtemis** CR for your broker, configure the details required to connect to the database as well as any custom settings for the connection.

When you persist message data in a database, the broker uses a Java Database Connectivity (JDBC) connection to store message and bindings data in database tables. The data in the tables is encoded using AMQ Broker journal encoding. For information about supported databases, see [Red Hat AMQ 7 Supported Configurations](#) on the Red Hat Customer Portal.



IMPORTANT

An administrator might choose to store message data in a database based on the requirements of an organization's wider IT infrastructure. However, use of a database can negatively effect the performance of a messaging system. Specifically, writing messaging data to database tables via JDBC creates a significant performance overhead for a broker.

Prerequisite

- A dedicated database for use with AMQ Broker.
- The required JDBC driver JAR file is available to the broker at runtime. For information on how to make a JAR file available to the broker at runtime, see [Section 4.4, "Adding third-party JAR files"](#).
- The deployment has a single broker instance. To ensure that the deployment has a single broker instance, ensure that the **deployment.size** attribute is not in the **ActiveMQArtemis** custom

resource (CR). When the **deployment.size** attribute is omitted from the CR, a single broker instance is deployed.

Procedure

1. Edit the **ActiveMQArtemis** CR for your broker deployment.
2. Enable JDBC database persistence by using the **brokerProperties** attribute. For example:

```
spec:
  ...
  brokerProperties:
    - storeConfiguration=DATABASE
    - storeConfiguration.jdbcDriverClassName=<class name>
    - storeConfiguration.jdbcConnectionUrl=jdbc:<URL>
    - HAPolicyConfiguration=SHARED_STORE_PRIMARY
  ...
```

storeConfiguration

Specify a value of **DATABASE** to persist messages to a JDBC database.

storeConfiguration.jdbcDriverClassName

Fully-qualified class name of the JDBC database driver. For example, **org.postgresql.Driver**.

For information about supported databases, see [Red Hat AMQ 7 Supported Configurations](#) on the Red Hat Customer Portal.

storeConfiguration.jdbcConnectionUrl

Full JDBC connection URL for your database server including the database name and all configuration parameters. For example:

```
jdbc:postgresql://postgresql-service.default.svc.cluster.local:5432/postgres?
user=postgres&password=postgres
```

In the example, the database name is **postgres**.

HAPolicyConfiguration

Set to **SHARED_STORE_PRIMARY** to ensure that the broker uses a JDBC lease lock to protect the database tables from concurrent access by multiple brokers. If a second broker instance is deployed unintentionally, the lease lock prevents the second broker from writing to the database.

3. (Optional) Change the default values for the following attributes, if required:

storeConfiguration.jdbcNetworkTimeout

JDBC network connection timeout, in milliseconds. The default value is 20000 milliseconds.

storeConfiguration.jdbcLockRenewPeriod

Length, in milliseconds, of the renewal period for the current JDBC lock. When this time elapses, the broker can renew the lock. Set a value that is several times smaller than the value of **storeConfiguration.jdbcLockExpiration** to give the broker sufficient time to extend the lease and also gives the broker time to try to renew the lock in the event of a connection problem. The default value is 2000 milliseconds.

storeConfiguration.jdbcLockExpiration

Time, in milliseconds, that the current JDBC lock is considered owned (that is, acquired or renewed), even if the value of `storeConfiguration.jdbcLockRenewPeriod` has elapsed. The broker periodically tries to renew a lock that it owns according to the value of `storeConfiguration.jdbcLockRenewPeriod`. If the broker fails to renew the lock, for example, due to a connection problem, the broker keeps trying to renew the lock until the value of `storeConfiguration.jdbcLockExpiration` has passed since the lock was last successfully acquired or renewed. An exception to the renewal behavior described above is when another broker acquires the lock. This can happen if there is a time misalignment between the Database Management System (DBMS) and the brokers, or if there is a long pause for garbage collection. In this case, the broker that originally owned the lock considers the lock lost and does not try to renew it. If the JDBC lock has not been renewed by the broker that currently owns it after the expiration time elapses, another broker can establish a JDBC lock. The default value of is 20000 milliseconds.

`storeConfiguration.jdbcJournalSyncPeriod`

Duration, in milliseconds, for which the broker journal synchronizes with JDBC. The default value is 5 milliseconds.

`storeConfiguration.jdbcMaxPageSizeBytes`

Maximum size, in bytes, of each page file when AMQ Broker persists messages to a JDBC database. The default value is 102400, which is 100KB. The value that you specify also supports byte notation such as "K", "MB", and "GB".

4. Save the CR.

4.6. CONFIGURING BROKER STORAGE REQUIREMENTS FOR MESSAGE PERSISTENCE

If you do not have container-native storage in your OpenShift cluster, you need to manually provision Persistent Volumes (PVs) and ensure that these are available to be claimed by the Operator by using a Persistent Volume Claim (PVC).



IMPORTANT

When you manually provision PVs in OpenShift Container Platform, ensure that you set the reclaim policy for each PV to **Retain**. If the reclaim policy for a PV is not set to **Retain** and the PVC that the Operator used to claim the PV is deleted, the PV is also deleted. Deleting a PV results in the loss of any data on the volume. For more information, about setting the reclaim policy, see [Understanding persistent storage](#) in the OpenShift Container Platform documentation.

By default, a PVC obtains 2 GiB of storage for each broker from the default storage class configured for the cluster. You can override the default size and storage class requested in the PVC, but only by configuring new values in the CR **before** deploying the CR for the first time.

4.6.1. Configuring storage size and storage class

Persistent Volume Claims (PVCs) act as requests for storage by pods. You can specify the size and class of storage to request in a PVC submitted by a broker pod.



NOTE

If you change the storage configuration in the CR after you deploy AMQ Broker, the updated configuration is not applied retrospectively to existing Pods. However, the updated configuration is applied to new Pods that are created if you scale up the deployment.

Prerequisites

- You should be familiar with how to use a CR instance to create a basic broker deployment. See [Section 3.3.1, “Deploying a single broker instance”](#).
- You must have already provisioned Persistent Volumes (PVs) and made these available to be claimed by the Operator. For example, if you want to create a cluster of two brokers with persistent storage, you need to have two PVs available.
For more information about provisioning persistent storage, see [Understanding persistent storage](#) in the OpenShift Container Platform documentation.

Procedure

1. Edit the **ActiveMQArtemis** (CR) instance for your broker deployment.
2. To specify the broker storage size, in the **deploymentPlan** section of the CR, add a **storage** section. Add a **size** property and specify a value. For example:

```
spec:
  deploymentPlan:
    size: 1
    image: placeholder
    requireLogin: false
    persistenceEnabled: true
    journalType: nio
    messageMigration: true
    storage:
      size: 4Gi
```

storage.size

Size, in bytes, of the Persistent Volume Claim (PVC) that each broker Pod requires for persistent storage. This property applies only when **persistenceEnabled** is set to **true**. The value that you specify **must** include a unit using byte notation (for example, K, M, G), or the binary equivalents (Ki, Mi, Gi).

3. To specify the storage class that each broker Pod requires for persistent storage, in the **storage** section, add a **storageClassName** property and specify a value. For example:

```
spec:
  deploymentPlan:
    size: 1
    image: placeholder
    requireLogin: false
    persistenceEnabled: true
    journalType: nio
    messageMigration: true
    storage:
      storageClassName: standard
```

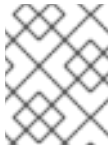


```
storage:
  size: 4Gi
  storageClassName: gp3
```

storage.storageClassName

The name of the storage class to request in the Persistent Volume Claim (PVC). Storage classes provide a way for administrators to describe and classify the available storage. For example, different storage classes might map to specific quality-of-service levels, backup policies and so on.

If you do not specify a storage class, a persistent volume with the default storage class configured for the cluster is claimed by the PVC.



NOTE

If you specify a storage class, a persistent volume is claimed by the PVC only if the volume's storage class matches the specified storage class.

4. Save the CR.

4.7. CONFIGURING RESOURCE LIMITS AND REQUESTS

You can set thresholds on the memory and CPU that can be requested and consumed by the broker container that runs in each Pod. Setting thresholds ensures that containers perform consistently, regardless of the number of pods running on a node.

- You must add configuration for limits and requests to the CR instance for your broker deployment **before** deploying the CR for the first time. You **cannot** add the configuration to a broker deployment that is already running.
- It is not possible for Red Hat to recommend values for limits and requests because these are based on your specific messaging system use-cases and the resulting architecture that you have implemented. However, it *is* recommended that you test and tune these values in a development environment before configuring them for your production environment.
- The Operator runs a type of container called an *Init Container* when initializing each broker Pod. Any resource limits and requests that you configure for each broker container also apply to each Init Container.

You can specify the following limit and request values:

CPU limit

For each broker container running in a Pod, this value is the maximum amount of host-node CPU that the container can consume. If a broker container attempts to exceed the specified CPU limit, OpenShift throttles the container. This ensures that containers have consistent performance, regardless of the number of Pods running on a node.

Memory limit

For each broker container running in a Pod, this value is the maximum amount of host-node memory that the container can consume. If a broker container attempts to exceed the specified memory limit, OpenShift stops the container. The broker Pod restarts.

CPU request

For each broker container running in a Pod, this value is the amount of host-node CPU that the container requests. The OpenShift scheduler considers the CPU request value during Pod placement, to bind the broker Pod to a node with sufficient compute resources.

The CPU request value is the *minimum* amount of CPU that the broker container requires to run. However, if there is no contention for CPU on the node, the container can use all available CPU. If you have specified a CPU limit, the container cannot exceed that amount of CPU usage. If there is CPU contention on the node, CPU request values provide a way for OpenShift to weigh CPU usage across all containers.

Memory request

For each broker container running in a Pod, this value is the amount of host-node memory that the container requests. The OpenShift scheduler considers the memory request value during Pod placement, to bind the broker Pod to a node with sufficient compute resources.

The memory request value is the *minimum* amount of memory that the broker container requires to run. However, the container can consume as much available memory as possible. If you have specified a memory limit, the broker container cannot exceed that amount of memory usage.

CPU is measured in units called millicores. Each node in an OpenShift cluster inspects the operating system to determine the number of CPU cores on the node. Then, the node multiplies that value by 1000 to express the total capacity. For example, if a node has two cores, the CPU capacity of the node is expressed as **2000m**. Therefore, if you want to use one-tenth of a single core, you specify a value of **100m**.

Memory is measured in bytes. You can specify the value by using byte notation (E, P, T, G, M, K) or the binary equivalents (Ei, Pi, Ti, Gi, Mi, Ki). The value that you specify must include a unit.

Additional resources

- [Section 4.1, “How the Operator generates the broker configuration”](#)

4.7.1. Configuring broker resource limits and requests

You can set thresholds on the memory and CPU requested and received by the broker container that runs in each Pod. Setting these thresholds ensures uniformity of resources among all pods running on a node.

- You must add configuration for limits and requests to the CR instance for your broker deployment **before** deploying the CR for the first time. You **cannot** add the configuration to a broker deployment that is already running.
- It is not possible for Red Hat to recommend values for limits and requests because these are based on your specific messaging system use-cases and the resulting architecture that you have implemented. However, it *is* recommended that you test and tune these values in a development environment before configuring them for your production environment.

Prerequisites

- You are familiar with how to use a CR instance to create a basic broker deployment. See [Section 3.3.1, “Deploying a single broker instance”](#).

Procedure

1. Edit the **ActiveMQArtemis** (CR) instance for your broker deployment.

2. In the **deploymentPlan** section of the CR, add a **resources** section. Add **limits** and **requests** sub-sections. In each sub-section, add a **cpu** and **memory** property and specify values. For example:

```
spec:
  deploymentPlan:
    size: 1
    image: placeholder
    requireLogin: false
    persistenceEnabled: true
    journalType: nio
    messageMigration: true
    resources:
      limits:
        cpu: "500m"
        memory: "1024M"
      requests:
        cpu: "250m"
        memory: "512M"
```

limits.cpu

Each broker container running in a Pod in the deployment cannot exceed this amount of host-node CPU usage.

limits.memory

Each broker container running in a Pod in the deployment cannot exceed this amount of host-node memory usage.

requests.cpu

Each broker container running in a Pod in the deployment requests this amount of host-node CPU. This value is the *minimum* amount of CPU required for the broker container to run.

requests.memory

Each broker container running in a Pod in the deployment requests this amount of host-node memory. This value is the *minimum* amount of memory required for the broker container to run.



NOTE

If you specify **limits** for a resource, but do not specify **requests**, a broker container requests the configured **limits** values for that resource. For example, in the following configuration, a broker container requests the configured **limits** values of **500m** cpu and **1024M** memory.

```
spec:
  deploymentPlan:
    size: 3
    ...
  resources:
    limits:
      cpu: "500m"
      memory: "1024M"
```



IMPORTANT

Set **limits** without setting **requests** to control the precise amount of memory and CPU requested and to ensure that the same values are requested for each broker container if there are multiple brokers in your deployment.

3. Save the CR.

4.8. ENABLING ACCESS TO AMQ MANAGEMENT CONSOLE

Each broker pod in a deployment hosts its own instance of AMQ Management Console at port 8161. You can expose the console in the custom resource (CR) instance for your broker deployment. After it is exposed, you can use the console to view and monitor brokers in a web browser.

Procedure

1. Edit the **ActiveMQArtemis** (CR) instance for your broker deployment.
2. In the **spec** section of the CR, add a **console** attribute. In the **console** section, add the **expose** attribute and set the value to **true**.

```
spec:
  ..
  console:
    expose: true
```

When you expose the console, the Operator automatically creates a dedicated service and Openshift route for the console on each broker pod in the deployment.

3. If you want to customize the host name of the routes that are exposed for the console to match the internal routing configuration on your Openshift cluster, you can do one or both of the following:
 - Use the **ingressHost** attribute to replace the default host name with a custom host name for the console routes.
 - Use the **ingressDomain** attribute to append a custom domain to the host name. The custom domain is also applied to all other routes, such as routes for acceptors, that are exposed by the CR configuration.
 - a. To set a custom host name specifically for the console routes, add the **ingressHost** attribute and specify the host string. For example:

```
spec:
  ..
  console:
    expose: true
    ingressHost: my-console-production.my-subdomain.com
  ..
```

**NOTE**

The **ingressHost** value must be unique on your Openshift cluster. If your broker cluster has multiple broker pods, you can make the **ingressHost** value unique by including the `$(BROKER_ORDINAL)` variable in the value. The Operator replaces this variable in the route it creates for each broker pod with the ordinal number the StatefulSet assigned to the pod. For example, an **ingressHost** value of **my-console-\$(BROKER_ORDINAL)-production.my-subdomain.com** sets the host name of the route to **my-console-0-production.my-subdomain.com** on the first pod, **my-console-1-production.my-subdomain.com** on the second pod and so on.

You can include any the following variables in the custom host name string for the console route:

Table 4.3. Variables in a custom host name string for the console route

Name	Description
<code>\$(CR_NAME)</code>	The value of the metadata.name attribute in the CR.
<code>\$(CR_NAMESPACE)</code>	The namespace of the custom resource.
<code>\$(BROKER_ORDINAL)</code>	The ordinal number assigned to the broker pod by the StatefulSet.
<code>\$(ITEM_NAME)</code>	The name of the acceptor.
<code>\$(RES_TYPE)</code>	The resource type. A route has a resource type of rte . An ingress has a resource type of ing .
<code>\$(INGRESS_DOMAIN)</code>	The value of the spec.ingressDomain attribute if it is configured in the CR.

- b. To append a custom domain to the host name in routes, add a **spec.ingressDomain** attribute and specify a custom string. For example:

```
spec:
  ...
  ingressDomain: my.domain.com
```

4. If your organization's network policy require that you expose the console by using an ingress instead of a route, complete the following steps:
 - a. Add the **exposeMode** attribute and set the value to **ingress**.

```
spec:
  ..
  console:
    expose: true
    exposeMode: ingress
  ..
```

- b. If you want to customize the host name of the ingresses that are exposed for the console to match the internal routing configuration on your OpenShift cluster, you can do one or both of the following:

- Use the **ingressHost** attribute to replace the default host name with a custom host name.
- Use the **ingressDomain** attribute to append a custom domain to the host name. The custom domain is also applied to all other ingresses, such as ingresses for acceptors, that are exposed by the CR configuration.
 - i. To set a custom host name specifically for the ingresses created for the console, add the **ingressHost** attribute and specify the host string. For example:

```
spec:
  ..
  console:
    expose: true
    exposeMode: ingress
    expose: true
    exposeMode: ingress
    ingressHost: my-console-production.my-subdomain.com
  ...
```

You can include the same variables to customize an ingress host as a route host, which are described earlier in this procedure.

- ii. To append a custom domain to the host name in ingresses, add a **spec.ingressDomain** attribute and specify a custom string.

```
spec:
  ...
  ingressDomain: my.domain.com
```

For the console, the default host name of an ingress is in the format **<cr-name>-wconsj-<ordinal>-svc-ing-<namespace>**. If, for example, you have a CR named **production** in the **amqbroker** name space, an **ingressDomain** value of **mydomain.com** gives a host value of **production-wconsj-0-svc-ing-mynamespace.amqbroker.com** for the ingress created on pod 0.

For more information on the **spec.ingressDomain** attribute, see [Section 8.1, “Custom resource configuration reference”](#).

5. If you want to enable secure connections to the console from clients outside of the OpenShift cluster, complete the following steps:

- a. Add the **sslEnabled** attribute and set the value to **true**.

```
spec:
  ..
  console:
    expose: true
```

```

    exposeMode: ingress
    sslEnabled: true
  ..

```

- b. Add the **sslSecret** attribute and specify the name of a secret that contains the certificate to secure the console. For example:

```

spec:
  ..
  console:
    expose: true
    exposeMode: ingress
    sslEnabled: true
    sslSecret: console-tls-secret
  ..

```

- c. Use the **spec.env** attribute to add an environment variable that configures the console to automatically load a new certificate each time the certificate is renewed. For example:

```

spec:
  ..
  env:
    - name: JAVA_ARGS_APPEND
      value: -Dwebconfig.bindings.artemis.sslAutoReload=true
  ..

```

6. Save the CR.

Additional resources

- [Chapter 5, Connecting to AMQ Management Console for an Operator-based broker deployment](#)

4.9. SETTING ENVIRONMENT VARIABLES FOR THE BROKER CONTAINERS

You can configure environment variables that are passed to an AMQ Broker container. You can set standard variables, such as **TZ** for the timezone. Or, you can set a custom AMQ Broker variable, **JAVA_ARGS_APPEND**, if you need to add to the command line arguments used by the Java launcher.

Procedure

1. Edit the Custom Resource (CR) instance for your broker deployment.
 - a. Using the OpenShift command-line interface:
 - i. Enter the following command:

```
oc edit ActiveMQArtemis <CR instance name> -n <namespace>
```

- b. Using the OpenShift Container Platform web console:
 - i. Log in to the console as a user that has privileges to deploy CRs in the project for the broker deployment.

- ii. In the left pane, click **Ecosystem > Installed Operators**
 - iii. Click the **Red Hat Integration - AMQ Broker for RHEL 9 (Multiarch)** operator.
 - iv. Click the **AMQ Broker** tab.
 - v. Click the name of the ActiveMQArtemis instance name.
 - vi. Click the **YAML** tab.
Within the console, a YAML editor opens, which enables you to configure the CR instance.
2. In the **spec** section of the CR, add an **env** element and add the environment variables that you want to set for the AMQ Broker container. For example:

```
apiVersion: broker.amq.io/v1beta1
kind: ActiveMQArtemis
metadata:
  name: ex-aao
spec:
  ...
  env:
    - name: TZ
      value: Europe/Vienna
    - name: JAVA_ARGS_APPEND
      value: --Hawtio.realm=console
    - name: JDK_JAVA_OPTIONS
      value: -XshowSettings:system
  ...
```

In the example, the CR configuration includes the following environment variables:

- **TZ** to set the time zone of the AMQ Broker container.
- **JAVA_ARGS_APPEND** to configure AMQ Management Console to use a realm named **console** for authentication.
- **JDK_JAVA_OPTIONS** to set the Java **-XshowSettings:system** parameter, which displays system property settings for the Java Virtual Machine.



NOTE

Values configured using the **JDK_JAVA_OPTIONS** environment variable are prepended to the command line arguments used by the Java launcher. Values configured using the **JAVA_ARGS_APPEND** environment variable are appended to the arguments used by the launcher. If an argument is duplicated, the rightmost argument takes precedence.

3. Save the CR.



NOTE

Red Hat recommends that you do not change AMQ Broker environment variables that have an **AMQ_** prefix and that you exercise caution if you want to change the **POD_NAMESPACE** variable.

Additional resources

- [Define Environment Variables for a Container](#)

4.10. CONFIGURING THE MAXIMUM JVM MEMORY AVAILABLE TO A BROKER

By default, the maximum memory available to a broker is half of the maximum memory that is available to the broker's Java Virtual Machine (JVM). You can override the default memory limit by specifying a new limit that is either a percentage of the maximum JVM memory or an absolute value.



NOTE

It is recommended that you set memory usage limits for individual addresses instead of relying on a much larger global limit for all addresses. If you have a global limit for all addresses, the number of objects in memory can reach a level that impacts the normal operation of the broker. To set a memory limit for an individual address, use the **maxSizeBytes** property. For more information, see [Section 8.1, "Custom resource configuration reference"](#).

Procedure

1. Edit the **ActiveMQArtemis** custom resource (CR) for your broker deployment.
2. If you want to set a memory limit for the broker that is a percentage of the maximum memory available to the JVM, add a **brokerProperties** section within the **spec** section of the CR. Within the **brokerProperties** section, add a **globalMaxSizePercentOfJvmMaxMemory** property and specify a memory limit. In the following example, the memory limit is 60 percent of the maximum memory available to the JVM.

```
spec:
  ...
  brokerProperties:
    - globalMaxSizePercentOfJvmMaxMemory=60
```

By specifying a percentage value, you avoid the need to change the value if the amount of JVM memory changes.

3. If you want to specify an absolute value for the new memory limit, add a **brokerProperties** section within the **spec** section of the CR. Within the **brokerProperties** section, add a **globalMaxSize** property and specify a memory limit. For example:

```
spec:
  ...
  brokerProperties:
    - globalMaxSize=500m
  ...
```

The default unit for the **globalMaxSize** property is bytes. To change the default unit, add a suffix of m (for MB) or g (for GB) to the value.

4. Configure the action that you want the broker to take for all further messages after a memory limit you set is reached. For example:

```
spec:
  ...
  brokerProperties:
    - addressSettings.<address match>.addressFullMessagePolicy=<action>
  ...
```

The valid actions are:

PAGE

The broker pages any further messages to disk.

DROP

The broker silently drops any further messages.

FAIL

The broker drops any further messages and issues exceptions to client message producers.

BLOCK

Client message producers block when they try to send further messages.

In the following example, messages for the **usa-news** address are paged when a memory limit is reached.

```
spec:
  ...
  brokerProperties:
    - addressSettings.usa-news.addressFullMessagePolicy=PAGE
  ...
```

5. Save the CR.

4.11. SPECIFYING A CUSTOM INIT CONTAINER IMAGE

You can use an Init Container resource to perform tasks before the rest of a pod is deployed. For example, you can build a custom Init container if you want to include extra runtime dependencies, **.jar** files, in the broker installation directory.

When you build a custom Init Container image, you must follow these important guidelines:

- In the build script (for example, a Docker Dockerfile or Podman Containerfile) that you create for the custom image, the **FROM** instruction must specify the latest version of the AMQ Broker Operator built-in Init Container as the base image. In your script, include the following line:

```
FROM registry.redhat.io/amq7/amq-broker-init-rhel9:7.14
```

- The custom image must include a script called **post-config.sh** that you include in a directory called **/amq/scripts**. The **post-config.sh** script is where you can modify or add to the initial configuration that the Operator generates. When you specify a custom Init Container, the Operator runs the **post-config.sh** script **after** it uses your CR instance to generate a configuration, but **before** it starts the broker application container.
- As described in [Section 4.1.2, "Directory structure of a broker Pod"](#), the path to the installation directory used by the Init Container is defined in an environment variable called **CONFIG_INSTANCE_DIR**. The **post-config.sh** script should use this environment variable

name when referencing the installation directory (for example, `${CONFIG_INSTANCE_DIR}/lib`) and **not** the actual value of this variable (for example, `/amq/init/config/lib`).

- If you want to include additional resources (for example, `.xml` or `.jar` files) in your custom broker configuration, you must ensure that these are included in the custom image and accessible to the `post-config.sh` script.

The following procedure describes how to specify a custom Init Container image.

Prerequisites

- You must have built a custom Init Container image that meets the guidelines described above. For a complete example of building and specifying a custom Init Container image for the ArtemisCloud Operator, see [custom Init Container image for JDBC-based persistence](#).
- To provide a custom Init Container image for the AMQ Broker Operator, you need to be able to add the image to a repository in a container registry such as the [Quay container registry](#).
- You should understand how the Operator uses an Init Container to generate the broker configuration. For more information, see [Section 4.1, “How the Operator generates the broker configuration”](#).
- You should be familiar with how to use a CR to create a broker deployment. For more information, see [Section 3.3, “Creating Operator-based broker deployments”](#).

Procedure

1. Edit the CR instance for the broker deployment.
 - a. Using the OpenShift command-line interface:
 - i. Log in to OpenShift Container Platform as a user that has privileges to deploy CRs in the project for the broker deployment.
 - ii. Edit the CR for your deployment.

```
oc edit ActiveMQArtemis <CR instance name> -n <namespace>
```

- b. Using the OpenShift Container Platform web console:
 - i. Log in to OpenShift Container Platform as a user that has privileges to deploy CRs in the project for the broker deployment.
 - ii. In the left pane, click **Administration > Custom Resource Definitions**
 - iii. Click the **ActiveMQArtemis** CRD.
 - iv. Click the **Instances** tab.
 - v. Click the instance for your broker deployment.
 - vi. Click the **YAML** tab.
Within the console, a YAML editor opens, enabling you to edit the CR instance.

2. In the **deploymentPlan** section of the CR, add an **initImage** attribute and set the value to the URL of your custom Init Container image.

```
apiVersion: broker.amq.io/v1beta1
kind: ActiveMQArtemis
metadata:
  name: ex-aao
spec:
  deploymentPlan:
    size: 1
    image: placeholder
    initImage: <custom_init_container_image_url>
    requireLogin: false
    persistenceEnabled: true
    journalType: nio
    messageMigration: true
```

initImage

Specifies the full URL for your custom Init Container image, which must be available from a container registry.



IMPORTANT

If a CR has a custom init container image specified in the **spec.deploymentPlan.initImage** attribute, Red Hat recommends that you also specify the URL of the corresponding broker container image in the **spec.deploymentPlan.image** attribute to prevent automatic upgrades of the broker image. If you do not specify the URL of a specific broker container image in the **spec.deploymentPlan.image** attribute, the broker image can be automatically upgraded. After the broker image is upgraded, the versions of the broker and custom init container image are different, which might prevent the broker from running.

If you have a working deployment that has a custom init container, you can prevent any further upgrades of the broker container image to eliminate the risk of a newer broker image not working with your custom init container image. For more information about preventing upgrades to the broker image, see, [Section 6.6.2, “Restricting automatic upgrades of images by using image URLs”](#).

3. Save the CR.

Additional resources

- [Custom Init Container image for JDBC-based persistence](#)

4.12. CONFIGURING BROKERS FOR CLIENT CONNECTIONS

You use acceptors to define how a broker pod accepts connections from clients.

4.12.1. Configuring acceptors

When you create an acceptor for inbound client connections, you specify information such as the messaging protocols to enable on the acceptor and the port on the broker pod to use for these protocols.

Procedure

1. Edit the **ActiveMQArtemis** custom resource (CR) for your broker deployment.
2. In the **acceptors** attribute, add a named acceptor. Add the **protocols** and **port** attributes. Set values to specify the messaging protocols to be used by the acceptor and the port on each broker pod to expose for those protocols. For example:

```
spec:
  ..
  acceptors:
    - name: my-acceptor
      protocols: amqp
      port: 5672
  ..
```

The configured acceptor exposes port 5672 to AMQP clients. The full set of values that you can specify for the **protocols** attribute is shown in the table.

Table 4.4. Acceptor protocols

Protocol	Value
Core Protocol	core
AMQP	amqp
OpenWire	openwire
MQTT	mqtt
STOMP	stomp
All supported protocols	all



NOTE

- For each broker pod in your deployment, the Operator also creates a default acceptor that uses port 61616. This default acceptor is required for broker clustering and has Core Protocol enabled.
- By default, the AMQ Broker management console uses port 8161 on the broker pod. Each broker pod in your deployment has a dedicated Service that provides access to the console. For more information, see [Chapter 5, Connecting to AMQ Management Console for an Operator-based broker deployment](#).

3. To use another protocol on the same acceptor, modify the **protocols** attribute. Specify a comma-separated list of protocols. For example:

```
spec:
  ..
  acceptors:
    - name: my-acceptor
      protocols: amqp,openwire
      port: 5672
  ...
```

The configured acceptor now exposes port 5672 to AMQP and OpenWire clients.

4. To specify the number of concurrent client connections that the acceptor allows, add the **connectionsAllowed** attribute and set a value. For example:

```
spec:
  ...
  acceptors:
    - name: my-acceptor
      protocols: amqp,openwire
      port: 5672
      connectionsAllowed: 5
  ...
```

5. By default, an acceptor is exposed only to clients in the same OpenShift cluster as the broker deployment. To also expose the acceptor to clients outside OpenShift, set both the **expose** attribute and the **sslEnabled** attribute to **true**.

```
spec:
  ...
  acceptors:
    - name: my-acceptor
      protocols: amqp,openwire
      port: 5672
      connectionsAllowed: 5
      expose: true
      sslEnabled: true
  ...
```

When you enable SSL (that is, Secure Sockets Layer) security on an acceptor (or connector), you can add related configuration, such as:

- The secret name used to store authentication credentials in your OpenShift cluster. A secret is required when you enable SSL on the acceptor.
- The Transport Layer Security (TLS) protocols to use for secure network communication. TLS is an updated, more secure version of SSL. You specify the TLS protocols in the **enabledProtocols** attribute.
- Whether the acceptor uses mTLS, also known as mutual authentication, between the broker and the client. You specify this by setting the value of the **needClientAuth** attribute to **true**.

For more information about these tasks, see [Section 4.12.2, "Securing broker-client connections"](#).

When you expose an acceptor to clients outside OpenShift, the Operator automatically creates a dedicated service and Openshift route for the acceptor on each broker pod in the deployment.

6. If you want to customize the host name of the route that is exposed for the acceptor on each pod to match the internal routing configuration on your Openshift cluster, you can do one or both of the following:
 - Use the **ingressHost** attribute to replace the default host name with a custom host name for a specific acceptor.
 - Use the **ingressDomain** attribute to append a custom domain to the host name. The custom domain is also applied to all other routes, such as routes for other acceptors and the console, that are exposed by the CR configuration.
 - a. To set a custom host name for the acceptor routes, add the **ingressHost** attribute and specify the host string. For example:

```
spec:
  ...
  acceptors:
  - name: my-acceptor
    protocols: amqp,openwire
    port: 5672
    connectionsAllowed: 5
    expose: true
    ingressHost: my-acceptor-production.my-subdomain.com
  ...
```

NOTE

The **ingressHost** value must be unique on your Openshift cluster. If your broker cluster has multiple broker pods, you can make the **ingressHost** value unique by including the `$(BROKER_ORDINAL)` variable in the value. The Operator replaces this variable on each broker pod with the ordinal number the StatefulSet assigned to the pod. For example, an **ingressHost** value of **my-acceptor-\$(BROKER_ORDINAL)-production.my-subdomain.com** sets the host name of the route to **my-acceptor-0-production.my-subdomain** on the first pod, **my-acceptor-1-production.my-subdomain** on the second pod and so on.

You can include any the following variables in the custom host name string for an acceptor route:

Table 4.5. Variables in a custom host name string for an acceptor route

Name	Description
<code>\$(CR_NAME)</code>	The value of the metadata.name attribute in the CR.
<code>\$(CR_NAMESPACE)</code>	The namespace of the custom resource.

Name	Description
\$(BROKER_ORDINAL)	The ordinal number assigned to the broker pod by the StatefulSet.
\$(ITEM_NAME)	The name of the acceptor.
\$(RES_TYPE)	The resource type. A route has a resource type of rte . An ingress has a resource type of ing .
\$(INGRESS_DOMAIN)	The value of the spec.ingressDomain attribute if it is configured in the CR.

- b. To append a custom domain to the host name in routes, add a **spec.ingressDomain** attribute and specify a custom string. For example:

```
spec:
  ...
  ingressDomain: my.domain.com
```

7. If your organization's network policy require that you expose acceptors by using an ingress instead of a route, complete the following steps:

- a. Add the **exposeMode** attribute and set the value to **ingress**.

```
spec:
  ...
  acceptors:
  - name: my-acceptor
    protocols: amqp,openwire
    port: 5672
    connectionsAllowed: 5
    expose: true
    exposeMode: ingress
  ...
```

- b. If you want to customize the host name of the ingresses that are exposed for the acceptor to match the internal routing configuration on your OpenShift cluster, you can do one or both of the following:

- Use the **ingressHost** attribute to replace the default host name with a custom host name.
- Use the **ingressDomain** attribute to append a custom domain to the host name. The custom domain is also applied to all other ingresses, such as ingresses for other acceptors and the console, that are exposed by the CR configuration.
 - i. To set a custom host name for the ingresses for the acceptor, add the **ingressHost** attribute and specify the host string. For example:

```
spec:
  ...
```



```

acceptors:
- name: my-acceptor
  protocols: amqp,openwire
  port: 5672
  connectionsAllowed: 5
  expose: true
  exposeMode: ingress
  ingressHost: my-acceptor-production.my-subdomain.com
...

```

You can include the same variables to customize an ingress host as a route host, which are described earlier in this procedure.

- ii. To append a custom domain to the host name in ingresses, add a **spec.ingressDomain** attribute and specify a custom string. For example:

```

spec:
...
  ingressDomain: my-subdomain.domain.com

```

For acceptors, the default host name of an ingress is in the format of **<cr-name>-<acceptor name>-<ordinal>-svc-ing-<namespace>**. If, for example, you have a CR named **production** in the **amqbroker** name space, an **ingressDomain** value of **mydomain.com** gives a host value of **production-wconsj-0-svc-ing-mynamespace.amqbroker.com** for the ingress created on pod 0.

Additional resources

- [Section 4.12.2, “Securing broker-client connections”](#)
- [Section 8.1, “Custom resource configuration reference”](#)

4.12.2. Securing broker-client connections

If you enable security on an acceptor, by setting the **sslEnabled** attribute to **true**, you must configure Transport Layer Security (TLS) to allow certificate-based authentication between the broker and clients. TLS is an updated, more secure version of SSL.

There are two primary TLS configurations:

TLS

Only the broker presents a certificate. The certificate is used by the client to authenticate the broker. This is the most common configuration.

mTLS

Both the broker and the client present certificates. This is sometimes called *mutual authentication*.

You can use various methods to generate a TLS certificate.

If the broker and clients are running on the same OpenShift cluster, you can use OpenShift to generate a service serving certificate for the broker.

If the broker and clients are not running on the same OpenShift cluster, you must generate a certificate by using a method that allows you to customize the certificate. You can generate custom certificates by using either of the following methods:

- cert-manager Operator for OpenShift
- Java keytool utility.

4.12.2.1. Using Openshift service serving certificates

If you want to secure connections between the broker and clients on the same OpenShift cluster, you can add an annotation to the acceptor service to request that OpenShift generates a service serving TLS certificate.

The generated certificate and key are in PEM format, stored in **tls.crt** and **tls.key** respectively, within a created secret.



NOTE

The service CA certificate, which issues the service serving certificates, is valid for 26 months and is automatically rotated when there is less than 13 months validity left. After rotation, the previous service CA configuration is still trusted until its expiration. This allows a grace period for all affected services to refresh their key material before the expiration. If you do not upgrade your cluster during this grace period, which restarts services and refreshes their key material, you might need to manually restart services to avoid failures after the previous service CA expires.

Procedure

1. Edit the **ActiveMQArtemis** custom resource (CR) for your broker deployment.
2. Use the **resourceTemplates** attribute to annotate the service that is created for an acceptor. For example:

```
spec:
  ...
  resourceTemplates:
    - selector:
        kind: Service
        name: amq-broker-myacceptor-0-svc
      annotations:
        service.beta.openshift.io/serving-cert-secret-name: myacceptor-ptls
    ...
```

resourceTemplates.selector.kind

Specify that the type of resource to which the customization applies is **Service**.

resourceTemplates.selector.name

Specify the name of the service to which you want to apply the annotation. An acceptor service has a name format of: **<CR name><acceptor name><ordinal>-svc**, where:

- **<CR name>** name is the value of the **metadata.name** attribute in the CR.
- **<acceptor name>** is the name of the acceptor. The example assumes that the name of the acceptor is **myacceptor**.
- **<ordinal>** is the ordinal number assigned to the broker pod by the StatefulSet.

resourceTemplates.annotations

Specify an annotation of **service.beta.openshift.io/serving-cert-secret-name: <secret>**, where <secret> is the name of the secret that Openshift creates for the service.



NOTE

The secret name must match the acceptor name and have a **-ptls** suffix. The specific suffix is required to allow the Operator to deploy the CR before the secret is created.

3. In the `sslSecret`` attribute in the CR, specify the secret that contains the broker certificate. For example:

```
spec:
  acceptors:
    - name: myacceptor
      protocols: CORE
      port: 61626
      sslEnabled: true
      sslSecret: myacceptor-ptls
```

4. In the **brokerProperties** attribute, configure the broker to automatically load a new certificate each time the certificate is renewed in Openshift. For example:

```
spec:
  ...
  brokerProperties
    - "acceptorConfigurations.myacceptor.params.sslAutoReload=true"
  ...
```

5. Add the public key of the service serving certificate to each client's trust store.
6. If you want to configure mTLS authentication between the broker and clients, complete the following steps.
 - a. Create a trust bundle that contains the certificate of each client that you want the broker to trust and add the trust bundle to a secret, for example, **trusted-clients-bundle**.
 - b. In the acceptors configured in the broker CR, add the **needClientAuth** attribute and set to **true** to require client authentication. For example:

```
spec:
  ..
  acceptors:
    - name: myacceptor
      protocols: all
      port: 62666
      sslEnabled: true
      sslSecret: myacceptor-ptls
      needClientAuth: true
  ..
```

- c. In the **trustSecret** attribute of each acceptor, specify the secret that contains the trust bundle of client certificates. For example:

```
spec:
  ..
  acceptors:
    - name: new-acceptor
      protocols: all
      port: 62666
      sslEnabled: true
      sslSecret: myacceptor-ptls
      needClientAuth: true
      trustSecret: trusted-clients-bundle
  ..
```

7. Save the CR.

Additional resources

- [Securing service traffic using service serving certificate secrets](#)

4.12.2.2. Using cert-manager Operator for Openshift

You can use cert-manager Operator for OpenShift to generate a TLS certificate to secure connections between a broker and clients. The cert-manager Operator for OpenShift is a cluster-wide service that provides application certificate lifecycle management.

The following example procedure describes how to configure Transport Layer Security (TLS) by using a self-signed certificate. If your policy requires certificates that are signed by a recognized certificate manager, you can request the certificates by using the cert-manager Operator for OpenShift.

Prerequisites

- cert-manager Operator for Red Hat OpenShift is installed. For more information, see [Installing the cert-manager Operator for Red Hat OpenShift](#).
- Trust Manager for Kubernetes is installed if you want to configure mTLS between the broker and clients.
For more information, see [Installing trust-manager](#).

Procedure

1. Create a YAML file, for example, **self-signed-issuer.yaml**, that defines a root self-signed issuer. An issuer is an Openshift resource that represents certificate authorities (CAs) that are able to generate signed certificates by honoring certificate signing requests. The following example yaml creates a self-signed issuer, which you can then use to create a Certification Authority(CA) certificate. Your CA certificate can be managed by the cert-manager Operator.

```
apiVersion: cert-manager.io/v1
kind: ClusterIssuer
metadata:
  name: root-issuer
spec:
  selfSigned: {}
```

2. Create a YAML file, for example, **root-ca.yaml**, that defines a root CA certificate.

In the **issuerRef.name** field, specify the name of the self-signed issuer, **root-issuer**, that you created. For example:

```
apiVersion: cert-manager.io/v1
kind: Certificate
metadata:
  name: root-ca
  namespace: cert-manager
spec:
  isCA: true
  commonName: "amq.io.root"
  secretName: root-ca-secret
  subject:
    organizations:
      - "www.amq.io"
  issuerRef:
    name: root-issuer
    kind: ClusterIssuer
```

The Certificate is created in Privacy Enhanced Mail (PEM) format in a secret named **root-ca-secret**.

3. Create a YAML file, for example, **root-ca-issuer.yaml**, that defines a CA issuer for issuing certificates that are signed by the root CA. For example:

```
apiVersion: cert-manager.io/v1
kind: ClusterIssuer
metadata:
  name: root-ca-issuer
spec:
  ca:
    secretName: root-ca-secret
```

4. Create a YAML file, for example, **broker-cert.yaml**, that defines a broker certificate.
In the **issuerRef.Name** field, specify the name of the issuer, **root-ca-issuer**, that you created to issue certificates that are signed by the root CA. For example:

```
apiVersion: cert-manager.io/v1
kind: Certificate
Metadata:
  name: broker-cert
spec:
  isCA: false
  commonName: "amq.io"
  dnsNames:
    - "amq-broker-ss-0.amq-broker-svc-rte-default.cluster.local"
    - "amq-broker-ss-1.amq-broker-svc-rte-default.cluster.local"
  secretName: broker-cert-secret
  subject:
    organizations:
      - "www.amq.io"
  issuerRef:
    name: root-ca-issuer
    kind: ClusterIssuer
```

5. Deploy the custom resources that you defined for issuers and certificates in YAML files to create the corresponding OpenShift objects. For example:

```
$ oc create -f self-signed-issuer.yaml
$ oc create -f root-ca.yaml
$ oc create -f root-ca-issuer.yaml
$ oc create -f broker-cert.yaml
```

6. Edit the **ActiveMQArtemis** CR for you broker deployment.
7. Specify the secret that contains the broker certificate in the **sslSecret** attribute of each acceptor that you want to secure. For example:

```
spec:
  ..
  acceptors:
    - name: new-acceptor
      protocols: all
      port: 62666
      sslEnabled: true
      needClientAuth: false
      sslSecret: broker-cert-secret
    ..
```

8. In the **brokerProperties** attribute, configure the broker to automatically load a new broker certificate for the acceptor each time the certificate is renewed by the cert-manager Operator for Openshift. For example:

```
spec:
  ...
  brokerProperties
    - "acceptorConfigurations.new-acceptor.params.sslAutoReload=true"
  ...
```

9. Add the root CA certificate that signed the broker certificate, which was create in a secret named **root-ca-secret** secret in this example procedure, to each client's trust store, so clients can trust the broker.
10. If you want to configure mTLS authentication between the broker and clients, complete the following steps.
 - a. Use Trust Manager for Kubernetes to create a trust bundle that contains the certificate of each client that you want the broker to trust and add the trust bundle to a secret, for example, **trusted-clients-bundle**. For information on how to create a trust bundle, see the [trust-manager documentation](#).
 - b. In the acceptors configured in the broker CR, add the **needClientAuth** attribute and set to **true** to require client authentication. For example:

```
spec:
  ..
  acceptors:
    - name: new-acceptor
      protocols: all
      port: 62666
```

```

    sslEnabled: true
    sslSecret: broker-cert-secret
    needClientAuth: true

```

```
..
```

- c. in the **trustSecret** attribute of each acceptor, specify the secret that contains the trust bundle of client certificates. For example:

```

spec:
  ..
  acceptors:
    - name: new-acceptor
      protocols: all
      port: 62666
      sslEnabled: true
      sslSecret: broker-cert-secret
      needClientAuth: true
      trustSecret: trusted-clients-bundle
  ..

```

11. Save the CR.

Additional resources

- [cert-manager Operator for Red Hat OpenShift documentation](#)

4.12.2.3. Using the Java keytool utility

You can use keytool, a certificate management utility included with Java, to generate a TLS certificate to secure connections between a broker and clients.

4.12.2.3.1. Configuring one-way TLS

You can secure communications between the broker and clients with one-way Transport Layer Security (TLS) authentication. In one-way TLS, only the broker presents a certificate which clients use to authenticate the broker.

Prerequisites

- You should understand the requirements for broker certificate generation when clients use host name verification. For more information, see [Section 4.12.2.4, “Configuring a broker certificate for host name verification”](#).

Procedure

1. Generate a self-signed certificate for the broker key store.

```
$ keytool -genkey -alias broker -keyalg RSA -keystore ~/broker.ks
```

2. Export the certificate from the broker key store, so that it can be shared with clients. Export the certificate in the Base64-encoded **.pem** format. For example:

```
$ keytool -export -alias broker -keystore ~/broker.ks -file ~/broker_cert.pem
```

- On the client, create a client trust store that imports the broker certificate.

```
$ keytool -import -alias broker -keystore ~/client.ts -file ~/broker_cert.pem
```

- Log in to OpenShift Container Platform as an administrator. For example:

```
$ oc login -u system:admin
```

- Switch to the project that contains your broker deployment. For example:

```
$ oc project <my_openshift_project>
```

- Create a secret to store the TLS credentials. For example:

```
$ oc create secret generic my-tls-secret \
--from-file=broker.ks=~/.broker.ks \
--from-file=client.ts=~/.broker.ks \
--from-literal=keyStorePassword=<password> \
--from-literal=trustStorePassword=<password>
```



NOTE

When generating a secret, OpenShift requires you to specify both a key store and a trust store. The trust store key is generically named **client.ts**. For one-way TLS between the broker and a client, a trust store is not actually required. However, to successfully generate the secret, you need to specify some valid store file as a value for **client.ts**. The preceding step provides a placeholder value for **client.ts** by reusing the previously-generated broker key store file. This is sufficient to generate a secret with all of the credentials required for one-way TLS.

- Link the secret to the service account that you created when installing the Operator. For example:

```
$ oc secrets link sa/amq-broker-operator secret/my-tls-secret
```

- Specify the secret name in the **sslSecret** parameter of your secured acceptor or connector. For example:

```
spec:
...
acceptors:
- name: my-acceptor
  protocols: amqp,openwire
  port: 5672
  sslEnabled: true
  sslSecret: my-tls-secret
  expose: true
  connectionsAllowed: 5
...
```

4.12.2.3.2. Configuring two-way TLS

You can secure communications between the broker and clients with two-way Transport Layer Security (TLS) authentication. In two-way TLS, both the broker and clients present certificates which they use to authenticate each other in a process called *mutual authentication*.

Prerequisites

- You should understand the requirements for broker certificate generation when clients use host name verification. For more information, see [Section 4.12.2.4, “Configuring a broker certificate for host name verification”](#).

Procedure

1. Generate a self-signed certificate for the broker key store.

```
$ keytool -genkey -alias broker -keyalg RSA -keystore ~/broker.ks
```

2. Export the certificate from the broker key store, so that it can be shared with clients. Export the certificate in the Base64-encoded **.pem** format. For example:

```
$ keytool -export -alias broker -keystore ~/broker.ks -file ~/broker_cert.pem
```

3. On the client, create a client trust store that imports the broker certificate.

```
$ keytool -import -alias broker -keystore ~/client.ts -file ~/broker_cert.pem
```

4. On the client, generate a self-signed certificate for the client key store.

```
$ keytool -genkey -alias broker -keyalg RSA -keystore ~/client.ks
```

5. On the client, export the certificate from the client key store, so that it can be shared with the broker. Export the certificate in the Base64-encoded **.pem** format. For example:

```
$ keytool -export -alias broker -keystore ~/client.ks -file ~/client_cert.pem
```

6. Create a broker trust store that imports the client certificate.

```
$ keytool -import -alias broker -keystore ~/broker.ts -file ~/client_cert.pem
```

7. Log in to OpenShift Container Platform as an administrator. For example:

```
$ oc login -u system:admin
```

8. Switch to the project that contains your broker deployment. For example:

```
$ oc project <my_openshift_project>
```

9. Create a secret to store the TLS credentials. For example:

```
$ oc create secret generic my-tls-secret \
--from-file=broker.ks=~/broker.ks \
--from-file=client.ts=~/broker.ts \
```

```
--from-literal=keyStorePassword=<password> \
--from-literal=trustStorePassword=<password>
```



NOTE

When generating a secret, OpenShift requires you to specify both a key store and a trust store. The trust store key is generically named **client.ts**. For two-way TLS between the broker and a client, you must generate a secret that includes the broker trust store, because this holds the client certificate. Therefore, in the preceding step, the value that you specify for the **client.ts** key is actually the **broker** trust store file.

10. Link the secret to the service account that you created when installing the Operator. For example:

```
$ oc secrets link sa/amq-broker-operator secret/my-tls-secret
```

11. Specify the secret name in the **sslSecret** parameter of your secured acceptor or connector. For example:

```
spec:
...
  acceptors:
  - name: my-acceptor
    protocols: amqp,openwire
    port: 5672
    sslEnabled: true
    sslSecret: my-tls-secret
    expose: true
    connectionsAllowed: 5
...
```

4.12.2.4. Configuring a broker certificate for host name verification

If clients are configured to use host verification, you must ensure that the Common Name (CN) of the broker's certificate matches the host name that clients use to connect to the broker.

When a client tries to connect to a broker pod in your deployment, the **verifyHost** option in the client connection URL determines whether the client compares the Common Name (CN) of the broker's certificate to its host name, to verify that they match. The client performs this verification if you specify **verifyHost=true** or similar in the client connection URL.

You might omit this verification in rare cases where you have no concerns about the security of the connection, for example, if the brokers are deployed on an OpenShift cluster in an isolated network. Otherwise, for a secure connection, it is advisable for a client to perform this verification. In this case, correct configuration of the broker key store certificate is essential to ensure successful client connections.

In general, when a client is using host verification, the CN that you specify when generating the broker certificate must match the full host name for the Route on the broker Pod that the client is connecting to. For example, if you have a deployment with a single broker pod, the CN might look like the following:

```
CN=my-broker-deployment-0-svc-rte-my-openshift-project.my-openshift-domain
```

To ensure that the CN can resolve to **any** broker pod in a deployment with multiple brokers, you can specify an asterisk (*) wildcard character in place of the ordinal of the broker pod. For example:

```
CN=my-broker-deployment-*-svc-rte-my-openshift-project.my-openshift-domain
```

The CN shown in the preceding example successfully resolves to any broker pod in the **my-broker-deployment** deployment.

In the Subject Alternative Name (SAN) that you specify when generating the broker certificate, you can specify a wildcard DNS name to match all of the brokers in the cluster. In the following example, an asterisk (*) wildcard character is specified in place of the ordinal of the broker pod.

```
"SAN=DNS:my-broker-deployment-*-svc-rte-my-openshift-project.my-openshift-domain"
```

If wildcard DNS names are not supported, you can include a comma-separated list of DNS names in the SAN field of the certificate for all of the broker pods in the cluster. For example:

```
"SAN=DNS:my-broker-deployment-0-svc-rte-my-openshift-project.my-openshift-domain,DNS:my-broker-deployment-1-svc-rte-my-openshift-project.my-openshift-domain,..."
```

4.12.3. Networking services in your broker deployments

Two network services, a headless service and a ping service, are automatically created by the Operator for your broker deployment.

On the **Networking** pane of the OpenShift Container Platform web console for your broker deployment, there are two running services; a *headless* service and a *ping* service. The default name of the headless service uses the format **<custom_resource_name>-hdls-svc**, for example, **my-broker-deployment-hdls-svc**. The default name of the ping service uses a format of **<custom_resource_name>-ping-svc**, for example, **my-broker-deployment-ping-svc**.

The headless service provides access to port 61616, which is used for internal broker clustering.

The ping service is used by the brokers for discovery, and enables brokers to form a cluster within the OpenShift environment. Internally, this service exposes port 8888.

4.12.4. Connecting to the broker from internal and external clients

Internal clients run on the same OpenShift cluster as the broker while external clients are external to the OpenShift cluster where the broker runs. The way in which clients connect to the broker differs between internal and external clients.

4.12.4.1. Connecting to the broker from internal clients

To connect an internal client to a broker, you specify the DNS resolvable name of the broker pod in the client connection details.

The following is an example connection for internal clients.

```
$ tcp://ex-aao-ss-0:<port>
```

If the internal client is using the Core protocol and the **useTopologyForLoadBalancing=false** key was not set in the connection URL, after the client connects to the broker for the first time, the broker can

inform the client of the addresses of all the brokers in the cluster. The client can then load balance connections across all brokers.

If your brokers have durable subscription queues or request/reply queues, be aware of the caveats associated with using these queues when client connections are load balanced.

Additional resources

- [Section 4.12.4.4, “Caveats to load balancing client connections when you have durable subscription queues or reply/request queues”](#)

4.12.4.2. Connecting to the broker from external clients

When you expose an acceptor on the broker, the Operator automatically creates a dedicated service and route for each broker pod in the deployment. External clients can connect to a broker by specifying the full host name of the route created for the broker pod.

You can use a basic **curl** command to test external access to the full host name of the route. For example:

```
$ curl https://my-broker-deployment-0-svc-rte-my-openshift-project.my-openshift-domain
```

The full host name of the route for the broker pod must resolve to the node that is hosting the OpenShift router. The OpenShift router uses the host name to determine where to send the traffic inside the OpenShift internal network. By default, the OpenShift router listens to port 80 for non-secured (that is, non-SSL) traffic and port 443 for secured (that is, SSL-encrypted) traffic. For an HTTP connection, the router automatically directs traffic to port 443 if you specify a secure connection URL (that is, **https**), or to port 80 if you specify a non-secure connection URL (that is, **http**).

In each external client’s connection URL, specify the following:

- The full host name of the route for each broker pod and the port number. The client attempts to connect to the first host name in the connection URL. However, if the first host name is unavailable, the client automatically connects to the next host name in the connection URL, and so on.
- For one-way TLS, the path to the trust store and the corresponding password.
- For two-way TLS, the key store and the corresponding password, also.
- If the external client uses the Core protocol, set the **useTopologyForLoadBalancing=false** key in the client’s connection URL.
Setting the **useTopologyForLoadBalancing=false** key prevents a client from using the AMQ Broker Pod DNS names that are in the cluster topology information provided by the broker. The Pod DNS names resolve to internal IP addresses, which an external client cannot access.

Some example client connection URLs, for supported messaging protocols, are shown below.

External Core client, using one-way TLS

```
tcp://my-broker-deployment-0-svc-rte-my-openshift-project.my-openshift-domain:443?
useTopologyForLoadBalancing=false&sslEnabled=true \
&trustStorePath=~/.client.ts&trustStorePassword=<password>
```

**NOTE**

The **useTopologyForLoadBalancing** key is explicitly set to **false** in the connection URL because an external Core client cannot use topology information returned by the broker. If this key is set to **true** or you do not specify a value, it results in a DEBUG log message.

External Core client, using two-way TLS

```
tcp://my-broker-deployment-0-svc-rte-my-openshift-project.my-openshift-domain:443?
useTopologyForLoadBalancing=false&sslEnabled=true \
&keyStorePath=~/.client.ks&keyStorePassword=<password> \
&trustStorePath=~/.client.ts&trustStorePassword=<password>
```

External OpenWire client, using one-way TLS

```
ssl://my-broker-deployment-0-svc-rte-my-openshift-project.my-openshift-domain:443"

# Also, specify the following JVM flags
-Djavax.net.ssl.trustStore=~/.client.ts -Djavax.net.ssl.trustStorePassword=<password>
```

External OpenWire client, using two-way TLS

```
ssl://my-broker-deployment-0-svc-rte-my-openshift-project.my-openshift-domain:443"

# Also, specify the following JVM flags
-Djavax.net.ssl.keyStore=~/.client.ks -Djavax.net.ssl.keyStorePassword=<password> \
-Djavax.net.ssl.trustStore=~/.client.ts -Djavax.net.ssl.trustStorePassword=<password>
```

External AMQP client, using one-way TLS

```
amqps://my-broker-deployment-0-svc-rte-my-openshift-project.my-openshift-domain:443?
transport.verifyHost=true \
&transport.trustStoreLocation=~/.client.ts&transport.trustStorePassword=<password>
```

External AMQP client, using two-way TLS

```
amqps://my-broker-deployment-0-svc-rte-my-openshift-project.my-openshift-domain:443?
transport.verifyHost=true \
&transport.keyStoreLocation=~/.client.ks&transport.keyStorePassword=<password> \
&transport.trustStoreLocation=~/.client.ts&transport.trustStorePassword=<password>
```

4.12.4.3. Connecting to the Broker using a NodePort

As an alternative to exposing routes on the broker for external client connections, you can configure a NodePort on the broker for these connections.

The NodePort should map to one of the protocol-specific ports specified by the acceptors configured for the broker.

By default, NodePorts are in the range 30000 to 32767, which means that a NodePort typically does not match the intended port on the broker pod.

To connect from a client outside OpenShift to the broker via a NodePort, you specify a URL in the format **<protocol>://<ocp_node_ip>:<node_port_number>**.

Additional resources

- [Configuring ingress cluster traffic overview](#)

4.12.4.4. Caveats to load balancing client connections when you have durable subscription queues or reply/request queues

If client connections are load balanced, a client is not guaranteed to connect to the same broker every time it initiates a connection. This can lead to issues with features that require connection persistence, such as the use of durable subscriptions and reply/request queues.

Durable subscriptions

A durable subscription is represented as a queue on a broker and is created when a durable subscriber first connects to the broker. This queue exists and receives messages until the client unsubscribes. If the client reconnects to a different broker, another durable subscription queue is created on that broker. This can cause the following issues.

Table 4.6. Issues with durable subscriptions

Issue	Mitigation
Messages may get stranded in the original subscription queue.	Enable message distribution by setting the redistributionDelay property for an address or set of addresses. You can set this property under the brokerProperties attribute in the ActiveMQArtemis CR. For example: addressSettings. <address>.redistributionDelay=5000. In the example, the broker waits 5000 milliseconds after a queue's final consumer closes before it redistributes messages to other brokers.
Messages may be received in the wrong order as there is a window during message redistribution when other messages are still routed.	None.

Issue	Mitigation
When a client unsubscribes, it deletes the queue only on the broker it last connected to. This means that the other queues can still exist and receive messages.	To delete other empty queues that may exist for a client that unsubscribed, configure both of the following properties for an address or set of addresses. You can set these properties under the brokerProperties attribute in the ActiveMQArtemis CR. addressSettings. <code><address>.autoDeleteQueuesMessageCount=0</code> addressSettings. <code><address>.autoDeleteQueuesDelay=5000</code> With the autoDeleteQueuesMessageCount property set to 0, a queue is deleted only if there are no messages in the queue. The value of the autoDeleteQueuesDelay property is the number of milliseconds after which a queue that has no messages is deleted.

Request/Reply queues

When a JMS Producer creates a temporary reply queue, the queue is created on the broker. If the client that is consuming from the work queue and replying to the temporary queue connects to a different broker, the following issues can occur.

Table 4.7. Issues with request/reply queues

Issue	Mitigation
Since the reply queue does not exist on the broker that the client is connected to, the client may generate an error.	Configure the broker to automatically create a queue when a client requests to connect to a queue that does not exist. To configure automatic queue creation, add the following property under the brokerProperties attribute in the ActiveMQArtemis CR. addressSettings. <code><address>.autoCreateQueues=true</code>

Issue	Mitigation
Messages sent to the work queue may not be distributed.	Enable load balancing on demand by adding the following property under the brokerProperties attribute in the ActiveMQArtemis CR: clusterConfigurations. <cluster>.messageLoadBalancingType=ON-DEMAND. Also, enable message distribution by setting the redistributionDelay property for an address or set of addresses. You can set this property under the brokerProperties attribute in the ActiveMQArtemis CR. For example: addressSettings<address>.redistributionDelay=5000

Additional resources

- [Enabling message redistribution.](#)
- [Configuring automatic creation and deletion of addresses and queues .](#)

4.12.5. Disabling FIPS mode

FIPS is a standard that specifies the security requirements for cryptographic modules used in applications. AMQ Broker runs on FIPS-enabled RHEL systems. If you need the broker to accept connections from clients that use non-FIPS compliant algorithms, you can disable FIPS mode on the broker.

Procedure

1. Edit the **ActiveMQArtemis** custom resource (CR) for your broker deployment.
2. Add an environment variable to set a Java system property that disables FIPS mode on the broker. For example:

```
spec:
  ...
  env:
    - name: JDK_JAVA_OPTIONS
      value: com.redhat.fips=false
  ..
```

3. Save the **ActiveMQArtemis** CR.

4.13. SECURING BROKER-BROKER CONNECTIONS IN A CLUSTER

Internal connections between brokers in a cluster use an internal connector and acceptor, both of which are named **artemis**. You can secure connections between brokers in a cluster by configuring Transport Layer Security (TLS) for the **artemis** connector and acceptor.

On the acceptor, you specify a secret that contains a common TLS certificate for all the brokers in the cluster. On the connector, you specify a truststore that contains the public key of the TLS certificate. The public key is required in each broker's truststore so a broker can trust the other brokers in the cluster when they establish a TLS connection.

The following example procedure describes how to secure the internal connections between the brokers in a cluster by using a self-signed certificate.

Procedure

1. Generate a self-signed TLS certificate and add it to a keystore file.

- In the **Subject Alternative Name** (SAN) field of the certificate, specify a wildcard DNS name to match all of the brokers in the cluster, as shown in the following example. The example is based on using a CR named **ex-aao** that is deployed in a **test** namespace.

```
$ keytool -storetype jks -keystore server-keystore.jks -storepass artemis -keypass artemis -alias server -genkey -keyalg "RSA" -keysize 2048 -dname "CN=AMQ Server, OU=Artemis, O=ActiveMQ, L=AMQ, S=AMQ, C=AMQ" -validity 365 -ext bc=ca:false -ext eku=sA -ext san=dns:*.ex-aao-hdls-svc.test.svc.cluster.local
```

- If the certificate does not support the use of wildcard DNS names, you can include a comma-separated list of DNS names in the **SAN** field of the certificate for all of the broker pods in the cluster. For example:

```
keytool -storetype jks -keystore server-keystore.jks -storepass artemis -keypass artemis -alias server -genkey -keyalg "RSA" -keysize 2048 -dname "CN=AMQ Server, OU=Artemis, O=ActiveMQ, L=AMQ, S=AMQ, C=AMQ" -validity 365 -ext bc=ca:false -ext eku=sA -ext san=dns:ex-aao-ss-0.ex-aao-hdls-svc.test.svc.cluster.local,dns:ex-aao-ss-1.ex-aao-hdls-svc.test.svc.cluster.local
```

- If the TLS certificate does not support the use of DNS names, you must disable host verification in the **ActiveMQArtemis** CR, as described below.

2. Export the public key of the TLS certificate from the keystore file so that it can be imported into a truststore file. For example:

```
$ keytool -storetype jks -keystore server-keystore.jks -storepass artemis -alias server -exportcert -rfc > server.crt
```

3. Import the public key of the TLS certificate into a truststore file so the other brokers in the clusters can trust the certificate. For example:

```
$ keytool -storetype jks -keystore server-truststore.jks -storepass artemis -keypass artemis -importcert -alias server -file server.crt -noprompt
```

4. Create a secret to store the keystore and truststore files and their associated passwords. For example:

```
oc create secret generic artemis-ssl-secret --namespace test --from-file=broker.ks=server-
keystore.jks --from-file=client.ts=server-truststore.jks --from-
literal=keyStorePassword=artemis --from-literal=trustStorePassword=artemis
```

5. Edit the **ActiveMQArtemis** CR for your broker deployment and add an internal acceptor named **artemis**. In the **artemis** acceptor, set the **sslEnabled** attribute to **true** and specify the name of the secret that you created in the **sslSecret** attribute. For example:

```
spec:
  ..
  deploymentPlan:
    size: 2
  acceptors:
  - name: artemis
    port: 61616
    sslEnabled: true
    sslSecret: artemis-ssl-secret
  ..
```

6. Enable SSL for the **artemis** connector, which is used by each broker in the cluster to connect to other brokers in the cluster. Use the **brokerProperties** attribute to enable SSL and specify the path and credentials of the truststore file that contains the public key of the TLS certificate.

```
spec:
  ..
  deploymentPlan:
    size: 2
  acceptors:
  - name: artemis
    port: 61616
    sslEnabled: true
    sslSecret: artemis-ssl-secret
  brokerProperties:
  - 'connectorConfigurations.artemis.params.sslEnabled=true'
  - 'connectorConfigurations.artemis.params.trustStorePath=/etc/artemis-ssl-secret-
volume/client.ts'
  - 'connectorConfigurations.artemis.params.trustStorePassword=artemis'
  ..
```

connectorConfigurations.artemis.params.trustStorePath

This value must match the location of the truststore file, **client.ts** on the broker pods. The truststore file and accompanying password file in the secret are mounted in a **/etc/<secret name>-volume** directory on each broker pod. The previous example specifies the location of a truststore that is in a secret named **artemis-ssl-secret**.

7. If the TLS certificate does not support the use of DNS names, use the **brokerProperties** attribute to disable host verification. For example:

```
spec:
  ..
  brokerProperties:
  ..
  - 'connectorConfigurations.artemis.params.verifyHost=false'
  ..
```

- 8. Save the CR.

4.14. CONFIGURING LARGE MESSAGE HANDLING FOR AMQP MESSAGES

It is possible for clients to send large AMQP messages that exceed the size of the broker's internal buffer, causing unexpected errors. To avoid this risk, you can configure the broker to store messages as files, instead of in memory, when the message size exceeds a specified minimum value.

For a broker deployment on OpenShift Container Platform, the large messages directory is `/opt/<custom_resource_name>/data/large-messages` on the Persistent Volume (PV) used by the broker for message storage. When the broker stores a message as a large message, the queue retains a reference to the file in the large messages directory.



NOTE

You can configure the large message size limit in the broker configuration for the AMQP protocol only. For the Core Protocol and OpenWire protocols, you can configure large message size limits in the client connection configuration.

Additional resources

- [Red Hat AMQ Clients documentation](#)

4.14.1. Configuring AMQP acceptors for large message handling

You can configure the broker to store messages as files, instead of in memory, when the message size exceeds a minimum threshold. Any message that exceeds the configured threshold is stored in a dedicated directory specifically for large message files.

Prerequisites

- You should be familiar with how to configure acceptors for Operator-based broker deployments. See [Section 4.12.1, "Configuring acceptors"](#).
- To store large AMQP messages in a dedicated large messages directory, your broker deployment must be using persistent storage (that is, **persistenceEnabled** is set to **true** in the Custom Resource (CR) instance used to create the deployment). For more information about configuring persistent storage, see:
 - [Section 2.9, "Operator deployment notes"](#)
 - [Section 8.1, "Custom resource configuration reference"](#)

Procedure

1. Open the Custom Resource (CR) instance in which you previously defined an AMQP acceptor.
 - a. Using the OpenShift command-line interface:

```
$ oc edit -f <path/to/custom_resource_instance>.yaml
```

- b. Using the OpenShift Container Platform web console:

- i. In the left navigation menu, click **Administration > Custom Resource Definitions**
- ii. Click the **ActiveMQArtemis** CRD.
- iii. Click the **Instances** tab.
- iv. Locate the CR instance that corresponds to your project namespace.

A previously-configured AMQP acceptor might resemble the following:

```
spec:
...
acceptors:
- name: my-acceptor
  protocols: amqp
  port: 5672
  connectionsAllowed: 5
  expose: true
  sslEnabled: true
...
```

2. Specify the minimum size, in bytes, of an AMQP message that the broker handles as a large message. For example:

```
spec:
...
acceptors:
- name: my-acceptor
  protocols: amqp
  port: 5672
  connectionsAllowed: 5
  expose: true
  sslEnabled: true
  amqpMinLargeMessageSize: 204800
...
...
```

In the preceding example, the broker is configured to accept AMQP messages on port 5672. Based on the value of **amqpMinLargeMessageSize**, if the acceptor receives an AMQP message with a body larger than or equal to 204800 bytes (that is, 200 kilobytes), the broker stores the message as a large message.

The broker stores the message in the large messages directory (**/opt/<custom_resource_name>/data/large-messages**, by default) on the persistent volume (PV) used by the broker for message storage.

If you do not explicitly specify a value for the **amqpMinLargeMessageSize** property, the broker uses a default value of 102400 (that is, 100 kilobytes).

If you set **amqpMinLargeMessageSize** to a value of **-1**, large message handling for AMQP messages is disabled.

4.15. CONFIGURING BROKER HEALTH CHECKS

You can configure readiness, liveness, and startup probes to monitor the health and operational status of your broker containers.

The following summary describes the check performed by each probe.

- A startup probe indicates whether the application within a container is started.
- A liveness probe determines if a container is still running.
- A readiness probe determines if a container is ready to accept service requests

If a startup probe or a liveness probe check of a Pod fails, the probe restarts the Pod.

AMQ Broker includes default readiness and liveness probes. The default liveness probe checks if the broker is running by pinging the broker's HTTP port. The default readiness probe checks if the broker can accept network traffic by opening a connection to each of the acceptor ports configured for the broker.

A limitation of using the default liveness and readiness probes is that they are unable to identify underlying issues, for example, issues with the broker's file system. You can create custom liveness and readiness probes that use the broker's command-line utility, **artemis**, to run more comprehensive health checks.

AMQ Broker does not include a default startup probe. You can configure a startup probe in the **ActiveMQArtemis** Custom Resource (CR).

4.15.1. Configuring a startup probe

You can configure a startup probe to check if the AMQ Broker application within the broker container has started.

Procedure

1. Edit the CR instance for the broker deployment.
 - a. Using the OpenShift command-line interface:
 - i. Log in to OpenShift Container Platform as a user that has privileges to deploy CRs in the project for the broker deployment.
 - ii. Edit the CR for your deployment.

```
oc edit ActiveMQArtemis <CR instance name> -n <namespace>
```

- b. Using the OpenShift Container Platform web console:
 - i. Log in to OpenShift Container Platform as a user that has privileges to deploy CRs in the project for the broker deployment.
 - ii. In the left pane, click **Administration > Custom Resource Definitions**
 - iii. Click the **ActiveMQArtemis** CRD.
 - iv. Click the **Instances** tab.
 - v. Click the instance for your broker deployment.

vi. Click the **YAML** tab.

Within the console, a YAML editor opens, enabling you to edit the CR instance.

2. In the **deploymentPlan** section of the CR, add a **startupProbe** section. For example:

```
spec:
  deploymentPlan:
    startupProbe:
      exec:
        command:
          - /bin/bash
          - '-c'
          - /opt/amq/bin/artemis
          - 'check'
          - 'node'
          - '--up'
          - '--url'
          - 'tcp://$HOSTNAME:61616'
      initialDelaySeconds: 5
      periodSeconds: 10
      timeoutSeconds: 3
      failureThreshold: 30
```

command

The startup probe command to run within the container. In the example, the startup probe uses the **artemis check node** command to verify that AMQ Broker has started in the container for a broker Pod.

initialDelaySeconds

The delay, in seconds, before the probe runs after the container starts. The default is **0**.

periodSeconds

The interval, in seconds, at which the probe runs. The default is **10**.

timeoutSeconds

Time, in seconds, that the startup probe command waits for a reply from the broker. If a response to the command is not received, the command stops. The default value is **1**.

failureThreshold

The minimum consecutive failures, including timeouts, of the startup probe after which the probe is deemed to have failed. When the probe is deemed to have failed, it restarts the Pod. The default value is **3**.

Depending on the resources of the cluster and the size of the broker journal, you might need to increase the failure threshold to allow the broker sufficient time to start and pass the probe check. Otherwise, the broker enters a loop condition whereby the failure threshold is reached repeatedly and the broker is restarted each time by the startup probe. For example, if you set the **failureThreshold** to **30** and the probe runs at the default interval of 10 seconds, the broker has 300 seconds to start and pass the probe check.

3. Save the CR.

Additional resources

- [Monitoring application health by using health checks](#)

4.15.2. Configuring liveness and readiness probes

You can configure a liveness probe to check if a container is running and a readiness probe to determine if a container is ready to accept service requests.

Prerequisites

- You should be familiar with how to use a CR instance to create a basic broker deployment. See [Section 3.3.1, “Deploying a single broker instance”](#).

Procedure

1. Edit the CR instance for the broker deployment.
 - a. Using the OpenShift command-line interface:
 - i. Log in to OpenShift Container Platform as a user that has privileges to deploy CRs in the project for the broker deployment.
 - ii. Edit the CR for your deployment.

```
oc edit ActiveMQArtemis <CR instance name> -n <namespace>
```

- b. Using the OpenShift Container Platform web console:
 - i. Log in to OpenShift Container Platform as a user that has privileges to deploy CRs in the project for the broker deployment.
 - ii. In the left pane, click **Administration > Custom Resource Definitions**
 - iii. Click the **ActiveMQArtemis** CRD.
 - iv. Click the **Instances** tab.
 - v. Click the instance for your broker deployment.
 - vi. Click the **YAML** tab.
2. To configure a liveness probe, in the **deploymentPlan** section of the CR, add a **livenessProbe** section. For example:

```
spec:
  deploymentPlan:
    livenessProbe:
      initialDelaySeconds: 5
      periodSeconds: 5
      failureThreshold: 30
```

initialDelaySeconds

The delay, in seconds, before the probe runs after the container starts. The default is **5**.



NOTE

If the deployment also has a startup probe configured, you can set the delay to 0 for both a liveness and a readiness probe. Both of these probes run only after the startup probe has passed. If the startup probe has already passed, it confirms that the broker has started successfully, so a delay in running the liveness and readiness probes is not required.

periodSeconds

The interval, in seconds, at which the probe runs. The default is **5**.

failureThreshold

The minimum consecutive failures, including timeouts, of the liveness probe that signify the probe has failed. When the probe fails, it restarts the Pod. The default value is 3.

If your deployment does not have a startup probe configured, which verifies that the broker application is started before the liveness probe runs, you might need to increase the failure threshold to allow the broker sufficient time to start and pass the liveness probe check. Otherwise, the broker can enter a loop condition whereby the failure threshold is reached repeatedly and the broker Pod is restarted each time by the liveness probe.

The time required by the broker to start and pass a liveness probe check depends on the resources of the cluster and the size of the broker journal. For example, if you set the **failureThreshold** to 30 and the probe runs at the default interval of 5 seconds, the broker has 150 seconds to start and pass the liveness probe check.



NOTE

If you do not configure a liveness probe or if the handler is missing from a configured probe, the AMQ Broker Operator creates a default TCP probe that has the following configuration. The default TCP probe attempts to open a socket to the broker container on the specified port.

```
spec:
  deploymentPlan:
    livenessProbe:
      tcpSocket:
        port: 8181
      initialDelaySeconds: 30
      timeoutSeconds: 5
```

3. To configure a readiness probe, in the **deploymentPlan** section of the CR, add a **readinessProbe** section. For example:

```
spec:
  deploymentPlan:
    readinessProbe:
      initialDelaySeconds: 5
      periodSeconds: 5
```

If you don't configure a readiness probe, a built-in [script](#) checks if all acceptors can accept connections.

4. If you want to configure more comprehensive health checks, add the **artemis check** command-line utility to the liveness or readiness probe configuration.
 - a. If you want to configure a health check that creates a full client connection to the broker, in the **livenessProbe** or **readinessProbe** section, add an **exec** section. In the **exec** section, add a **command** section. In the **command** section, add the **artemis check node** command syntax. For example:

```
spec:
  deploymentPlan:
    readinessProbe:
      exec:
        command:
          - bash
          - '-c'
          - /home/jboss/amq-broker/bin/artemis
          - check
          - node
          - '--silent'
          - '--acceptor'
          - <acceptor name>
          - '--user'
          - $AMQ_USER
          - '--password'
          - $AMQ_PASSWORD
        initialDelaySeconds: 30
        timeoutSeconds: 5
```

By default, the **artemis check node** command uses the URI of an acceptor called **artemis**. If the broker has an acceptor called **artemis**, you can exclude the **--acceptor <acceptor name>** option from the command.



NOTE

\$AMQ_USER and **\$AMQ_PASSWORD** are environment variables that are configured by the AMQ Operator.

- b. If you want to configure a health check that produces and consumes messages, which also validates the health of the broker's file system, in the **livenessProbe** or **readinessProbe** section, add an **exec** section. In the **exec** section, add a **command** section. In the **command** section, add the **artemis check queue** command syntax. For example:

```
spec:
  deploymentPlan:
    readinessProbe:
      exec:
        command:
          - bash
          - '-c'
          - /home/jboss/amq-broker/bin/artemis
          - check
          - queue
          - '--name'
          - livenessqueue
          - '--produce'
```

```

- "1"
- '--consume'
- "1"
- '--silent'
- '--user'
- $AMQ_USER
- '--password'
- $AMQ_PASSWORD
initialDelaySeconds: 30
timeoutSeconds: 5

```

**NOTE**

The queue name that you specify must be configured on the broker and have a **routingType** of **anycast**. For example:

```

apiVersion: broker.amq.io/v1beta1
kind: ActiveMQArtemisAddress
metadata:
  name: livenessqueue
  namespace: activemq-artemis-operator
spec:
  addressName: livenessqueue
  queueConfiguration:
    purgeOnNoConsumers: false
    maxConsumers: -1
    durable: true
    enabled: true
  queueName: livenessqueue
  routingType: anycast

```

5. Save the CR.

Additional resources

- [Monitoring application health by using health checks](#)

4.16. ENABLING MESSAGE MIGRATION TO SUPPORT CLUSTER SCALEDOWN

AMQ Broker allows you to scale a cluster up or down by editing the cluster deployment size.

If you want to be able to scale down the number of brokers in a cluster that is underutilized, you must enable message migration to avoid message loss. When message migration is enabled, messages are moved from a broker that is being removed to a remaining broker during the scaledown operation.

Starting in AMQ Broker 7.14, you can configure the AMQ Broker Operator to manage scaledown. The default method of managing scaledown by using a scaledown controller is deprecated in 7.14.

4.16.1. Managing scaledown by using the Operator

When the AMQ Broker Operator manages scaledown, the broker remains in the StatefulSet until all messages are migrated.

A benefit of using the Operator to manage scaledown is that you can view the queues on the broker that is being removed to monitor the progress of the message migration. In addition, a target broker for the migrated messages is automatically chosen.

Procedure

1. Edit the **ActiveMQArtemis** custom resource (CR) instance for the broker deployment.
2. Add configuration to manage scaledown by using the AMQ Broker Operator. For example:

```
spec:
  ...
  deploymentPlan:
    size: 3
    messageMigration: true
    persistenceEnabled: true
    brokerProperties:
      - "HAPolicyConfiguration.scaleDownConfiguration.enabled=false"
      - "HAPolicyConfiguration=PRIMARY_ONLY"
      - "HAPolicyConfiguration.scaleDownConfiguration.discoveryGroup=my-discovery-group"
    ...
```

messageMigration: true

Ensures that messages are moved from a broker that is being removed from the cluster to a remaining broker when a scaledown is initiated.

persistenceEnabled: true

Configures a broker to persist messages, which is a prerequisite for message migration.

HAPolicyConfiguration.scaleDownConfiguration.enabled=false

Disables the default method of managing scaledown which uses the scaledown controller. With the default method disabled, the AMQ Broker Operator manages scaledown requests. If this property is set to **true** or is excluded from the CR, the default scaledown method is used.

HAPolicyConfiguration=PRIMARY_ONLY

Specifies that the broker is a standalone broker instance without any backup or replication partners.

HAPolicyConfiguration.scaleDownConfiguration.discoveryGroup=my-discovery-group

Uses the **my-discovery-group** default cluster discovery group to identify a target broker for messages that must be migrated from the broker that is being removed from the cluster.

3. Save the CR.
4. After your request a scale down, check the **status** section of the CR. The **scaledown** condition displays the following reason values, which correspond to the stages of a scaledown operation.

Table 4.8. Reason values for a scaledown operation

Reason	Description
PendingEmpty	Waiting for all messages to be moved to another broker in the cluster.

Reason	Description
PendingConfigApplied	Waiting for the Operator to change the value of the HAPolicyConfiguration.scaleDownConfiguration.enabled attribute to true for the pod that is going to be removed from the cluster.
PendingDelete	Waiting removal of the broker from the cluster.

Check the **message** field for the **scaledown** condition to view any additional information for the scaledown operation.



NOTE

The broker that is being removed from the cluster as a result of the scaledown remains in the StatefulSet until it has no pending messages. If message migration fails to complete before the pod is shut down, the pod is restarted and the procedure is repeated until all of the messages are migrated.

4.16.2. Managing scaledown by using the scaledown controller (deprecated)

When you scale down a cluster that has message migration enabled, a scaledown controller manages the message migration process by default.

4.16.2.1. Steps in message migration process

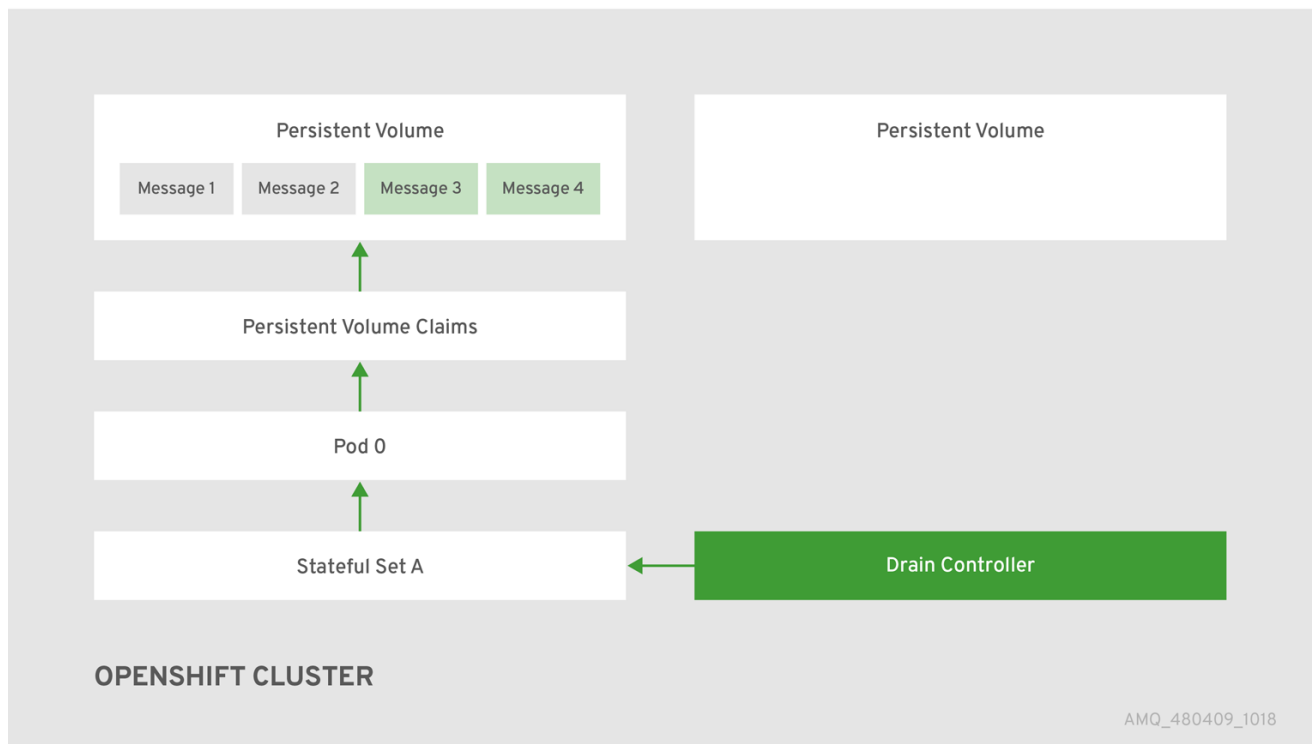
If you scale down the number of brokers in a cluster that has message migration enabled, a temporary drainer pod migrates messages to another broker in the cluster.

The message migration process follows these steps.

1. When a broker pod in the deployment shuts down due to an intentional scale down of the deployment, the Operator automatically deploys a scaledown custom resource to prepare for message migration.
2. To check for Persistent Volumes (PVs) that have been orphaned, the scaledown controller looks at the ordinal on the volume claim. The controller compares the ordinal on the volume claim to that of the broker pods that are still running in the StatefulSet (that is, the broker cluster) in the project.
If the ordinal on the volume claim is higher than the ordinal on any of the broker Pods still running in the broker cluster, the scaledown controller determines that the broker Pod at that ordinal has been shut down and that messaging data must be migrated to another broker Pod.
3. The scaledown controller starts a drainer pod. The drainer pod connects to one of the other live broker pods in the cluster and migrates messages to that live broker pod.

The following figure illustrates how the scaledown controller (also known as a *drain controller*) migrates messages to a running broker pod.

Figure 4.1. Message migration using the scaledown controller



After the messages are migrated successfully to an operational broker pod, the drainer pod shuts down and the scaledown controller removes the PVC for the orphaned PV. The PV is returned to a "Released" state.



NOTE

If the reclaim policy for the PV is set to **retain**, the PV cannot be used by another pod until you delete and recreate the PV. For example, if you scale the cluster up after scaling it down, the PV is not available to a pod started until you delete and recreate the PV.

Additional resources

- [Section 4.16.2.2, "Enabling message migration"](#)

4.16.2.2. Enabling message migration

If you want to use the scaledown controller, which is the default method, to manage scaledown requests, you must enable message migration.

Prerequisites

- You already have a basic broker deployment. See [Section 3.3.1, "Deploying a single broker instance"](#).
- You understand how message migration works. For more information, see [Section 4.16.2.1, "Steps in message migration process"](#).



NOTE

- A scaledown controller operates only within a single OpenShift project. The controller cannot migrate messages between brokers in separate projects.
- If you scale a broker deployment down to 0 (zero), message migration does not occur, since there is no running broker pod to which messaging data can be migrated. However, if you scale a deployment down to zero and then back up to a size that is smaller than the original deployment, drainer pods are started for the brokers that remain shut down.

Procedure

1. Edit the **ActiveMQArtemis** CR instance for the broker deployment.
2. In the **deploymentPlan** section of the CR, add a **messageMigration** attribute and set to **true**. If not already configured, add a **persistenceEnabled** attribute and also set to **true**. For example:

```
spec:
  deploymentPlan:
    messageMigration: true
    persistenceEnabled: true
  ...
```

These settings mean that when you later scale down the size of your clustered broker deployment, the Operator automatically starts a scaledown controller and migrates messages to a broker pod that is still running.

3. Save the CR.
4. (Optional) Complete the following steps to scale down the cluster and view the message migration process.
 - a. In your existing broker deployment, verify which pods are running.

```
$ oc get pods
```

You see output that looks like the following.

```
activemq-artemis-operator-8566d9bf58-9g25l 1/1 Running 0 3m38s
ex-aa0-ss-0                                1/1 Running 0 112s
ex-aa0-ss-1                                1/1 Running 0 8s
```

The preceding output shows that there are three pods running; one for the broker Operator itself, and a separate pod for each broker in the deployment.

- b. Log into each pod and send some messages to each broker.
 - i. Supposing that pod **ex-aa0-ss-0** has a cluster IP address of **172.17.0.6**, run the following command:

```
$ /opt/amq/bin/artemis producer --url tcp://172.17.0.6:61616 --user admin --password admin
```

- c. Supposing that pod **ex-aao-ss-1** has a cluster IP address of **172.17.0.7**, run the following command:

```
$ /opt/amq/bin/artemis producer --url tcp://172.17.0.7:61616 --user admin --password admin
```

The preceding commands create a queue called **TEST** on each broker and add 1000 messages to each queue.

- d. Scale the cluster down from two brokers to one.
 - i. Open the main broker CR, **broker_activemqartemis_cr.yaml**.
 - ii. In the CR, set **deploymentPlan.size** to **1**.
 - iii. At the command line, apply the change:

```
$ oc apply -f deploy/crs/broker_activemqartemis_cr.yaml
```

You see that the pod **ex-aao-ss-1** starts to shut down. The scaledown controller starts a new drainer pod of the same name. This drainer pod also shuts down after it migrates all messages from broker pod **ex-aao-ss-1** to the other broker pod in the cluster, **ex-aao-ss-0**.

- e. When the drainer pod is shut down, check the message count on the **TEST** queue of broker pod **ex-aao-ss-0**. You see that the number of messages in the queue is 2000, indicating that the drainer pod successfully migrated 1000 messages from the broker pod that shut down.

4.17. CONTROLLING PLACEMENT OF BROKER PODS ON OPENSIFT CONTAINER PLATFORM NODES

You can control the placement of AMQ Broker pods on OpenShift Container Platform nodes to, for example, provide greater resilience to node failures.

The control of pod placement can be implemented by using node selectors, tolerations, or affinity and anti-affinity rules.

Node selectors

A node selector allows you to schedule a broker pod on a specific node.

Tolerations

A toleration enables a broker pod to be scheduled on a node if the toleration matches a taint configured for the node. Without a matching pod toleration, a taint allows a node to refuse to accept a pod.

Affinity and anti-affinity

Node affinity rules control which nodes a pod can be scheduled on based on the node's labels. Pod affinity and anti-affinity rules control which nodes a pod can be scheduled on based on the pods already running on that node.

4.17.1. Placing pods on specific nodes using node selectors

In a node selector, you configure a key-value pair that OpenShift uses to schedule the broker pod on a node that has a matching key-value pair.

The following example shows how to configure a node selector to schedule a broker pod on a specific node.

Prerequisites

- You should be familiar with how to use a CR instance to create a basic broker deployment. See [Section 3.3.1, “Deploying a single broker instance”](#).
- Add a label to the OpenShift Container Platform node on which you want to schedule the broker pod. For more information about adding node labels, see [Using node selectors to control pod placement](#) in the OpenShift Container Platform documentation.

Procedure

1. Create a Custom Resource (CR) instance based on the main broker CRD.

- a. Using the OpenShift command-line interface:

- i. Log in to OpenShift as a user that has privileges to deploy CRs in the project for the broker deployment.

```
oc login -u <user> -p <password> --server=<host:port>
```

- ii. Open the sample CR file called **broker_activemqartemis_cr.yaml** that was included in the **deploy/crs** directory of the Operator installation archive that you downloaded and extracted.

- b. Using the OpenShift Container Platform web console:

- i. Log in to the console as a user that has privileges to deploy CRs in the project for the broker deployment.

- ii. Start a new CR instance based on the main broker CRD. In the left pane, click **Administration > Custom Resource Definitions**

- iii. Click the **ActiveMQArtemis** CRD.

- iv. Click the **Instances** tab.

- v. Click **Create ActiveMQArtemis**.

Within the console, a YAML editor opens, enabling you to configure a CR instance.

2. In the **deploymentPlan** section of the CR, add a **nodeSelector** section and add the node label that you want to match to select a node for the pod. For example:

```
spec:
  deploymentPlan:
    nodeSelector:
      app: broker1
```

In this example, the broker pod is scheduled on a node that has a **app: broker1** label.

3. Deploy the CR instance.

- a. Using the OpenShift command-line interface:

- i. Save the CR file.
- ii. Switch to the project in which you are creating the broker deployment.

```
$ oc project <project_name>
```

- iii. Create the CR instance.

```
$ oc create -f <path/to/custom_resource_instance>.yaml
```

- b. Using the OpenShift web console:

- i. When you have finished configuring the CR, click **Create**.

Additional resources

- [Placing pods on specific nodes using node selectors](#)

4.17.2. Controlling pod placement using tolerations

You can use taints and tolerations to control whether broker pods can or cannot be scheduled on specific nodes. A taint allows a node to refuse to schedule a pod unless the pod has a matching toleration.

You can use taints to exclude pods from a node so the node is reserved for specific pods, such as broker pods, that have a matching toleration.

Having a matching toleration permits a broker pod to be scheduled on a node but does not guarantee that the pod is scheduled on that node. To guarantee that the broker pod is scheduled on the node that has a taint configured, you can configure affinity rules. For more information, see [Section 4.17.3, "Controlling pod placement by using affinity and anti-affinity rules"](#)

The following example shows how to configure a toleration to match a taint that is configured on a node.

Prerequisites

- You should be familiar with how to use a CR instance to create a basic broker deployment. See [Section 3.3.1, "Deploying a single broker instance"](#).
- Apply a taint to the nodes which you want to reserve for scheduling broker pods. A taint consists of a key, value, and effect. The taint effect determines if:
 - existing pods on the node are evicted
 - existing pods are allowed to remain on the node but new pods cannot be scheduled unless they have a matching toleration
 - new pods can be scheduled on the node if necessary, but preference is to not schedule new pods on the node.

Procedure

1. Create a Custom Resource (CR) instance based on the main broker CRD.
 - a. Using the OpenShift command-line interface:

- i. Log in to OpenShift as a user that has privileges to deploy CRs in the project for the broker deployment.

```
oc login -u <user> -p <password> --server=<host:port>
```

- ii. Open the sample CR file called **broker_activemqartemis_cr.yaml** that was included in the **deploy/crs** directory of the Operator installation archive that you downloaded and extracted.

- b. Using the OpenShift Container Platform web console:

- i. Log in to the console as a user that has privileges to deploy CRs in the project for the broker deployment.

- ii. Start a new CR instance based on the main broker CRD. In the left pane, click **Administration > Custom Resource Definitions**

- iii. Click the **ActiveMQArtemis** CRD.

- iv. Click the **Instances** tab.

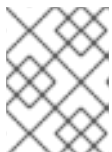
- v. Click **Create ActiveMQArtemis**.

Within the console, a YAML editor opens, enabling you to configure a CR instance.

2. In the **deploymentPlan** section of the CR, add a **tolerations** section. In the **tolerations** section, add a toleration for the node taint that you want to match. For example:

```
spec:
  deploymentPlan:
    tolerations:
      - key: "app"
        value: "amq-broker"
        effect: "NoSchedule"
```

In this example, the toleration matches a node taint of **app=amq-broker:NoSchedule**, so the pod can be scheduled on a node that has this taint configured.



NOTE

To ensure that the broker pods are scheduled correctly, do not specify a **tolerationsSeconds** attribute in the **tolerations** section of the CR.

3. Deploy the CR instance.

- a. Using the OpenShift command-line interface:

- i. Save the CR file.

- ii. Switch to the project in which you are creating the broker deployment.

```
$ oc project <project_name>
```

- iii. Create the CR instance.

```
$ oc create -f <path/to/custom_resource_instance>.yaml
```

- b. Using the OpenShift web console:
 - i. When you have finished configuring the CR, click **Create**.

Additional resources

- [Controlling pod placement using node taints](#)

4.17.3. Controlling pod placement by using affinity and anti-affinity rules

You can control the placement of broker pods on nodes by using node affinity, pod affinity, or pod anti-affinity rules.

4.17.3.1. Controlling pod placement using node affinity rules

A broker pod can be scheduled on any node that has a matching key-value pair to that specified in an affinity rule.

The following example shows how to configure a broker to control pod placement by using node affinity rules.

Prerequisites

- You should be familiar with how to use a CR instance to create a basic broker deployment. See [Section 3.3.1, “Deploying a single broker instance”](#).
- Assign a common label to the nodes in your OpenShift Container Platform cluster that can schedule the broker pod, for example, **zone: emea**.

Procedure

1. Create a Custom Resource (CR) instance based on the main broker CRD.
 - a. Using the OpenShift command-line interface:
 - i. Log in to OpenShift as a user that has privileges to deploy CRs in the project for the broker deployment.


```
oc login -u <user> -p <password> --server=<host:port>
```
 - ii. Open the sample CR file called **broker_activemqartemis_cr.yaml** that was included in the **deploy/crs** directory of the Operator installation archive that you downloaded and extracted.
 - b. Using the OpenShift Container Platform web console:
 - i. Log in to the console as a user that has privileges to deploy CRs in the project for the broker deployment.
 - ii. Start a new CR instance based on the main broker CRD. In the left pane, click **Administration > Custom Resource Definitions**
 - iii. Click the **ActiveMQArtemis** CRD.
 - iv. Click the **Instances** tab.

- v. Click **Create ActiveMQArtemis**.
Within the console, a YAML editor opens, enabling you to configure a CR instance.
2. In the **deploymentPlan** section of the CR, add the following sections: **affinity**, **nodeAffinity**, **requiredDuringSchedulingIgnoredDuringExecution**, and **nodeSelectorTerms**. In the **nodeSelectorTerms** section, add the **- matchExpressions** parameter and specify the key-value string of a node label to match. For example:

```
spec:
  deploymentPlan:
    affinity:
      nodeAffinity:
        requiredDuringSchedulingIgnoredDuringExecution:
          nodeSelectorTerms:
            - matchExpressions:
                - key: zone
                  operator: In
                  values:
                    - emea
```

In this example, the affinity rule allows the pod to be scheduled on any node that has a label with a key of **zone** and a value of **emea**.

3. Deploy the CR instance.
 - a. Using the OpenShift command-line interface:
 - i. Save the CR file.
 - ii. Switch to the project in which you are creating the broker deployment.

```
$ oc project <project_name>
```

- iii. Create the CR instance.

```
$ oc create -f <path/to/custom_resource_instance>.yaml
```

- b. Using the OpenShift web console:
 - i. When you have finished configuring the CR, click **Create**.

Additional resources

- [Controlling pod placement on nodes using node affinity rules](#)

4.17.3.2. Placing pods relative to other pods using anti-affinity rules

Anti-affinity rules allow you to restrict which Openshift node a broker pod can be scheduled on based on the labels of pods already running on the node. You can use anti-affinity rules to ensure that multiple broker pods are not scheduled on the same Openshift node.

Prerequisites

- You created two separate broker deployments. See [Section 3.3.1, “Deploying a single broker instance”](#).

Procedure

1. Edit the **ActiveMQArtemis** CR instance for your first broker deployment.
2. In the **deploymentPlan** section of the CR, add a **labels** section. Create an identifying label for the broker so that you can create an anti-affinity rule based on the label in your second deployment. For example:

```
spec:
  ...
  deploymentPlan:
    labels:
      name: broker1
```

3. Save the CR.
4. Edit the **ActiveMQArtemis** CR instance for your second broker deployment.
5. In the **deploymentPlan** section of the CR, add the following sections: **affinity**, **podAntiAffinity**, **requiredDuringSchedulingIgnoredDuringExecution**, and **labelSelector**. In the **labelSelector** section, add the **matchExpressions** parameter and specify the key-value string of the label to match. A pod in this deployment cannot be scheduled on a node that contains a pod with the matching label.

```
spec:
  deploymentPlan:
    affinity:
      podAntiAffinity:
        requiredDuringSchedulingIgnoredDuringExecution:
          - labelSelector:
              matchExpressions:
                - key: name
                  operator: In
                  values:
                    - broker1
            topologyKey: topology.kubernetes.io/zone
```

In this example, the pod anti-affinity rule prevents a pod from being placed on the same node as a pod that has a label with a key of **name** and a value of **broker1**, which is the label assigned to the broker in your first deployment.

6. Save the CR.

Additional resources

- [Controlling pod placement on nodes using node affinity rules](#)

4.18. CONFIGURING LOGGING FOR BROKERS

AMQ Broker uses the Log4j 2 logging utility to provide message logging. You can override the default Log4j 2 configuration by creating a new configuration file stored in a secret or configMap. The Operator mounts this file to configure logging for each broker pod.

Prerequisite

- You are familiar with the Log4j 2 configuration options.

Procedure

1. Prepare a file that contains the log4j 2 configuration that you want to use with AMQ Broker. The default Log4j 2 configuration file that is used by a broker is located in the **/home/jboss/amq-broker/etc/log4j2.properties** file on each broker pod. You can use the contents of the default configuration file as the basis for creating a new Log4j 2 configuration in a secret or configMap. To get the contents of the default Log4j 2 configuration file, complete the following steps.
 - a. Using the OpenShift Container Platform web console:
 - i. Click **Workloads > Pods**
 - ii. Click the **ex-aao-ss** Pod.
 - iii. Click the **Terminal** tab.
 - iv. Use the **cat** command to display the contents of the **/home/jboss/amq-broker/etc/log4j2.properties** file on a broker pod and copy the contents.
 - v. Paste the contents into a local file, where the OpenShift Container Platform CLI is installed, and save the file as **logging.properties**.
 - b. Using the OpenShift command-line interface:
 - i. Get the name of a pod in your deployment.

```
$ oc get pods -o wide
```

NAME	STATUS	IP
amq-broker-operator-54d996c	Running	10.129.2.14
ex-aao-ss-0	Running	10.129.2.15

- ii. Use the **oc cp** command to copy the log configuration file from a pod to your local directory.

```
$ oc cp <pod name>:/home/jboss/amq-broker/etc/log4j2.properties  
logging.properties -c <name>-container
```

Where the *<name>* part of the container name is the prefix before the **-ss** string in the pod name. For example:

```
$ oc cp ex-aao-ss-0:/home/jboss/amq-broker/etc/log4j2.properties logging.properties  
-c ex-aao-container
```



NOTE

When you create a configMap or secret from a file, the key in the configMap or secret defaults to the file name and the value defaults to the file content. By creating a secret from a file named **logging.properties**, the required key for the new logging configuration is inserted in the secret or configMap.

2. Edit the **logging.properties** file and create the Log4j 2 configuration that you want to use with AMQ Broker.
For example, with the default configuration, AMQ Broker logs messages to the console only. You might want to update the configuration so that AMQ Broker logs messages to disk also.

3. Add the updated Log4j 2 configuration to a secret or a ConfigMap.

- a. Log in to OpenShift as a user that has privileges to create secrets or ConfigMaps in the project for the broker deployment.

```
oc login -u <user> -p <password> --server=<host:port>
```

- b. If you want to configure the log settings in a secret, use the **oc create secret** command. For example:

```
oc create secret generic newlog4j-logging-config --from-file=logging.properties
```

- c. If you want to configure the log settings in a ConfigMap, use the **oc create configmap** command. For example:

```
oc create configmap newlog4j-logging-config --from-file=logging.properties
```

The configMap or secret name must have a suffix of **-logging-config**, so the Operator can recognize that the secret contains new logging configuration.

4. Add the secret or ConfigMap to the Custom Resource (CR) instance for your broker deployment.

- a. Using the OpenShift command-line interface:

- i. Log in to OpenShift as a user that has privileges to deploy CRs in the project for the broker deployment.

```
oc login -u <user> -p <password> --server=<host:port>
```

- ii. Edit the CR.

```
oc edit ActiveMQArtemis <CR instance name> -n <namespace>
```

- b. Using the OpenShift Container Platform web console:

- i. Log in to the console as a user that has privileges to deploy CRs in the project for the broker deployment.

- ii. In the left pane, click **Ecosystem > Installed Operators**

- iii. Click the **Red Hat Integration - AMQ Broker for RHEL 9 (Multiarch)** operator.

- iv. Click the **AMQ Broker** tab.

- v. Click the name of the ActiveMQArtemis instance name

- vi. Click the **YAML** tab.

Within the console, a YAML editor opens, enabling you to configure a CR instance.

- c. Add the secret or configMap that contains the Log4j 2 logging configuration to the CR. The following examples show a secret and a configMap added to the CR.

```
apiVersion: broker.amq.io/v1beta1
kind: ActiveMQArtemis
metadata:
  name: ex-aa0
spec:
  deploymentPlan:
    ...
  extraMounts:
    secrets:
      - "newlog4j-logging-config"
    ...
```

```
apiVersion: broker.amq.io/v1beta1
kind: ActiveMQArtemis
metadata:
  name: ex-aa0
spec:
  deploymentPlan:
    ...
  extraMounts:
    configMaps:
      - "newlog4j-logging-config"
    ...
```

5. Save the CR.

In each broker pod, the Operator mounts a **logging.properties** file that contains the logging configuration in the secret or configMap that you created. In addition, the Operator configures each broker to use the mounted log configuration file instead of the default log configuration file.



NOTE

If you update the logging configuration in a configMap or secret, each broker automatically uses the updated logging configuration.

4.19. CONFIGURING A POD DISRUPTION BUDGET

A pod disruption budget specifies the minimum number of pods in a cluster that must be available simultaneously during a voluntary disruption, such as a maintenance window.

Procedure

1. Edit the CR instance for the broker deployment.
 - a. Using the OpenShift command-line interface:
 - i. Log in to OpenShift Container Platform as a user that has privileges to deploy CRs in the project for the broker deployment.
 - ii. Edit the CR for your deployment.


```
oc edit ActiveMQArtemis <CR instance name> -n <namespace>
```

b. Using the OpenShift Container Platform web console:

- i. Log in to OpenShift Container Platform as a user that has privileges to deploy CRs in the project for the broker deployment.
- ii. In the left pane, click **Administration > Custom Resource Definitions**
- iii. Click the **ActiveMQArtemis** CRD.
- iv. Click the **Instances** tab.
- v. Click the instance for your broker deployment.
- vi. Click the **YAML** tab.

Within the console, a YAML editor opens, enabling you to edit the CR instance.

2. In the **spec** section of the CR, add a **podDisruptionBudget** element and specify the minimum number of pods in your deployment that must be available during a voluntary disruption. In the following example, a minimum of one pod must be available:

```
spec:
  ...
  podDisruptionBudget:
    minAvailable: 1
  ...
```

3. Save the CR.

Additional resources

- [Understanding how to use pod disruption budgets to specify the number of pods that must be up](#)

4.20. CONFIGURING ROLE-BASED ACCESS CONTROL FOR MANAGEMENT OPERATIONS

Role-based access control (RBAC) is used to restrict access to the attributes and methods of MBeans. MBeans are the way the management API is exposed by AMQ Broker to support management operations.

Previously, you could restrict access to MBeans by setting the RBAC configuration in the **ActiveMQArtemisSecurity** custom resource (CR) and restarting the broker for the changes to take effect. Starting in 7.12, you can restrict access to MBeans in the **ActiveMQArtemis** CR and a broker restart is not required for the changes to take effect.

Procedure

1. Edit the **ActiveMQArtemis** CR instance for your broker deployment.
2. Add the following environment variable to configure the broker to use the RBAC configuration that you specify in the **ActiveMQArtemis** CR.

```
spec:
  ..
  env:
    - name: JAVA_ARGS_APPEND
      value: "-Dhawtio.roles=* -
Djavax.management.builder.initial=org.apache.activemq.artemis.core.server.management.Arte
misRbacMBeanServerBuilder"
  ..
```

3. In the **brokerProperties** attribute, add the role based access control configuration for management operations.

The format of the match addresses for management operations is:

mops.<resource type>.<resource name>.<operation>

For example, the following configuration grants a **manager** role **view** and **edit** permission to an **activemq.management** address. The asterisk (*) in the operation position grants access to all operations.

```
spec:
  ..
  brokerProperties:
    - securityRoles."mops.address.activemq.management.*".manager.view=true
    - securityRoles."mops.address.activemq.management.*".manager.edit=true
```

In the following example, the number sign (#) after the **mops** prefix grants the **amq** role **view** and **edit** permissions to all MBeans.

```
spec:
  ..
  brokerProperties:
    - securityRoles."mops.#".amq.view=true
    - securityRoles."mops.#".amq.edit=true
  ..
```

4. Use the **resourceTemplates** attribute to define an init container that runs a script to remove the default RBAC configuration in the **/amq/init/config/amq-broker/etc/management.xml** file in each broker container, as shown in the following example. You must remove the default RBAC configuration so the broker uses the new RBAC configuration that you created in the **ActiveMQArtemis** CR.

```
spec:
  ..
  resourceTemplates:
    - selector:
        kind: "StatefulSet"
      patch:
        kind: "StatefulSet"
      spec:
        template:
          spec:
            initContainers:
              - name: "${CR_NAME}-container-init"
                args:
```

```

- '-c'
- '/opt/amq/bin/launch.sh && /opt/amq-broker/script/default.sh; echo "Empty
management.xml";echo "<management-context
xmlns=\"http://activemq.apache.org/schema\" />" > /amq/init/config/amq-
broker/etc/management.xml'

```

\$(CR_NAME) is the value of the **metadata.name** attribute in your broker CR instance.

5. Save the CR.

4.21. CUSTOMIZING OPENSIFT RESOURCES CREATED BY THE OPERATOR

An AMQ Broker deployment creates OpenShift resources, such as deployments, pods, statefulSets and service resources, which are managed by the AMQ Broker Operator. You can customize these Operator-managed OpenShift resources.

Customizing an operator-managed OpenShift resource can be useful if you want to perform certain tasks, such as:

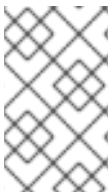
- Adding custom annotations that control how resources are treated by other services.
- Modifying attributes that are not exposed in the broker custom resource.

You can use the **resourceTemplates** attribute to customize resources created by the AMQ Broker Operator. If you want to add an annotation or label to a resource, configure the **resourceTemplates** attribute to include the **annotations** or **labels** attribute. In the following example, the **annotations** attribute is used to add an annotation to all the services managed by the Operator.

```

spec:
  ..
  resourceTemplates:
    - selector:
        kind: "Service"
      annotations:
        name: "amq-operator-managed"
    ..

```



NOTE

The **selector** attribute determines which Operator-managed resources are customized. For example, a **selector** value of **kind: "Service"**, customizes all **service** resources. If the **selector** attribute is empty, changes are applied to all Operator-managed resources.

If you want to customize items other than annotations or labels for resources, you must use the **patch** attribute with the **resourceTemplates** attribute. When you specify the **patch** attribute, the Operator uses a strategic merge to update resources.



NOTE

If you use the **patch** attribute, you must populate the **selector** attribute to identify specific resources to update.

In the following example, the **patch** attribute is used to change the default value of the **minReadySeconds** property in the **StatefulSet** resource.

```
spec:
  ..
  resourceTemplates:
  - selector:
      kind: "StatefulSet"
    patch:
      kind: "StatefulSet"
      spec:
        template:
          spec:
            minReadySeconds: 10
    ..
```

Additional resources

- [Use a strategic merge patch to update a Deployment](#)

4.22. REGISTERING PLUGINS WITH AMQ BROKER

You can extend the functionality of AMQ Broker by registering plugins in the **brokerProperties** attribute in the CR.

Procedure

1. Edit the custom resource (CR) for your broker deployment.
2. In the **brokerProperties** attribute, specify the class name of the plugin and include a comma-separated string of `<key>=<value>` pairs that define the properties for the plugin.
In the following example, the **LoggingActiveMQServerPlugin** plugin, which is provided with AMQ Broker, is registered.

```
spec:
  ...
  brokerProperties:
  -
    brokerPlugins: "org.apache.activemq.artemis.core.server.plugin.impl.LoggingActiveMQServerPlugin.class".init=LOG_CONNECTION_EVENTS=true,LOG_SESSION_EVENTS=true,LOG_CONSUMER_EVENTS=true
    ...
```

3. Save the CR.
After an instance of the plugin is created, the `init` method is passed a string that contains the `<key>=<value>` pairs, which are used to set properties for the plugin.



NOTE

If you create a custom plugin, ensure that the JAR files for the plugin class are in the Java classpath of the broker. For more information, see [Section 4.4, "Adding third-party JAR files"](#).

4.23. SEPARATING THE `BROKERPROPERTIES` CONFIGURATION

If the broker custom resource reaches the 1 MB size limit, separate **`brokerProperties`** entries into Java properties or JSON files. Store these files in an OpenShift secret referenced by the CR.

Individual Java properties files and JSON files are each limited to 1 MB in size, which is the size limit for secrets in OpenShift.

Procedure

1. Create a file that contains the **`brokerProperties`** configuration that you want to apply to the broker. You can add the properties in text or JSON format.

Text format

In text format, you add each property in a separate line in the properties file. For example:

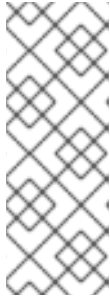
```
addressConfigurations.usa-news.queueConfigs.usa-news.routingType=ANYCAST
addressConfigurations.usa-news.queueConfigs.usa-news.purgeOnNoConsumers=true
addressConfigurations.usa-news.queueConfigs.usa-news.maxConsumers=30
```

JSON format

JSON format is more compact than text format for writing properties. Therefore, you can store more properties in an individual file without exceeding the 1 MB size limit. The following is the JSON equivalent of the previous text format example:

```
{
  "addressConfigurations": {
    "usa-news": {
      "routingTypes": "MULTICAST",
      "queueConfigs": {
        "usa-news": {
          "routingType": "MULTICAST",
          "purgeOnNoConsumers": "true",
          "maxConsumers": "30"
        }
      }
    }
  }
}
```

2. Save the file with one of the following extensions depending on the format of the properties in the file:
 - **`.properties`** if the properties are in text format. For example, **`addresses.properties`**.
 - **`.json`** if the properties are in JSON format. For example, **`addresses.json`**.

**NOTE**

If you have multiple brokers in your deployment, you can add addresses to individual brokers by naming the file with a prefix of **broker-*<ordinal>***. Replace *<ordinal>* with the ordinal number assigned to the specific broker pod by the StatefulSet. For example, a file name of **broker-0.addresses.properties** adds the addresses to the first pod, **broker-1.addresses.properties** to the second pod, and so on.

3. Create a secret that contains the file you created. For example:

```
oc create secret generic address-settings-bp --from-file=addresses.properties
```

**NOTE**

The secret name must have a suffix of **-bp**. When a secret has a **-bp** suffix, the broker finds **.properties** and **.json** files in the directory where the secret is mounted on the broker pod.

4. Add a reference to the secret in the **extraMounts** attribute so the Operator mounts the properties files that are in the secret on each broker pod:

```
deploymentPlan:
  ...
  extraMounts:
    secrets:
      - "address-settings-bp"
  ...
```

The Operator mounts the file or files that are in the secret in a **/amq/extra/secrets/*<secret name>*** directory on each broker pod.

At startup, the broker searches each mounted directory for files that have a **.properties** or **.json** extension, sorts the files alphabetically, and applies the configuration in the files one after another. Within a file, the broker applies the properties in the order in which they are listed.

4.24. CONFIGURING LEADER-FOLLOWER DEPLOYMENTS FOR HIGH AVAILABILITY

You can configure a high availability (HA) leader-follower topology for two standalone brokers. Both brokers must be configured to persist messages to the same journal or JDBC database. Only the leader broker processes client requests at any given time.

A leader-follower configuration has separate deployments that each have a single broker instance. High availability is achieved by the brokers competing to acquire a lock for the database or the shared volume. The broker that acquires a lock becomes the leader broker, which serves client requests. The broker that is unable to acquire a lock becomes a follower. A follower is a passive broker that continuously tries to obtain a lock for the database or the shared volume. If the follower obtains a lock, it immediately becomes the leader and starts to serve clients.

Leader-follower deployments provide a faster mean time to repair (MTTR) for a node failure than that provided by OpenShift for a single deployment. In leader-follower deployments, the brokers can be on separate clusters to protect against a cluster failure. These clusters can be in different data centers to

also make the broker service resilient to a data center outage, provided the storage used to persist messages is available across data centers.

4.24.1. Configuring leader-follower broker deployments that use a shared Java Database Connectivity (JDBC) database

You can configure leader-follower broker deployments that persist messages to a shared JDBC database.

Prerequisite

You have a container image that contains the JAR file for the JDBC database you want to use with AMQ Broker. For information about creating container images, see [Creating images](#) in the OpenShift documentation. In the configuration for each broker, you can specify an init container to copy the JAR file from the container image to a location that is available to the broker at runtime.

Procedure

1. Configure two **ActiveMQArtemis** custom resource instances to create separate broker deployments.
In each custom resource, specify a unique name and ensure that the **clustered** and **persistenceEnabled** attributes are set to **false**. Set the **size** attribute to **1** to create a single broker in each deployment. For example:

```
apiVersion: broker.amq.io/v1beta1
kind: ActiveMQArtemis
metadata:
  name: peer-broker-a
spec:
  deploymentPlan:
    size: 1
    clustered: false
    persistenceEnabled: false
```

```
apiVersion: broker.amq.io/v1beta1
kind: ActiveMQArtemis
metadata:
  name: peer-broker-b
spec:
  deploymentPlan:
    size: 1
    clustered: false
    persistenceEnabled: false
```



NOTE

If both broker deployments are on the same OpenShift cluster, ensure that the broker pods are provisioned on separate nodes to avoid an outage of both brokers if a node failure occurs. For more information about controlling the placement of pods on nodes, see [Section 4.17, “Controlling placement of broker pods on OpenShift Container Platform nodes”](#).

2. Add a new liveness probe to each CR.
If you do not configure a liveness probe, a default probe is enabled to check the health of a

broker. The default probe checks that the AMQ Management Console is reachable. In a leader-follower configuration, the AMQ Management Console is not reachable on the broker that is a follower at any given time, which causes the liveness probe to fail on that broker. Each time the liveness probe fails, it restarts the broker, which puts the broker in a persistent restart loop. As a result, the follower broker enters a **CrashLoopBackOff** state and is not available to become the leader if the current leader fails.

To prevent the default liveness probe from running, you must configure a new liveness probe that can run successfully when a broker is either a leader or a follower. The following is an example of a liveness probe that meets this requirement. In this example, the liveness probe checks that the command to run the broker was executed, which is indicated by the presence of the **cli.lock** lock file.

```
spec:
  ..
  livenessProbe:
    exec:
      command:
        - test
        - -f
        - /home/jboss/amq-broker/lock/cli.lock
    ..
```

For more information about configuring liveness probes, see [Section 4.15.2, “Configuring liveness and readiness probes”](#).

3. In each broker configuration, enable JDBC database persistence by using the **brokerProperties** attribute. For example:

```
spec:
  ..
  brokerProperties:
    - storeConfiguration=DATABASE
    - storeConfiguration.jdbcDriverClassName=<class name>
    - storeConfiguration.jdbcConnectionUrl=jdbc:<Database URL>
    - HAPolicyConfiguration=SHARED_STORE_PRIMARY
    - storeConfiguration.jdbcLockRenewPeriodMillis=2000
    - storeConfiguration.jdbcLockExpirationMillis=6000
```

For more information about enabling JDBC database persistence, see [Section 4.5.2, “Configuring database persistence”](#).

4. In each broker configuration, configure the broker to load the JAR file required to connect to the JDBC database.
 - Use the **resourceTemplates** attribute to customize the **StatefulSet** resource for each broker. In the customization, use the **patch** attribute to specify an init container that copies the JAR file from the custom container image you prepared to the broker pod.
 - Use the **env** attribute to create an **ARTEMIS_EXTRA_LIBS** environment variable to extend the broker’s Java classpath to include the directory to which the JAR file for the JDBC database is copied. By extending the Java classpath, the broker can load the JAR file from the specified directory on the pod at runtime.

```
spec:
```



```

..
env:
- name: ARTEMIS_EXTRA_LIBS
  value: '/amq/init/config/extra-libs'
resourceTemplates:
- selector:
  kind: StatefulSet
  patch:
  kind: StatefulSet
  spec:
    template:
      spec:
        initContainers:
        - name: jdbc-driver-init
          image: <custom container image with JAR>
          volumeMounts:
          - name: amq-cfg-dir
            mountPath: /amq/init/config
        command:
        - "bash"
        - "-c"
        - "mkdir -p /amq/init/config/extra-libs && cp <__JAR file__> /amq/init/config/extra-libs"

```

For more information about customizing OpenShift resources created by the Operator, see [Section 4.21, “Customizing Openshift resources created by the Operator”](#).

5. Save each custom resource.

The following example shows the configuration for leader-follower broker deployments that use an Oracle database.

```

apiVersion: broker.amq.io/v1beta1
kind: ActiveMQArtemis
metadata:
  name: peer-broker-a
spec:
  deploymentPlan:
    size: 1
    clustered: false
    persistenceEnabled: false
    livenessProbe:
      exec:
        command:
        - test
        - -f
        - /home/jboss/amq-broker/lock/cli.lock
  env:
  - name: ARTEMIS_EXTRA_LIBS
    value: '/amq/init/config/extra-libs'
  brokerProperties:
  - criticalAnalyser=false
  - storeConfiguration=DATABASE
  - storeConfiguration.jdbcDriverClassName=oracle.jdbc.OracleDriver
  - storeConfiguration.jdbcConnectionUrl=jdbc:<Database URL>
  - storeConfiguration.jdbcLockRenewPeriodMillis=2000
  - storeConfiguration.jdbcLockExpirationMillis=6000

```

```

- HAPolicyConfiguration=SHARED_STORE_PRIMARY
acceptors:
- name: ext-acceptor
  protocols: CORE
  port: 61626
  expose: true
  sslEnabled: true
  sslSecret: ext-acceptor-ssl-secret
console:
  expose: true
resourceTemplates:
- selector:
  kind: StatefulSet
  patch:
  kind: StatefulSet
  spec:
    template:
      spec:
        initContainers:
        - name: oracle-database-jdbc-driver-init
          image: <custom container image with JAR>
          volumeMounts:
          - name: amq-cfg-dir
            mountPath: /amq/init/config
          command:
          - "bash"
          - "-c"
          - "mkdir -p /amq/init/config/extra-libs && cp <JAR file> /amq/init/config/extra-libs"

```

```

apiVersion: broker.amq.io/v1beta1
kind: ActiveMQArtemis
metadata:
  name: peer-broker-b
spec:
  deploymentPlan:
    size: 1
    clustered: false
    persistenceEnabled: false
    livenessProbe:
      exec:
        command:
        - test
        - -f
        - /home/jboss/amq-broker/lock/cli.lock
  env:
  - name: ARTEMIS_EXTRA_LIBS
    value: '/amq/init/config/extra-libs'
  brokerProperties:
  - criticalAnalyser=false
  - storeConfiguration=DATABASE
  - storeConfiguration.jdbcDriverClassName=oracle.jdbc.OracleDriver
  - storeConfiguration.jdbcConnectionUrl=jdbc:<Database URL>
  - storeConfiguration.jdbcLockRenewPeriodMillis=2000
  - storeConfiguration.jdbcLockExpirationMillis=6000
  - HAPolicyConfiguration=SHARED_STORE_PRIMARY
acceptors:

```

```

- name: ext-acceptor
  protocols: CORE
  port: 61626
  expose: true
  sslEnabled: true
  sslSecret: ext-acceptor-ssl-secret
console:
  expose: true
resourceTemplates:
- selector:
    kind: StatefulSet
  patch:
    kind: StatefulSet
  spec:
    template:
      spec:
        initContainers:
        - name: oracle-database-jdbc-driver-init
          image: <custom container image with JAR>
          volumeMounts:
          - name: amq-cfg-dir
            mountPath: /amq/init/config
          command:
          - "bash"
          - "-c"
          - "mkdir -p /amq/init/config/extra-libs && cp <JAR file> /amq/init/config/extra-libs"

```

4.24.2. Configuring leader-follower broker deployments that use a shared journal

You can configure leader-follower broker deployments that persist messages to a journal on a volume that is shared by both broker instances.

Prerequisite

You created a Persistent Volume Claim (PVC) to request a shared volume to store messaging data for both brokers. Give careful consideration to ensure that the capacity you request is sufficient to store the messaging data on the shared volume. In the following PVC example, 2 gigabytes storage capacity is requested.

```

oc apply -f - <<EOF
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: shared-volume
  namespace: activemq
spec:
  accessModes:
  - ReadWriteMany
  volumeMode: Filesystem
resources:
  requests:
    storage: 2Gi
EOF

```

**NOTE**

After you create the PVC, the output of the **oc get pvc** command shows that the shared volume has a **pending** status until the brokers are deployed and bind to the volume. For example:

NAME	STATUS	VOLUME	CAPACITY	ACCESS MODES	STORAGECLASS
VOLUMEATTRIBUTESCLASS AGE					
shared-volume	Pending			gp3-csi	<unset> 7s

Procedure

1. Prepare and apply a YAML configuration to create leader-follower broker deployments. Ensure that you set the **persistenceEnabled** attribute to **false** so the Operator does not create the resources to attach a separate persistent volume to each broker container. In a leader-follower configuration, both brokers must share the same persistent volume, which must be provisioned before you apply the YAML configuration. The Operator uses the **extraVolumes** and **extraVolumeMount** attributes in the YAML to mount that persistent volume on both broker containers. In addition, the Operator uses the **brokerProperties** attributes in the YAML to create the journal and other data files on that volume.

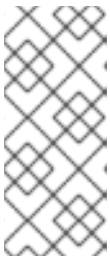
For example:

```
oc apply -f - <<EOF
apiVersion: broker.amq.io/v1beta1
kind: ActiveMQArtemis
metadata:
  name: peer-broker-a
spec:
  deploymentPlan:
    size: 1
    persistenceEnabled: false
    clustered: false
    extraVolumes:
      - name: extra-volume
        persistentVolumeClaim:
          claimName: shared-volume
    extraVolumeMounts:
      - name: extra-volume
        mountPath: /opt/amq-broker/data
    livenessProbe:
      exec:
        command:
          - test
          - -f
          - /home/jboss/amq-broker/lock/cli.lock
    brokerProperties:
      - HAPolicyConfiguration=SHARED_STORE_PRIMARY
      - journalDirectory=/opt/amq-broker/data/journal
      - pagingDirectory=/opt/amq-broker/data/paging,
      - bindingsDirectory=/opt/amq-broker/data/bindings
      - largeMessagesDirectory=/opt/amq-broker/data/largemessages
    acceptors:
      - name: ext-acceptor
```

```

    protocols: CORE
    port: 61626
    expose: true
    sslEnabled: true
    sslSecret: ext-acceptor-ssl-secret
  ---
  apiVersion: broker.amq.io/v1beta1
  kind: ActiveMQArtemis
  metadata:
    name: peer-broker-b
  spec:
    deploymentPlan:
      size: 1
      persistenceEnabled: false
      clustered: false
      extraVolumes:
        - name: extra-volume
          persistentVolumeClaim:
            claimName: shared-volume
      extraVolumeMounts:
        - name: extra-volume
          mountPath: /opt/amq-broker/data
      livenessProbe:
        exec:
          command:
            - test
            - -f
            - /home/jboss/amq-broker/lock/cli.lock
    brokerProperties:
      - HAPolicyConfiguration=SHARED_STORE_PRIMARY
      - journalDirectory=/opt/amq-broker/data/journal
      - pagingDirectory=/opt/amq-broker/data/paging,
      - bindingsDirectory=/opt/amq-broker/data/bindings
      - largeMessagesDirectory=/opt/amq-broker/data/largemessages
    acceptors:
      - name: ext-acceptor
        protocols: CORE
        port: 61626
        expose: true
        sslEnabled: true
        sslSecret: ext-acceptor-ssl-secret
  EOF

```



NOTE

If both broker deployments are on the same OpenShift cluster, ensure that the broker pods are provisioned on separate nodes to avoid an outage of both brokers if a node failure occurs. For more information about controlling the placement of pods on nodes, see [Section 4.17, “Controlling placement of broker pods on OpenShift Container Platform nodes”](#).

The following table describes the YAML configuration in the example that is specific to a leader-follower deployment.

Table 4.9. Leader-follower configuration

Attribute	Description
metadata.name	Specify a unique name for the broker in each deployment.
size	Specify a value of 1 to create a single broker in each deployment.
persistenceEnabled	Set to false so the Operator does not create the resources to attach a separate persistent volume to each broker container. In a leader-follower configuration, both brokers must share the same persistent volume, which must be provisioned before you apply the leader-follower YAML configuration.
clustered	Set to false to disable clustering.
extraVolumes	Specify the details of the PVC you created to request a shared persistent volume for the brokers, which is a prerequisite to creating leader-follower deployments. name Use a string value to assign a name to the volume. claimName Ensure that this matches the value of the metadata.name attribute in the PVC you created.
extraVolumeMounts	Specify the details to mount the shared persistent volume on each broker container. name Ensure that this matches the extraVolumes.name value. mountPath The path to which the shared volume is mounted on each broker container.

Attribute	Description
livenessProbe	A default liveness probe is enabled to check the health of a broker. The default probe checks that the AMQ Management Console is reachable. In a leader-follower configuration, the AMQ Management Console is not reachable on the broker that is a follower at any given time, which causes the liveness probe to fail on that broker. Each time the liveness probe fails, it restarts the broker, which puts the broker in a persistent restart loop. As a result, the follower broker enters a CrashLoopBackOff state and is not available to become the leader if the current leader fails. Configure a new liveness probe that can run successfully when a broker is either a leader or a follower, as shown in the example. In the example, the liveness probe checks that the command to run the broker was executed, which is indicated by the presence of the cli.lock lock file. For more information about configuring liveness probes, see Section 4.15.2, “Configuring liveness and readiness probes”.
brokerProperties	Specify the directories on the mounted persistent volume to store the journal and other messaging data files. HAPolicyConfiguration Set to SHARED_STORE_PRIMARY to ensure that the broker uses a file lock to protect the broker from concurrent access by both brokers. journalDirectory File system location to store the message journal. pagingDirectory File system location to store paged messages. bindingsDirectory File system location to store the bindings journal. largeMessagesDirectory File system location to store large messages.

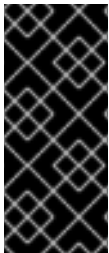
- Use the **oc get pods** command to identify which broker is the leader and which is the follower. In the following example, **peer-broker-a-ss-0** is the leader, which is indicated by the **1/1** entry in the **READY** column.

NAME	READY	STATUS	RESTARTS	AGE
peer-broker-a-ss-0	1/1	Running	0	72m
peer-broker-b-ss-0	0/1	Running	0	72m

4.25. CONFIGURING DATA MIRRORING FOR DISASTER RECOVERY

Mirroring is the process of copying data from a broker to one or more other brokers for disaster recovery. The source and target brokers in a mirror can be on separate OpenShift clusters in different data centers to protect against a data center outage.

Mirroring can also be used for data backup or to create a failover broker for use during maintenance windows. Messages that existed before a mirror is created are not mirrored.



IMPORTANT

Ensure that you do not implement a mirror topology that creates a loop, which cause message duplication. For example, if you have a 3-broker deployment where broker 1 mirrors data to broker 2 and broker 2 mirrors data to broker 3, avoid creating a loop by mirroring data from broker 3 to broker 1. Similarly, do not create a topology of 3 or more brokers where each broker mirrors data to all of the other brokers.



NOTE

You can implement a dual mirror topology where both brokers are mirrors of each other.

Procedure

1. Configure two **ActiveMQArtemis** custom resource (CR) instances to create a source broker and a target broker for the mirrored data. Specify a unique name for each. For example:

```
apiVersion: broker.amq.io/v1beta1
kind: ActiveMQArtemis
metadata:
  name: production-broker
  namespace: production
spec:
  deploymentPlan:
    size: 1
```

```
apiVersion: broker.amq.io/v1beta1
kind: ActiveMQArtemis
metadata:
  name: mirror-broker
  namespace: dr
spec:
  deploymentPlan:
    size: 1
```

2. In the CR of the target broker, add an acceptor for the mirror connection. For example:

```
metadata:
  name: mirror-broker
  namespace: dr
spec:
  ...
  acceptors:
  - expose: true
    name: amqp
```



```
port: 5672
protocols: amqp
...
```

After you create the acceptor, a route is exposed for the acceptor in the following format:

<broker name>-<acceptor name>-<ordinal>-svc-rte.<namespace>.<hostname>

When you add the mirror configuration to the source broker later in this procedure, you use this route to create a connection to the target broker.

The *<ordinal>* is the ordinal assigned to the broker pod by the StatefulSet. The first broker pod in a cluster is assigned an ordinal of 0. The second pod is assigned an ordinal of 1 and so on. The ordinal value of a pod is stored in a **STATEFUL_SET_ORDINAL** variable. You can use this variable instead of the ordinal value in the mirror connection details on the source broker. For example:

<broker name>-<acceptor name>-\${STATEFUL_SET_ORDINAL}-svc-rte.<namespace>.<hostname>

By using the **STATEFUL_SET_ORDINAL** variable, you ensure that a source creates a mirror connection to a target that has the same ordinal if you scale up the number of brokers on the source and target clusters.

3. If you want to send data securely over the mirror connection, configure Transport Layer Security (TLS) for the connection. Depending on your requirements, you can use various methods to generate SSL/TLS certificates. For example, you can use a trusted Certificate Authority (CA), cert-manager Operator for OpenShift or Secure Sockets Layer (SSL) tools.

The following example provides a summary of the steps to configure mutual TLS authentication (mTLS) between the brokers by using SSL/TLS tools to manually generate self-signed certificates.

- a. Generate a self-signed SSL/TLS certificate for the target broker. For example:
keytool -genkey -trustcacerts -alias broker -keyalg RSA -keystore broker.keystore -keypass password -storepass password
- b. Export the public key of the SSL/TLS certificate you created to a file, so you can import the key into a truststore file for use on the source broker. For example:
keytool -export -noprompt -alias broker -keystore broker.keystore -file for_source_truststore -storepass password
- c. Import the public key of the SSL/TLS certificate that you exported into a truststore file for use on the source broker. For example:
keytool -import -noprompt -trustcacerts -alias broker -keystore client.truststore -file for_source_truststore -storepass password
- d. Generate a self-signed SSL/TLS certificate for the source broker. For example:
keytool -genkey -trustcacerts -alias broker -keyalg RSA -keystore broker.keystore -keypass password -storepass password
- e. Export the public key of the SSL/TLS certificate you created, so you can import the key into a truststore file for use on the target broker.
keytool -export -noprompt -alias broker -keystore broker.keystore -file for_target_truststore -storepass password

- f. Import the public key of the SSL/TLS certificate that you exported into a truststore file for use on the target broker. For example:

```
keytool -import -noprompt -trustcacerts -alias broker -keystore client.ts -file  
for_target_truststore -storepass password
```

- g. Add the keystore and truststore files you created for the source broker to a secret in the namespace of the source broker. For example:

```
oc create secret generic mirror --from-file=broker.ks=broker.ks --from-  
file=client.ts=client.ts --from-literal=keyStorePassword=password --from-  
literal=trustStorePassword=password
```

Repeat this step to add the keystore and truststore files you created for the target broker to a secret in the namespace of the target broker.

- h. In the acceptor configured for the target broker, set the **sslEnabled** attribute to **true** and specify the name of the secret that you created in the namespace of the target broker. For example:

```
metadata:
  name: mirror-broker
  namespace: dr
spec:
  ...
  acceptors:
  - expose: true
    name: amqp
    port: 5672
    protocols: amqp
    sslEnabled: true
    sslSecret: mirror
  ...
```

- i. In the CR of the source broker, add a reference to the secret you created in the namespace of the source broker to the **extraMounts** attribute. This step is required so the Operator mounts the keystore and truststore files in the secret on each broker pod. For example:

```
spec:
  ...
  deploymentPlan:
    extraMounts:
      secrets:
      - mirror
  ...
```

The keystore and truststore files in the secret are mounted in a **/amq/extra/secrets/<secret name>** directory on a broker pod.

4. In the CR of the source broker, under the **brokerProperties** attribute, configure the mirror connection details. For the connection URI, specify the route exposed for the acceptor that you created on the target broker. If you want to use SSL/TLS to secure the mirror connection, also include the following in the URI:

- a port number of 443
- **sslEnabled=true** to enable SSL/TLS

- the paths and credentials for the keystore and truststore files

For example:

```
spec:
  ...
  brokerProperties:
    - AMQPConnections.datacenter1.uri=tcp://broker-dr-amqp-${STATEFUL_SET_ORDINAL}-
      svc-rte-
      dr.apps.lab.redhat.com:443?;sslEnabled=true;trustStorePath=/amq/extra/secrets/mirror/client.t
      s;trustStorePassword=password;keyStorePath=/amq/extra/secrets/mirror/broker.ks;keyStorePa
      ssword=password
    - AMQPConnections.datacenter1.connectionElements.mirror.type=MIRROR
  ...
```



NOTE

If required, you can configure multiple mirror targets for the source broker. For example:

```
spec:
  ...
  brokerProperties:
    - AMQPConnections.datacenter1.uri=tcp://primary-mirror-broker-amqp-
      ${STATEFUL_SET_ORDINAL}-svc.dr.svc.cluster.local:61616
    - AMQPConnections.datacenter1.connectionElements.mirror.type=MIRROR
    - AMQPConnections.datacenter2.uri=tcp://backup-mirror-broker-amqp-
      ${STATEFUL_SET_ORDINAL}-svc.dr.svc.cluster.local:61616
    - AMQPConnections.datacenter2.connectionElements.mirror.type=MIRROR
  ...
```

5. In the CR of the source broker, configure additional mirror configuration properties as required. For example:

```
- AMQPConnections.datacenter1.user=admin
- AMQPConnections.datacenter1.password=admin
- AMQPConnections.datacenter1.retryInterval=5000
-
AMQPConnections.datacenter1.connectionElements.mirror.messageAcknowledgements=true
- AMQPConnections.datacenter1.connectionElements.mirror.queueCreation=true
- AMQPConnections.datacenter1.connectionElements.mirror.queueRemoval=true
- AMQPConnections.datacenter1.connectionElements.mirror.addressFilter=addresses
```



NOTE

You can use any alphanumeric string to name the AMQP connection. In the previous example, the AMQP connection name is **datacenter1**.

AMQPConnections.<name>.user

The name of the user on the target broker with the permissions to mirror the required events.

AMQPConnections.<name>.password

The password of the user on the target broker.

AMQPConnections.<name>.retryInterval

The interval, in milliseconds, between retry attempts to connect to the target broker.

AMQPConnections.<name>.connectionElements.mirror.messageAcknowledgements

Specifies whether message acknowledgments are mirrored. The default value is **true**.

AMQPConnections.<name>.connectionElements.mirror.queueCreation

Specifies whether queue or address creation events are mirrored. The default value is **true**.

AMQPConnections.<name>.connectionElements.mirror.queueRemoval

Specifies whether queue or address removal events are mirrored. The default value is **true**.

AMQPConnections.<name>.connectionElements.mirror.addressFilter

A filter that the source broker can use to include or exclude addresses for which events are mirrored. For example, you might want to exclude temporary queues from being mirrored. You specify the filter as a comma-separated list of addresses. If you want to specify a list of addresses to exclude, prefix each address with an exclamation mark (!). In the following example, events for addresses that start with **us.** and **europe.** are not mirrored.

AMQPConnections.<name>.connectionElements.mirror.addressFilter=!us.,!europe.

**NOTE**

If you specify one or more addresses to include, events for all other addresses are not mirrored. If you specify one or more addresses to exclude, events for all other addresses are mirrored.

6. In the **status** section of the CR for the source broker, verify that the status of the **BrokerPropertiesApplied** condition is **true** to confirm that all the properties you specified in the CR were applied. For more information, see [Section 3.6, “Viewing status information for your broker deployment”](#).
7. Check the logs of the source broker pod for a line similar to the following to confirm that the mirror connection was established.
broker-prod-ss-0 broker-prod-container Connected on Server AMQP Connection dr on broker-dr-amqp-0-svc-rte-dr.lab.redhat.com:443 after 0 retries

CHAPTER 5. CONNECTING TO AMQ MANAGEMENT CONSOLE FOR AN OPERATOR-BASED BROKER DEPLOYMENT

Each broker pod in a deployment hosts its own instance of AMQ Management Console at port 8161. You can expose the console in the custom resource (CR) instance for your broker deployment. After it is exposed, you can use the console to view and monitor brokers in a web browser.

5.1. CONNECTING TO AMQ MANAGEMENT CONSOLE

When you expose AMQ Management Console in the custom resource (CR) instance for your broker deployment, the Operator creates a dedicated service and route for each broker pod to provide access to AMQ Management Console.

The default name of the service created when you expose AMQ Management Console is in the form **<custom-resource-name>wconsj-<broker-pod-ordinal>svc**. For example, **my-broker-deployment-wconsj-0-svc**. The default name of the route created when you expose AMQ Management Console is in the form **<custom-resource-name>wconsj-<broker-pod-ordinal>svc-rte**. For example, **my-broker-deployment-wconsj-0-svc-rte**.

Prerequisites

- [Section 3.3.1, “Deploying a single broker instance”](#).
- [Section 4.8, “Enabling access to AMQ Management Console”](#).

Procedure

1. In the OpenShift Container Platform web console, click **Networking > Routes**.
On the **Routes** page, identify the **wconsj** Route for the given broker Pod. For example, **my-broker-deployment-wconsj-0-svc-rte**.
2. Under **Location**, click the link that corresponds to the Route.
A new tab opens in your web browser.
3. Click the **Management Console** link.
The AMQ Management Console login page opens.



NOTE

Credentials are required to log in to AMQ Management Console **only** if the **requireLogin** property of the CR is set to **true**. This property specifies whether login credentials are required to log in to the broker **and** AMQ Management Console. By default, the **requireLogin** property is set to **false**. If **requireLogin** is set to **false**, you can log in to AMQ Management Console without supplying a valid username and password by entering any text when prompted for a username and password.

4. If the **requireLogin** property is set to **true**, enter a username and password.
You can enter the credentials for a preconfigured user that is available for connecting to the broker and AMQ Management Console. You can find these credentials in the **adminUser** and **adminPassword** properties if these properties are configured in the Custom Resource (CR)

instance. If these properties are not configured in the CR, the Operator automatically generates the credentials. To obtain the automatically generated credentials, see [Section 5.2, "Accessing AMQ Management Console login credentials"](#).

If you want to log in as any other user, note that a user must belong to a security role specified for the **hawtio.roles** system property to have the permissions required to log in to AMQ Management Console. The default role for the **hawtio.roles** system property is **admin**, which the preconfigured user belongs to.

Additional resources

- [Managing brokers using AMQ Management Console](#)

5.2. ACCESSING AMQ MANAGEMENT CONSOLE LOGIN CREDENTIALS

If the **requireLogin** attribute is set to **true** in the broker custom resource (CR), you must specify a username and password to log in to AMQ Management Console. You can configure the **adminUser** and **adminPassword** attributes in the CR and use the configured values to log in to AMQ Management Console.

If you do not configure the **adminUser** and **adminPassword** in the CR, the Operator automatically generates values for these credentials and stores them in a secret. You can retrieve these values from the secret and use them to log in to AMQ Management Console.

The default name of the generated secret is in the form **<custom-resource-name>credentials-secret**, for example, **my-broker-deployment-credentials-secret**.



NOTE

If **requireLogin** is set to **false**, you can log in to the console without supplying a valid username password by entering any text when prompted for username and password.

Procedure

1. See the complete list of secrets in your OpenShift project.
 - a. From the OpenShift Container Platform web console, click **Workload > Secrets**
 - b. From the command line:
2. Open the appropriate secret to reveal the Base64-encoded console login credentials.
 - a. From the OpenShift Container Platform web console, click the secret that includes your broker Custom Resource instance in its name. Click the **YAML** tab.
 - b. From the command line:

```
$ oc get secrets
```

```
$ oc edit secret <my-broker-deployment-credentials-secret>
```

3. To decode a value in the secret, use a command such as the following:

```
■
```

```
$ echo 'dXNlcl9uYW11' | base64 --decode  
console_admin
```

CHAPTER 6. UPGRADING AN OPERATOR-BASED BROKER DEPLOYMENT

Upgrading an Operator-based broker deployment involves upgrading both the Operator and the broker container images.

6.1. BEFORE YOU BEGIN

Review planning considerations and best practices before upgrading your Operator-based broker deployment.

The following are some important considerations before you upgrade the Operator and broker container images for an Operator-based broker deployment.

- Upgrade the Operator and the broker images in two separate steps, beginning with the Operator upgrade, to ensure that the upgrade runs smoothly.
To separate the upgrade of the broker images from the Operator upgrade, you must prevent the upgraded Operator from automatically upgrading the broker container images to the latest version supported by the new Operator. You can prevent this automatic upgrade by setting the **version** attribute in the CR. For example, you can set the value of the **version** attribute to the version of the broker images currently deployed, which is displayed in the **status** section of the CR.
- If you originally used the CLI to *install* the Operator, you should also use the CLI to *upgrade* the Operator. If you originally used the OpenShift web console to install the Operator (that is, it appears under **Ecosystem > Installed Operators** for your project in the OpenShift web console), you should also use the console to upgrade the Operator.
- Upgrading the Operator using either the OpenShift command-line interface (CLI) or the OpenShift web console requires cluster administrator privileges for your OpenShift cluster.
- If the **redeliveryDelayMultiplier** and the **redeliveryCollisionAvoidanceFactor** attributes are configured in the main broker CR in a 7.8.x or 7.9.x deployment, the new Operator is unable to reconcile any CR after you upgrade to 7.10.x or later. The reconcile fails because the data type of both attributes changed from float to string in 7.10.x.
You can work around this issue by deleting the **redeliveryDelayMultiplier** and the **redeliveryCollisionAvoidanceFactor** attributes from the **spec.deploymentPlan.addressSettings.addressSetting** attribute. Then, configure the attributes under the **brokerProperties** attribute. For example:

```
spec:
  ...
  brokerProperties:
    - "addressSettings.#.redeliveryMultiplier=2.1"
    - "addressSettings.#.redeliveryCollisionAvoidanceFactor=1.2"
```



NOTE

Under the **brokerProperties** attribute, use the **redeliveryMultiplier** attribute name instead of the **redeliveryDelayMultiplier** attribute name that you deleted.

Additional resources

- [Section 6.4, “Manually upgrading the Operator by using the OpenShift command-line interface \(CLI\)”](#)
- [Section 6.3, “Manually upgrading the Operator by using the OpenShift web console”](#)
- [Section 6.6.1, “Restricting automatic upgrades of broker images by using version numbers”](#)

6.2. AUTOMATIC OPERATOR UPGRADES

If you select automatic update approval when you install an Operator by using the OpenShift web console, the Operator is automatically updated to new micro versions that become available. For example, if the installed Operator version is 7.12.3, the Operator is automatically upgraded to version 7.12.4.

Automatic upgrades between **minor** Operator versions are not supported. For example, if the current Operator is version 7.12.4, an automatic upgrade to version 7.13.x is not possible. To upgrade between minor versions of the Operator, you must manually uninstall the current Operator and install the new Operator from the channel where Operators for that minor version are made available. You can manually upgrade between minor versions of the Operator by using the OpenShift web console or the OpenShift command-line interface (CLI).



NOTE

Uninstalling the Operator has no effect on the brokers that are managed by the Operator provided that you do not select the **Delete all operand instances for this operator** checkbox during the uninstall procedure.

Additional resources

- [Section 6.3.2, “Upgrading the Operator”](#)

6.3. MANUALLY UPGRADING THE OPERATOR BY USING THE OPENSIFT WEB CONSOLE

Use the OpenShift web console to upgrade the Operator only if you originally used the console to *install* the Operator (that is, the Operator is displayed under **Ecosystem > Installed Operators** for your project in the OpenShift web console).

If you originally used the OpenShift command-line interface (CLI) to install the Operator, use the CLI to upgrade the Operator.



NOTE

If you are upgrading from 7.10.0 or 7.10.1, see the specific sections that describe how to complete the upgrade of these versions of the Operator.

6.3.1. Before you begin

Review planning considerations and best practices before you upgrade your Operator-based broker deployment by using the OpenShift web console.

Before you use the OpenShift web console to upgrade an instance of the AMQ Broker Operator, consider the following:

- The Operator Lifecycle Manager **automatically** updates the CRDs in your OpenShift cluster when you install the latest Operator version by using the OpenShift web console. You must not remove existing CRDs. If you remove existing CRDs, all CRs and broker instances are also removed.
- When you update your cluster with the CRDs for the latest Operator version, this update affects **all** projects in the cluster. Any broker pods deployed from previous versions of the Operator might become unable to update their status in the OpenShift Container Platform web console. When you click the **Logs** tab of a running broker pod, you see messages indicating that 'UpdatepodStatus' has failed. However, the broker pods and Operator in that project continue to work as expected. To fix this issue for an affected project, you must also upgrade that project to use the latest version of the Operator.
- If you are upgrading from 7.10.0 or 7.10.1, see the specific sections that describe how to complete the upgrade of these versions of the Operator. Upgrading these versions requires additional steps to prevent the Operator upgrade from restarting the broker pods in the deployment.

Additional resources

- [Section 6.3.3, "Upgrading the Operator from 7.10.0"](#)
- [Section 6.3.4, "Upgrading the Operator from 7.10.1"](#)

6.3.2. Upgrading the Operator

To complete a manual upgrade of the Operator, you must uninstall the Operator that is currently installed and install the new Operator.

Procedure

1. Log in to the OpenShift Container Platform web console as a cluster administrator.
2. Uninstall the existing AMQ Broker Operator from your project.
3. In the left navigation menu, click **Ecosystem > Installed Operators**
4. From the Project drop-down menu at the top of the page, select the project in which you want to uninstall the Operator.
5. Locate the **Red Hat Integration - AMQ Broker** instance that you want to uninstall.
6. For your Operator instance, click the **More Options** icon (three vertical dots) on the right-hand side. Select **Uninstall Operator**.



WARNING

Ensure that the **Delete all operand instances for this operator** checkbox is **NOT** selected. If this checkbox is selected, the broker instances that are managed by the Operator are deleted when the Operator is uninstalled.

7. On the confirmation dialog box, click **Uninstall**.
8. Use the OpenShift web console install the latest version of the Operator for AMQ Broker 7.14. For more information, see [Section 3.2.1, “Deploying the Operator by using the OpenShift web console”](#).
9. In the left navigation menu, click **Ecosystem > Installed Operators**
10. Verify that the correct version number is displayed under the **Red Hat Integration - AMQ Broker** instance that is installed.

6.3.3. Upgrading the Operator from 7.10.0

You must uninstall the 7.10.0 Operator and install the new Operator to complete the upgrade. During this procedure you must complete additional steps to prevent the new Operator from restarting the broker pods in the deployment and causing an outage.

Procedure

1. Log in to the OpenShift Container Platform web console as a cluster administrator.
2. Uninstall the existing AMQ Broker Operator from your project.
 - a. In the left navigation menu, click **Ecosystem > Installed Operators**
 - b. From the Project drop-down menu at the top of the page, select the project in which you want to uninstall the Operator.
 - c. Locate the **Red Hat Integration - AMQ Broker** instance that you want to uninstall.
 - d. For your Operator instance, click the **More Options** icon (three vertical dots) on the right-hand side. Select **Uninstall Operator**.
 - e. On the confirmation dialog box, click **Uninstall**.
3. When you upgrade a 7.10.0 Operator, the new Operator deletes the StatefulSet to remove custom and Operator metering labels, which were incorrectly added to the StatefulSet selector by the Operator in 7.10.0. When the Operator deletes the StatefulSet, it also deletes the existing broker pods, which causes a temporary broker outage. If you want to avoid the outage, complete the following steps to delete the StatefulSet and orphan the broker pods so that they continue to run.
 - i. Log in to OpenShift Container Platform CLI as an administrator for the project that contains your existing Operator deployment:

```
$ oc login -u <user>
```

- ii. Switch to the OpenShift project in which you want to upgrade your Operator version.

```
$ oc project <project-name>
```

- iii. Delete the StatefulSet with the **--cascade=orphan** option to orphan the broker pods. The orphaned broker pods continue to run after the StatefulSet is deleted.

```
$ oc delete statefulset <statefulset-name> --cascade=orphan
```

4. Check if your main broker CR has labels called **application** or **ActiveMQArtemis** configured in the **deploymentPlan.labels** attribute.

In 7.10.0, it was possible to configure these custom labels in the CR. These labels are reserved for the Operator to assign labels to pods and cannot be added as custom labels after 7.10.0. If these custom labels were configured in the main broker CR in 7.10.0, the Operator-assigned labels on the pods were overwritten by the custom labels. If the CR has either of these labels, complete the following steps to restore the correct labels on the pods and remove the labels from the CR.

- a. In the OpenShift command-line interface (CLI), run the following command to restore the correct pod labels. In the following example, 'ex-aa0' is the name of the StatefulSet deployed.

```
$ for pod in $(oc get pods | grep -o '^ex-aa0[^\ ]*'); do oc label --overwrite pods $pod
ActiveMQArtemis=ex-aa0 application=ex-aa0-app; done
```

- b. Delete the **application** and **ActiveMQArtemis** labels from the **deploymentPlan.labels** attribute in the CR.
 - i. Using the OpenShift command-line interface:
 - A. Log in to OpenShift as a user that has privileges to deploy CRs in the project for the broker deployment.

```
oc login -u <user> -p <password> --server=<host:port>
```

- B. Edit the CR for your deployment.

```
oc edit ActiveMQArtemis <statefulset name> -n <namespace>
```

- C. In the **deploymentPlan.labels** element in the CR, delete any custom labels called **application** or **ActiveMQArtemis**.
 - D. Save the CR.

- ii. Using the OpenShift Container Platform web console:

- A. Log in to the console as a user that has privileges to deploy CRs in the project for the broker deployment.
 - B. In the left pane, click **Administration > Custom Resource Definitions**
 - C. Click the **ActiveMQArtemis** CRD.
 - D. Click the **Instances** tab.
 - E. Click the instance for your broker deployment.
 - F. Click the **YAML** tab.
Within the console, a YAML editor opens, enabling you to configure a CR instance.
 - G. In the **deploymentPlan.labels** element in the CR, delete any custom labels called **application** or **ActiveMQArtemis**.
 - H. Click **Save**.

5. Use the OpenShift web console to install the latest version of the Operator for AMQ Broker 7.14. For more information, see [Section 3.2.1, “Deploying the Operator by using the OpenShift web console”](#).

The new Operator can recognize and manage your previous broker deployments. If you set values in the **image** or **version** field in the CR, the Operator’s reconciliation process upgrades the broker pods to the corresponding images when the Operator starts. For more information, see [Section 6.6, “Restricting automatic upgrades of broker container images”](#). Otherwise, the Operator upgrades each broker pod to the latest container image.



NOTE

If the reconciliation process does not start, you can start the process by scaling the deployment. For more information, see [Section 3.3.1, “Deploying a single broker instance”](#).

6. Add attributes to the CR for the new features that are available in the upgraded broker, as required.

6.3.4. Upgrading the Operator from 7.10.1

You must uninstall the 7.10.1 Operator and install the new Operator to complete the upgrade. Depending on your configuration, you might need to complete additional steps during the upgrade to prevent the new Operator from restarting the broker pods and causing an outage.

Procedure

1. Log in to the OpenShift Container Platform web console as a cluster administrator.
2. Check if your main broker CR has labels called **application** or **ActiveMQArtemis** configured in the **deploymentPlan.labels** attribute.
These labels are reserved for the Operator to assign labels to pods and cannot be used after 7.10.1. If these custom labels were configured in the main broker CR, the Operator-assigned labels on the pods were overwritten by the custom labels.
3. If these custom labels are not configured in the main broker CR, install the latest version of the Operator for AMQ Broker 7.14 by using the OpenShift web console. For more information, see [Section 3.2.1, “Deploying the Operator by using the OpenShift web console”](#).
4. If either of these custom labels are configured in the main broker CR, complete the following steps to uninstall the existing Operator, restore the correct pod labels and remove the labels from the CR, before you install the new Operator.



NOTE

By uninstalling the Operator, you can remove the custom labels without the Operator deleting the StatefulSet, which also deletes the existing broker pods and causes a temporary broker outage.

- a. Uninstall the existing AMQ Broker Operator from your project.
 - i. In the left navigation menu, click **Ecosystem > Installed Operators**
 - ii. From the Project drop-down menu at the top of the page, select the project from which you want to uninstall the Operator.

- iii. Locate the **Red Hat Integration - AMQ Broker** instance that you want to uninstall.
 - iv. For your Operator instance, click the **More Options** icon (three vertical dots) on the right-hand side. Select **Uninstall Operator**.
 - v. On the confirmation dialog box, click **Uninstall**.
- b. In the OpenShift command-line interface (CLI), run the following command to restore the correct pod labels. In the following example, 'ex-aa0' is the name of the StatefulSet deployed.

```
$ for pod in $(oc get pods | grep -o '^ex-aa0[^\ ]*'); do oc label --overwrite pods $pod
ActiveMQArtemis=ex-aa0 application=ex-aa0-app; done
```

- c. Delete the **application** and **ActiveMQArtemis** labels from the **deploymentPlan.labels** attribute in the CR.
- i. Using the OpenShift command-line interface:
 - A. Log in to OpenShift as a user that has privileges to deploy CRs in the project for the broker deployment.

```
oc login -u <user> -p <password> --server=<host:port>
```

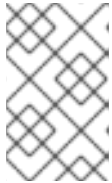
 - B. Edit the CR for your deployment.

```
oc edit ActiveMQArtemis <statefulset name> -n <namespace>
```

 - C. In the **deploymentPlan.labels** attribute in the CR, delete any custom labels called **application** or **ActiveMQArtemis**.
 - D. Save the CR file.
- ii. Using the OpenShift Container Platform web console:
 - A. Log in to the console as a user that has privileges to deploy CRs in the project for the broker deployment.
 - B. In the left pane, click **Administration > Custom Resource Definitions**
 - C. Click the **ActiveMQArtemis** CRD.
 - D. Click the **Instances** tab.
 - E. Click the instance for your broker deployment.
 - F. Click the **YAML** tab.
Within the console, a YAML editor opens, enabling you to configure a CR instance.
 - G. In the **deploymentPlan.labels** attribute in the CR, delete any custom labels called **application** or **ActiveMQArtemis**.
 - H. Click **Save**.

5. Use the OpenShift web console to install the latest version of the Operator for AMQ Broker 7.14. For more information, see [Section 3.2.1, “Deploying the Operator by using the OpenShift web console”](#).

The new Operator can recognize and manage your previous broker deployments. If you set values in the **image** or **version** field in the CR, the Operator’s reconciliation process upgrades the broker pods to the corresponding images when the Operator starts. For more information, see [Section 6.6, “Restricting automatic upgrades of broker container images”](#). Otherwise, the Operator upgrades each broker pod to the latest container image.



NOTE

If the reconciliation process does not start, you can start the process by scaling the deployment. For more information, see [Section 3.3.1, “Deploying a single broker instance”](#).

6. Add attributes to the CR for the new features that are available in the upgraded broker, as required.

6.4. MANUALLY UPGRADING THE OPERATOR BY USING THE OPENSIFT COMMAND-LINE INTERFACE (CLI)

You can use the CLI to upgrade the Operator if you originally used the CLI to *install* the Operator.

Procedure

1. In your web browser, navigate to the **Software Downloads** page for [AMQ Broker 7.14.0](#).
2. Ensure that the value of the **Version** drop-down list is set to **7.14.0** and the **Releases** tab is selected.
3. Next to **AMQ Broker 7.14.0 Operator Installation and Example Files** click **Download**. Download of the **amq-broker-operator-7.14.0-ocp-install-examples-rhel9.zip** compressed archive automatically begins.
4. When the download has completed, move the archive to your chosen installation directory. The following example moves the archive to a directory called **~/broker/operator**.

```
$ mkdir ~/broker/operator
$ mv amq-broker-operator-7.14.0-ocp-install-examples-rhel9.zip ~/broker/operator
```

5. In your chosen installation directory, extract the contents of the archive. For example:

```
$ cd ~/broker/operator
$ unzip amq-broker-operator-operator-7.14.0-ocp-install-examples-rhel9.zip
```

6. Log in to OpenShift Container Platform as an administrator for the project that contains your existing Operator deployment.

```
$ oc login -u <user>
```

7. Switch to the OpenShift project in which you want to upgrade your Operator version.

```
$ oc project <project-name>
```

8. In the **deploy** directory of the latest Operator archive that you downloaded and extracted, open the **operator.yaml** file.



NOTE

In the **operator.yaml** file, the Operator uses an image that is represented by a *Secure Hash Algorithm* (SHA) value. The comment line, which begins with a number sign (**#**) symbol, denotes that the SHA value corresponds to a specific container image tag.

9. Open the **operator.yaml** file for your **previous** Operator deployment. Check that any non-default values that you specified in your previous configuration are replicated in the **new operator.yaml** configuration file.
10. In the **new operator.yaml** file, the Operator is named **amq-broker-controller-manager** by default. If the name of the Operator in your previous deployment is not **amq-broker-controller-manager**, replace all instances of **amq-broker-controller-manager** with the previous Operator name. For example:

```
spec:
  ...
  selector
    matchLabels
      name: amq-broker-operator
  ...
```

11. In the **new operator.yaml** file, the service account for the Operator is named **amq-broker-controller-manager**. In previous versions, the service account for the Operator was named **amq-broker-operator**.
 - a. If you want to use the service account name in your previous deployment, replace the name of the service account in the **new operator.yaml** file with the name used in the previous deployment. For example:

```
spec:
  ...
  serviceAccountName: amq-broker-operator
  ...
```

- b. If you want to use the new service account name, **amq-broker-controller-manager** for the Operator, update the service account, role, and role binding in your project.

```
$ oc apply -f deploy/service_account.yaml
```

```
$ oc apply -f deploy/role.yaml
```

```
$ oc apply -f deploy/role_binding.yaml
```

12. Update the CRDs that are included with the Operator.

- a. Update the main broker CRD.

```
$ oc apply -f deploy/crds/broker_activemqartemis_crd.yaml
```


- b. Update the address CRD.

```
$ oc apply -f deploy/crds/broker_activemqartemisaddress_crd.yaml
```

- c. Update the scaledown controller CRD.

```
$ oc apply -f deploy/crds/broker_activemqartemisscaledown_crd.yaml
```

- d. Update the security CRD.

```
$ oc apply -f deploy/crds/broker_activemqartemissecurity_crd.yaml
```

13. If you are upgrading from AMQ Broker Operator 7.10.0 only, delete the Operator and the StatefulSet.

By default, the new Operator deletes the StatefulSet to remove custom and Operator metering labels, which were incorrectly added to the StatefulSet selector by the Operator in 7.10.0. When the Operator deletes the StatefulSet, it also deletes the existing broker pods, which causes a temporary broker outage. If you want to avoid an outage, complete the following steps to delete the Operator and the StatefulSet without deleting the broker pods.

- a. Delete the Operator.

```
$ oc delete -f deploy/operator.yaml
```



NOTE

Deleting the Operator does not remove the broker instances that are managed by the Operator.

- b. Delete the StatefulSet with the **--cascade=orphan** option to orphan the broker pods. The orphaned broker pods continue to run after the StatefulSet is deleted.

```
$ oc delete statefulset <statefulset-name> --cascade=orphan
```

14. If you are upgrading from AMQ Broker Operator 7.10.0 or 7.10.1, check if your main broker CR has labels called **application** or **ActiveMQArtemis** configured in the **deploymentPlan.labels** attribute.

These labels are reserved for the Operator to assign labels to pods and are not permitted as custom labels after 7.10.1. If these custom labels were configured in the main broker CR, the Operator-assigned labels on the pods were overwritten by the custom labels. If either of these custom labels are configured in the main broker CR, complete the following steps to restore the correct labels on the pods and remove the labels from the CR.

- a. If you are upgrading from 7.10.0, you deleted the Operator in the previous step. If you are upgrading from 7.10.1, delete the Operator.

```
$ oc delete -f deploy/operator.yaml
```

- b. Run the following command to restore the correct pod labels. In the following example, 'ex-aao' is the name of the StatefulSet deployed.

```
$ for pod in $(oc get pods | grep -o '^ex-aa[^ ]*'); do oc label --overwrite pods $pod
ActiveMQArtemis=ex-aa application=ex-aa-app; done
```

- c. Delete the **application** and **ActiveMQArtemis** labels from the **deploymentPlan.labels** attribute in the CR.

- i. Log in to OpenShift as a user that has privileges to deploy CRs in the project for the broker deployment.

```
oc login -u <user> -p <password> --server=<host:port>
```

- ii. Open the sample CR file called **broker_activemqartemis_cr.yaml** that was included in the **deploy/crs** directory of the Operator installation archive that you downloaded and extracted.
 - iii. In the **deploymentPlan.labels** attribute in the CR, delete any custom labels called **application** or **ActiveMQArtemis**.
 - iv. Save the CR file.
 - v. Deploy the CR instance.

- A. Switch to the project for the broker deployment.

```
$ oc project <project_name>
```

- B. Apply the CR.

```
$ oc apply -f <path/to/broker_custom_resource_instance>.yaml
```

- d. If you deleted the previous Operator, deploy the new Operator.

```
$ oc create -f deploy/operator.yaml
```

15. Apply the updated Operator configuration.

```
$ oc apply -f deploy/operator.yaml
```

16. Add attributes to the CR for the new features that are available in the upgraded broker, as required.

6.5. UPGRADING BROKER CONTAINER IMAGES

After the Operator is upgraded, it can manage multiple versions of AMQ Broker. You can find the precise list of versions supported by the new Operator in the Operator log.

The following is an example of the log entry showing the list of broker versions supported by the Operator.

```
INFO setup Version of the operator: 7.14.0.OPR.1 2024-08-14T15:04:17 INFO setup Supported
ActiveMQArtemis Kubernetes Image Versions: 7.12.0 7.12.1 7.12.2 7.12.3 7.12.4 7.12.5 7.12.6
7.13.0 7.13.1 7.13.2 7.13.3 7.13.4 7.14.0 ....
```

If, as recommended, you configured the **version** attribute in the custom resource to prevent the new Operator from automatically upgrading the broker images at the same time as the Operator upgrade, you can change the version number to upgrade the broker images.



NOTE

If the **version** attribute is not configured in the custom resource, the Operator automatically upgrades each broker it manages to the latest available version of the broker container image that the Operator supports. For a 7.14.0 Operator, the latest version of the broker container images supported is 7.14.0.

Additional resources

- [Section 6.6, “Restricting automatic upgrades of broker container images”](#)

6.6. RESTRICTING AUTOMATIC UPGRADES OF BROKER CONTAINER IMAGES

By default, the Operator automatically upgrades each broker it manages to the latest available images that the Operator supports. You can restrict the ability of the Operator to upgrade the images by configuring a version number or the URLs of specific container images in the broker custom resource.

6.6.1. Restricting automatic upgrades of broker images by using version numbers

You can prevent the Operator from automatically upgrading to the latest broker container images by specifying a version number in the custom resource (CR).



NOTE

When you restrict upgrades based on version numbers, the Operator continues to automatically upgrade the brokers to use any new images that contain security fixes for the version deployed.

Procedure

1. Edit the main broker CR instance for the broker deployment.
 - a. Using the OpenShift command-line interface:
 - i. Log in to OpenShift as a user that has privileges to edit and deploy CRs in the project for the broker deployment.

```
$ oc login -u <user> -p <password> --server=<host:port>
```

- ii. Edit the CR.

```
$ oc edit ActiveMQArtemis <CR instance name> -n <namespace>
```

- b. Using the OpenShift Container Platform web console:
 - i. Log in to the console as a user that has privileges to deploy CRs in the project for the broker deployment.

- ii. In the left pane, click **Ecosystem > Installed Operators**
- iii. Click the **Red Hat Integration - AMQ Broker for RHEL 9 (Multiarch)** operator.
- iv. Click the **AMQ Broker** tab.
- v. Click the name of the ActiveMQArtemis instance name.
- vi. Click the **YAML** tab.
Within the console, a YAML editor opens, enabling you to edit the CR instance.



NOTE

In the **status** section of the CR, the **.status.version.brokerVersion** field shows the version of AMQ Broker that is currently deployed.

2. In the **spec.version** attribute, specify the version to which the Operator can upgrade the broker and init container images in your deployment. The following are examples of values that you can specify.

Examples

In the following example, the Operator upgrades the current container images in your deployment to 7.13.0.

```
spec:
  version: '7.13.0'
  ...
```

In the following example, the Operator upgrades the current container images in your deployment to the latest available 7.12.x images. For example, if your deployment is using 7.12.1 container images, the Operator automatically upgrades the images to 7.12.4 but not to 7.14.0.

```
spec:
  version: '7.12'
  ...
```

In the following example, the Operator upgrades the current container images in your deployment to the latest 7.x.x images. For example, if your deployment is using 7.12.4 images, the Operator automatically upgrades the images to 7.14.0.

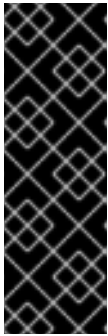
```
spec:
  version: '7'
  ...
```



NOTE

To upgrade between minor versions of the container images, for example, from 7.12.x to 7.13.x, you require an Operator that has the same minor version as that of the new container images. For example, to upgrade from 7.12.4 to 7.14.0, a 7.14.x Operator must be installed.

3. Save the CR.

**IMPORTANT**

If you use the **spec.version** attribute in the CR to restrict automatic upgrades of broker container images, ensure that the CR does not also contain a **spec.deploymentPlan.image** or a **spec.deploymentPlan.initImage** attribute. Both of these attributes override the **spec.version** attribute. If the CR has one of these attributes as well as the **spec.version** attribute, the versions of the broker and init images deployed can diverge, which might prevent the broker from running.

When you save the CR, the Operator first validates that an upgrade to the AMQ Broker version specified for **spec.version** is available for your existing deployment. If you specified an invalid version of AMQ Broker to which to upgrade, for example, a version that is not yet available, the Operator logs a warning message, and takes no further action.

However, if an upgrade to the specified version **is** available, then the Operator upgrades each broker in the deployment to use the broker container images that correspond to the new AMQ Broker version.

The broker container image that the Operator uses is defined in an environment variable in the **operator.yaml** configuration file of the Operator deployment. The environment variable name includes an identifier for the AMQ Broker version. For example, the environment variable **RELATED_IMAGE_ActiveMQ_Artemis_Broker_Kubernetes_7140** corresponds to AMQ Broker 7.14.0.

When the Operator applies the CR change, it restarts each broker pod in your deployment so that each pod uses the specified image version. If you have multiple brokers in your deployment, only one broker pod shuts down and restarts at a time.

Additional resources

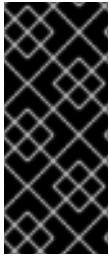
- [Section 2.7, “How the Operator chooses container images when a precise image is not specified in the CR”](#)
- [Section 2.8, “Validation of image and version configuration in a custom resource \(CR\)”](#)

6.6.2. Restricting automatic upgrades of images by using image URLs

You can prevent the Operator from automatically upgrading to the latest broker images by specifying the registry URLs of the images to deploy in the custom resource (CR). After the Operator deploys the specified images, no further upgrades occur until you replace the image URLs in the CR.

**NOTE**

If you restrict updates by specifying broker image URLs in the CR, the Operator does not automatically upgrade the brokers to use newer images that contain security fixes for the images deployed.



IMPORTANT

If you want to restrict automatic upgrades by using image URLs, specify URLs for both the **spec.deploymentPlan.image** and the **spec.deploymentPlan.initImage** attributes in the CR to ensure that the broker and init container images match. If you specify the URL of one container image only, the broker and init container image can diverge, which might prevent the broker from running.



NOTE

If a CR has a **spec.version** attribute in addition to **spec.deploymentPlan.image** and **spec.deploymentPlan.initImage** attributes, the Operator ignores the **spec.version** attribute.

Procedure

1. Obtain the URLs of the broker and init container images to which the Operator can upgrade the current images.
 - a. In the Red Hat Catalog, open the broker container component page: [AMQ Broker for RHEL 9 \(Multiarch\)](#).
 - b. In the **Architecture** drop-down, select your architecture.
 - c. In the **Tag** drop-down, select the tag that corresponds to the image you want to install. Tags are displayed in chronological order based on the release date. A tag consists of the release version and an assigned tag.
 - d. Open the **Get this image** tab.
 - e. In the **Manifest field**, click the **Copy** icon.
 - f. Paste the URL into a text file.
 - g. In the Red Hat Catalog, open the init container component page: [AMQ Broker Init for RHEL 9 \(Multiarch\)](#).
 - h. To obtain the URL of the init container image, repeat the steps that you followed to obtain the URL of the broker container image.
2. Edit the main broker CR instance for the broker deployment.
 - a. Using the OpenShift command-line interface:
 - i. Log in to OpenShift as a user that has privileges to edit and deploy CRs in the project for the broker deployment.


```
$ oc login -u <user> -p <password> --server=<host:port>
```
 - ii. Edit the CR.


```
oc edit ActiveMQArtemis <CR instance name> -n <namespace>
```
 - b. Using the OpenShift Container Platform web console:

- i. Log in to the console as a user that has privileges to deploy CRs in the project for the broker deployment.
- ii. In the left pane, click **Ecosystem > Installed Operators**
- iii. Click the **Red Hat Integration - AMQ Broker for RHEL 9 (Multiarch)** operator.
- iv. Click the **AMQ Broker** tab.
- v. Click the name of the ActiveMQArtemis instance name.
- vi. Click the **YAML** tab.
Within the console, a YAML editor opens, which enables you to configure the CR instance
- c. Copy the URLs of the broker and init container images that you recorded in the text file and insert them in the **spec.deploymentPlan.image** and **spec.deploymentPlan.initImage** fields in the CR. For example:

```
spec:
  ...
  deploymentPlan:
    image: registry.redhat.io/amq7/amq-broker-
rhel9@e44742f66183e225a77f5171da5db95e51f92b6848dea9ed02895a81b1966857
    initImage: registry.redhat.io/amq7/amq-broker-init-
rhel9@d87b1cea2eb804e0f115e51cb15821054d8c2791a8ef09e997840cc31270afb1
  ...
```

3. Save the CR.

When you save the CR, the Operator upgrades the brokers to use the new images and uses these images until you update the values of the **spec.deploymentPlan.image** and **spec.deploymentPlan.initImage** attributes again.

4. If you want to prevent future Operator upgrades from restarting the brokers in your deployment, edit the CR and specify the version number of the brokers that are deployed in the **spec.version** attribute.

If the **spec.version** attribute is not configured in the CR, subsequent upgrades of the Operator cause the broker pods to restart. The pod restart is required because the new Operator adds the latest supported broker version to a label in the StatefulSet unless a version number is explicitly set in the **spec.version** attribute.

You can find the version number value to specify for the **spec.version** attribute in the **status** section of the CR after the brokers start. For more information, see [Viewing status information for your broker deployment](#).



NOTE

If you already deployed AMQ Broker without setting image URLs, you can set the image URLs retrospectively to prevent the Operator from upgrading the current images deployed. You can find the registry URLs for the images deployed in the **.status.version.image** and **.status.version.initImage** attributes, which are in the **status** section of the CR.

If you copy the image URLs from the **.status.version.image** and **.status.version.initImage** attributes and insert them in the **spec.deploymentPlan.image** and the **spec.deploymentPlan.initImage** attributes, the Operator does not upgrade the images currently deployed.

Additional resources

- [Section 2.8, “Validation of image and version configuration in a custom resource \(CR\)”](#)

CHAPTER 7. MONITORING YOUR BROKERS

You can use Fuse Console for OpenShift instead of AMQ Management Console to view runtime data and perform basic management operations. You can also use the Prometheus plugin for AMQ Broker to collect runtime metrics, which you can use monitor the health and performance of brokers.

7.1. VIEWING BROKERS IN FUSE CONSOLE

You can configure an Operator-based broker deployment to use Fuse Console for OpenShift instead of AMQ Management Console. After you configure your broker deployment, Fuse Console discovers the brokers and displays them on a dedicated **Artemis** tab.

You can view the same broker runtime data in Fuse Console for OpenShift as in AMQ Management Console. You can also perform the same basic management operations, such as creating addresses and queues.

The following procedure describes how to configure the Custom Resource (CR) instance for a broker deployment to enable Fuse Console for OpenShift to discover and display brokers in the deployment.

Prerequisites

- Fuse Console for OpenShift must be deployed to an OCP cluster, or to a specific namespace on that cluster. If you have deployed the console to a specific namespace, your broker deployment must be in the **same** namespace, to enable the console to discover the brokers. Otherwise, it is sufficient for Fuse Console and the brokers to be deployed on the same OCP cluster.
- You must have already created a broker deployment. For example, to use a Custom Resource (CR) instance to create a basic Operator-based deployment, see [Section 3.3.1, “Deploying a single broker instance”](#).

Procedure

1. Open the CR instance that you used for your broker deployment. For example, the CR for a basic deployment might resemble the following:

```
apiVersion: broker.amq.io/v1beta1
kind: ActiveMQArtemis
metadata:
  name: ex-aao
spec:
  deploymentPlan:
    size: 4
    image: registry.redhat.io/amq7/amq-broker-rhel9:7.14
    ...
```

2. In the **deploymentPlan** section, add the **jolokiaAgentEnabled** and **managementRBACEnabled** properties and specify values, as shown below.

```
apiVersion: broker.amq.io/v1beta1
kind: ActiveMQArtemis
metadata:
  name: ex-aao
spec:
  deploymentPlan:
```

```
size: 4
image: registry.redhat.io/amq7/amq-broker-rhel9:7.14
...
jolokiaAgentEnabled: true
managementRBACEnabled: false
```

jolokiaAgentEnabled

Specifies whether Fuse Console can discover and display runtime data for the brokers in the deployment. To use Fuse Console, set the value to **true**.

managementRBACEnabled

Specifies whether role-based access control (RBAC) is enabled for the brokers in the deployment. You **must** set the value to **false** to use Fuse Console because Fuse Console uses its own role-based access control.

**IMPORTANT**

If you set the value of **managementRBACEnabled** to **false** to enable use of Fuse Console, management MBeans for the brokers no longer require authorization. You **should not** use AMQ Management Console while **managementRBACEnabled** is set to **false** because this potentially exposes all management operations on the brokers to unauthorized use.

3. Save the CR instance.
4. Switch to the project in which you previously created your broker deployment.

```
$ oc project <project_name>
```

5. At the command line, apply the change.

```
$ oc apply -f <path/to/custom_resource_instance>.yaml
```

6. In Fuse Console, click the **Online** tab to view Fuse applications. To view running brokers, in the left navigation menu, click **Artemis**.

Additional resources

- [Monitoring and managing Red Hat Fuse applications on OpenShift](#)
- [Managing brokers using AMQ Management Console](#)

7.2. MONITORING BROKER RUNTIME METRICS USING PROMETHEUS

You can use the Prometheus metrics plugin for AMQ Broker on OpenShift Container Platform to collect and store broker runtime metrics. You might also use a graphical tool such as Grafana to configure more advanced visualizations and dashboards of the data that the Prometheus plugin collects.



NOTE

The Prometheus metrics plugin enables you to collect and export broker metrics in Prometheus **format**. However, Red Hat **does not** provide support for installation or configuration of Prometheus itself, nor of visualization tools such as Grafana. If you require support with installing, configuring, or running Prometheus or Grafana, visit the product websites for resources such as community support and documentation.

7.2.1. Metrics overview

You can use the Prometheus plugin for AMQ Broker to collect runtime metrics, which you can use monitor the health and performance of brokers. The AMQ Broker Prometheus plugin exports the runtime metrics in Prometheus format, enabling you to use Prometheus itself to visualize and query the data.

You can also use a graphical tool, such as Grafana, to configure more advanced visualizations and dashboards for the metrics that the Prometheus plugin collects.

The metrics that the plugin exports to Prometheus format are described below.

For higher-level broker metrics that are not listed below, you can calculate these by aggregating lower-level metrics. For example, to calculate total message count, you can aggregate the **artemis_message_count** metrics from all queues in your broker deployment.

For a standalone deployment of AMQ Broker on RHEL, metrics for the Java Virtual Machine (JVM) hosting the broker are also exported to Prometheus format. This does not apply to a deployment of AMQ Broker on OpenShift Container Platform.

7.2.1.1. Broker metrics

artemis_address_memory_usage

Number of bytes used by all addresses on this broker for in-memory messages.

artemis_address_memory_usage_percentage

Memory used by all the addresses on this broker as a percentage of the **global-max-size** parameter.

artemis_connection_count

Number of clients connected to this broker.

artemis_session_count

Number of sessions on this broker.

artemis_total_session_count

Total number of sessions created on this broker since it was started.

artemis_total_connection_count

Number of clients that have connected to this broker since it was started.

artemis_disk_store_usage

The fraction of the total disk store used.

artemis_replica_sync

Is the initial replication synchronization process between primary and backup brokers complete.

artemis_active

Is the broker active in a primary-backup group configuration.

artemis_authentication_count

Number of successful and failed authentication attempts.

artemis_authorization_count

Number of successful and failed authorization attempts.

7.2.1.2. Address metrics

artemis_routed_message_count

Number of messages routed to one or more queue bindings.

artemis_unrouted_message_count

Number of messages *not* routed to any queue bindings.

artemis_address_size

The size in bytes used by all queues bound to this address.

artemis_number_of_pages

The number of page files used to store messages for this address.

artemis_limit_percent

The percentage of a configured memory limit that is used by this address. If a memory limit is set for the address by using the **max-size-bytes** attribute, the percentage reported is based on that value. If the **max-size-bytes** attribute is not set, the percentage reported is based on the value of the **global-max-size** attribute. If neither attribute is set, a value of 0 is reported.

7.2.1.3. Queue metrics

artemis_consumer_count

Number of clients consuming messages from a given queue.

artemis_delivering_durable_message_count

Number of durable messages that a given queue is currently delivering to consumers.

artemis_delivering_durable_persistent_size

Persistent size of durable messages that a given queue is currently delivering to consumers.

artemis_delivering_message_count

Number of messages that a given queue is currently delivering to consumers.

artemis_delivering_persistent_size

Persistent size of messages that a given queue is currently delivering to consumers.

artemis_durable_message_count

Number of durable messages currently in a given queue. This includes scheduled, paged, and in-delivery messages.

artemis_durable_persistent_size

Persistent size of durable messages currently in a given queue. This includes scheduled, paged, and in-delivery messages.

artemis_messages_acknowledged

Number of messages acknowledged from a given queue since the queue was created.

artemis_messages_added

Number of messages added to a given queue since the queue was created.

artemis_message_count

Number of messages currently in a given queue. This includes scheduled, paged, and in-delivery messages.

artemis_messages_killed

Number of messages removed from a given queue since the queue was created. The broker kills a message when the message exceeds the configured maximum number of delivery attempts.

artemis_messages_expired

Number of messages expired from a given queue since the queue was created.

artemis_persistent_size

Persistent size of all messages (both durable and non-durable) currently in a given queue. This includes scheduled, paged, and in-delivery messages.

artemis_scheduled_durable_message_count

Number of durable, scheduled messages in a given queue.

artemis_scheduled_durable_persistent_size

Persistent size of durable, scheduled messages in a given queue.

artemis_scheduled_message_count

Number of scheduled messages in a given queue.

artemis_scheduled_persistent_size

Persistent size of scheduled messages in a given queue.

7.2.2. Enabling the Prometheus plugin using a CR

When you install AMQ Broker, a Prometheus metrics plugin is included in your installation. You must enable the plugin to collect runtime metrics for the broker and exports these to Prometheus format.

See [Section 7.2.3, “Enabling the Prometheus plugin for a running broker deployment using an environment variable”](#) for an alternative procedure with running brokers.

Procedure

1. Edit the **ActiveMQArtemis** CR instance that you use for your broker deployment.
2. In the **deploymentPlan** section, add the **enableMetricsPlugin** attribute and set the value to **true**, as shown below.

```
apiVersion: broker.amq.io/v1beta1
kind: ActiveMQArtemis
metadata:
  name: amq-with-metrics
spec:
  deploymentPlan:
    enableMetricsPlugin: true
  ...
```

3. Save the CR instance.
The metrics plugin starts to gather broker runtime metrics in Prometheus format.

**NOTE**

After you enable the Prometheus plugin, you must create a **ServiceMonitor** custom resource (CR) for the Prometheus Operator. Prometheus uses the CR to find the endpoint where the broker metrics are exposed. For example:

```
apiVersion: monitoring.coreos.com/v1
kind: ServiceMonitor
metadata:
  name: amq-with-metrics-monitor
spec:
  selector:
    matchLabels:
      application: amq-with-metrics-app
  endpoints:
    - port: console-jolokia
```

**NOTE**

For the **application** attribute in the **ServiceMonitor** CR, specify the value of the **metadata.name** attribute that is in the **ActiveMQArtemis** CR and append a suffix of **-app**.

Additional resources

- [Section 3.3.3, “Applying Custom Resource \(CR\) changes to running broker deployments”](#)

7.2.3. Enabling the Prometheus plugin for a running broker deployment using an environment variable

You can enable the Prometheus metrics plugin by setting the **AMQ_ENABLE_METRICS_PLUGIN** environment variable to **true** in the custom resource (CR) for your broker.

See [Section 7.2.2, “Enabling the Prometheus plugin using a CR”](#) for an alternative procedure.

Prerequisites

- You can enable the Prometheus plugin for a broker Pod created with the AMQ Broker Operator. However, your deployed broker must use the broker container image for AMQ Broker 7.7 or later.

Procedure

1. Log in to the OpenShift Container Platform web console with administrator privileges for the project that contains your broker deployment.
2. In the web console, click **Home > Projects**. Choose the project that contains your broker deployment.
3. To see the StatefulSets or DeploymentConfigs in your project, click **Workloads > StatefulSets** or **Workloads > DeploymentConfigs**
4. Click the StatefulSet or DeploymentConfig that corresponds to your broker deployment.

5. To access the environment variables for your broker deployment, click the **Environment** tab.
6. Add a new environment variable, **AMQ_ENABLE_METRICS_PLUGIN**. Set the value of the variable to **true**.

When you set the **AMQ_ENABLE_METRICS_PLUGIN** environment variable, OpenShift restarts each broker Pod in the StatefulSet or DeploymentConfig. When there are multiple Pods in the deployment, OpenShift restarts each Pod in turn. When each broker Pod restarts, the Prometheus plugin for that broker starts to gather broker runtime metrics.

7.2.4. Accessing Prometheus metrics for a running broker Pod

You can view Prometheus metrics directly in a web browser by appending **/metrics** to the URL of the route that was created when you exposed AMQ Management Console.

Prerequisites

- You must have already enabled the Prometheus plugin for your broker pod. See [Section 7.2.3, “Enabling the Prometheus plugin for a running broker deployment using an environment variable”](#).

Procedure

1. For the broker pod whose metrics you want to access, you need to identify the route you previously created to connect the Pod to the AMQ Broker management console. The route name forms part of the URL needed to access the metrics.
 - a. Click **Networking > Routes**
 - b. For your chosen broker pod, identify the route created to connect the pod to the AMQ Broker management console. Under **Hostname**, note the complete URL that is shown. For example:

`http://rte-console-access-pod1.openshiftdomain`

2. To access Prometheus metrics, in a web browser, enter the previously noted route name appended with **“/metrics”**. For example:

`http://rte-console-access-pod1.openshiftdomain/metrics`



NOTE

If your console configuration does not use SSL, specify **http** in the URL. In this case, DNS resolution of the host name directs traffic to port 80 of the OpenShift router. If your console configuration uses SSL, specify **https** in the URL. In this case, your browser defaults to port 443 of the OpenShift router. This enables a successful connection to the console if the OpenShift router also uses port 443 for SSL traffic, which the router does by default.

7.3. MONITORING BROKER RUNTIME DATA USING JMX

View an example of how to monitor a broker by using the Jolokia REST interface to JMX.

Prerequisites

- Completion of [Deploying a basic broker](#) is recommended.

Procedure

1. Get the list of running pods:

```
$ oc get pods
```

```
NAME          READY   STATUS    RESTARTS   AGE
ex-aao-ss-1   1/1     Running   0           14d
```

2. Run the **oc logs** command:

```
$ oc logs -f ex-aao-ss-1
```

```
...
Running Broker in /home/jboss/amq-broker
...
2021-09-17 09:35:10,813 INFO [org.apache.activemq.artemis.integration.bootstrap]
AMQ101000: Starting Apache Artemis Server
2021-09-17 09:35:10,882 INFO [org.apache.activemq.artemis.core.server] AMQ221000: live
Message Broker is starting with configuration Broker Configuration
(clustered=true,journalDirectory=data/journal,bindingsDirectory=data/bindings,largeMessagesDi
rectory=data/large-messages,pagingDirectory=data/paging)
2021-09-17 09:35:10,971 INFO [org.apache.activemq.artemis.core.server] AMQ221013:
Using NIO Journal
2021-09-17 09:35:11,114 INFO [org.apache.activemq.artemis.core.server] AMQ221057:
Global Max Size is being adjusted to 1/2 of the JVM max size (-Xmx). being defined as
2,566,914,048
2021-09-17 09:35:11,369 WARNING [org.jgroups.stack.Configurator] JGRP000014:
BasicTCP.use_send_queues has been deprecated: will be removed in 4.0
2021-09-17 09:35:11,385 WARNING [org.jgroups.stack.Configurator] JGRP000014:
Discovery.timeout has been deprecated: GMS.join_timeout should be used instead
2021-09-17 09:35:11,480 INFO [org.jgroups.protocols.openshift.DNS_PING] serviceName
[ex-aao-ping-svc] set; clustering enabled
2021-09-17 09:35:24,540 INFO [org.openshift.ping.common.Utils] 3 attempt(s) with a
1000ms sleep to execute [GetServicePort] failed. Last failure was
[javax.naming.CommunicationException: DNS error]
...
2021-09-17 09:35:25,044 INFO [org.apache.activemq.artemis.core.server] AMQ221034:
Waiting indefinitely to obtain live lock
2021-09-17 09:35:25,045 INFO [org.apache.activemq.artemis.core.server] AMQ221035:
Live Server Obtained live lock
2021-09-17 09:35:25,206 INFO [org.apache.activemq.artemis.core.server] AMQ221080:
Deploying address DLQ supporting [ANYCAST]
2021-09-17 09:35:25,240 INFO [org.apache.activemq.artemis.core.server] AMQ221003:
Deploying ANYCAST queue DLQ on address DLQ
2021-09-17 09:35:25,360 INFO [org.apache.activemq.artemis.core.server] AMQ221080:
Deploying address ExpiryQueue supporting [ANYCAST]
2021-09-17 09:35:25,362 INFO [org.apache.activemq.artemis.core.server] AMQ221003:
Deploying ANYCAST queue ExpiryQueue on address ExpiryQueue
2021-09-17 09:35:25,656 INFO [org.apache.activemq.artemis.core.server] AMQ221020:
Started EPOLL Acceptor at ex-aao-ss-1.ex-aao-hdls-svc.broker.svc.cluster.local:61616 for
protocols [CORE]
2021-09-17 09:35:25,660 INFO [org.apache.activemq.artemis.core.server] AMQ221007:
```



```

Server is now live
2021-09-17 09:35:25,660 INFO [org.apache.activemq.artemis.core.server] AMQ221001:
Apache Artemis Message Broker version 2.16.0.redhat-00022 [amq-broker,
nodeID=8d886031-179a-11ec-9e02-0a580ad9008b]
2021-09-17 09:35:26,470 INFO [org.apache.amq.hawtio.branding.PluginContextListener]
Initialized amq-broker-redhat-branding plugin
2021-09-17 09:35:26,656 INFO [org.apache.activemq.hawtio.plugin.PluginContextListener]
Initialized artemis-plugin plugin
...

```

3. Run your query to monitor your broker for **MaxConsumers**:

```

$ curl -k -u admin:admin http://console-broker.amq-
demo.apps.example.com/console/jolokia/read/org.apache.activemq.artemis:broker=%22amq-
broker%22,component=addresses,address=%22TESTQUEUE%22,subcomponent=queues,ro-
uting-type=%22anycast%22,queue=%22TESTQUEUE%22/MaxConsumers

{"request":{"mbean":"org.apache.activemq.artemis:address=\"TESTQUEUE\",broker=\"amq-
broker\",component=addresses,queue=\"TESTQUEUE\",routing-
type=\"anycast\",subcomponent=queues\",\"attribute\":\"MaxConsumers\",\"type\":\"read\"},\"value\":-
1,\"timestamp\":1528297825,\"status\":200}

```

CHAPTER 8. REFERENCE

Review reference information that describes the configuration options for each custom resource definition included with AMQ Broker, JAAS login module examples, as well as instructions for viewing logs output to the container's console.

8.1. CUSTOM RESOURCE CONFIGURATION REFERENCE

A Custom resource definition (CRD) is a schema of configuration items for a custom OpenShift object deployed by using an Operator. By deploying a corresponding custom resource (CR) instance, you specify values for configuration items present in the CRD.

The following sub-sections detail the configuration items that you can set in custom resource instances that is based on the **ActiveMQArtemis** broker CRD.

8.1.1. Broker custom resource configuration reference

A CR instance based on the **ActiveMQArtemis** broker CRD enables you to configure brokers for deployment in an OpenShift project. The following table describes the items that you can configure in the CR instance.



IMPORTANT

Configuration items marked with an asterisk (*) are required in any corresponding custom resource (CR) that you deploy. If you do not explicitly specify a value for a non-required item, the configuration uses the default value.

Table 8.1. Broker CR reference

Entry	Sub-entry	Description and usage
-------	-----------	-----------------------

Entry	Sub-entry	Description and usage
adminUser*		Administrator user name required for connecting to the broker and management console. If you do not specify a value, the value is automatically generated and stored in a secret. The default secret name has a format of <custom_resource_name>credentials-secret. For example, my-broker-deployment-credentials-secret. Type: string Example: my-user Default value: Automatically-generated, random value
adminPassword*		Administrator password required for connecting to the broker and management console. If you do not specify a value, the value is automatically generated and stored in a secret. The default secret name has a format of <custom_resource_name>credentials-secret. For example, my-broker-deployment-credentials-secret. Type: string Example: my-password Default value: Automatically-generated, random value

Entry	Sub-entry	Description and usage
ingressDomain		Append a custom domain to the host name in routes and ingresses that are created for acceptors, connectors and the management console. Type: string Example: mydomain.com
deploymentPlan*		Broker deployment configuration
	image*	Full path of the broker container image used for each broker in the deployment. You do not need to explicitly specify a value for image in your CR. The default value of placeholder indicates that the Operator has not yet determined the appropriate image to use. To learn how the Operator chooses a broker container image to use, see Section 2.7, "How the Operator chooses container images when a precise image is not specified in the CR". Type: string Example: registry.redhat.io/amq7/a mq-broker- rhel9@sha256:e44742f661 83e225a77f5171da5db95e 51f92b6848dea9ed02895 a81b1966857 Default value: placeholder

Entry	Sub-entry	Description and usage
	size*	Number of broker Pods to create in the deployment. If you specify a value of 2 or greater, your broker deployment is clustered by default. The cluster user name and password are automatically generated and stored in the same secret as adminUser and adminPassword, by default. Type: int Example: 1 Default value: 1
	requireLogin	Specify whether login credentials are required to connect to the broker. Type: Boolean Example: false Default value: true
	persistenceEnabled	Specify whether to use journal storage for each broker Pod in the deployment. If set to true, each broker Pod requires an available Persistent Volume (PV) that the Operator can claim using a Persistent Volume Claim (PVC). Type: Boolean Example: false Default value: true

Entry	Sub-entry	Description and usage
	initImage	Init Container image used to configure the broker. You do not need to explicitly specify a value for initImage in your CR, unless you want to provide a custom image. To learn how the Operator chooses a built-in Init Container image to use, see Section 2.7, “How the Operator chooses container images when a precise image is not specified in the CR”. To learn how to specify a <i>custom</i> Init Container image, see Section 4.11, “Specifying a custom Init container image”. Type: string Example: registry.redhat.io/amq7/a mq-broker-init- rhel9@sha256:d87b1cea2e b804e0f115e51cb15821054 d8c2791a8ef09e997840c c31270afb1 Default value: Not specified
	journalType	Specify whether to use asynchronous I/O (AIO) or non-blocking I/O (NIO). Type: string Example: aio Default value: nio

Entry	Sub-entry	Description and usage
	messageMigration	When a broker Pod shuts down due to an intentional scaledown of the broker deployment, specify whether to migrate messages to another broker Pod that is still running in the broker cluster. Type: Boolean Example: false Default value: true
	resources.limits.cpu	Maximum amount of host-node CPU, in millicores, that each broker container running in a Pod in a deployment can consume. Type: string Example: "500m" Default value: Uses the same default value that your version of OpenShift Container Platform uses. Consult a cluster administrator.

Entry	Sub-entry	Description and usage
	resources.limits.memory	Maximum amount of host-node memory, in bytes, that each broker container running in a Pod in a deployment can consume. Supports byte notation (for example, K, M, G), or the binary equivalents (Ki, Mi, Gi). Type: string Example: "1024M" Default value: Uses the same default value that your version of OpenShift Container Platform uses. Consult a cluster administrator.
	resources.requests.cpu	Amount of host-node CPU, in millicores, that each broker container running in a Pod in a deployment explicitly requests. Type: string Example: "250m" Default value: Uses the same default value that your version of OpenShift Container Platform uses. Consult a cluster administrator.

Entry	Sub-entry	Description and usage
	resources.requests.memory	Amount of host-node memory, in bytes, that each broker container running in a Pod in a deployment explicitly requests. Supports byte notation (for example, K, M, G), or the binary equivalents (Ki, Mi, Gi). Type: string Example: "512M" Default value: Uses the same default value that your version of OpenShift Container Platform uses. Consult a cluster administrator.
	storage.size	Size, in bytes, of the Persistent Volume Claim (PVC) that each broker in a deployment requires for persistent storage. This property applies only when persistenceEnabled is set to true. The value that you specify must include a unit. Supports byte notation (for example, K, M, G), or the binary equivalents (Ki, Mi, Gi). Type: string Example: 4Gi Default value: 2Gi

Entry	Sub-entry	Description and usage
	jolokiaAgentEnabled	Specifies whether the Jolokia JVM Agent is enabled for the brokers in the deployment. If the value of this property is set to true, Fuse Console can discover and display runtime data for the brokers. Type: Boolean Example: true Default value: false
	managementRBACEnabled	Specifies whether role-based access control (RBAC) is enabled for the brokers in the deployment. To use Fuse Console, you must set the value to false, because Fuse Console uses its own role-based access control. Type: Boolean Example: false Default value: true
	affinity	Specifies scheduling constraints for pods. For information about affinity properties, see properties in the OpenShift Container Platform documentation.
	tolerations	Specifies the pod's tolerations. For information about tolerations properties, see properties in the OpenShift Container Platform documentation.

Entry	Sub-entry	Description and usage
	nodeSelector	Specify a label that matches a node's labels for the pod to be scheduled on that node.
	storageClassName	Specifies the name of the storage class to use for the Persistent Volume Claim (PVC). Storage classes provide a way for administrators to describe and classify the available storage. For example, a storage class might have specific quality-of-service levels, backup policies, or other administrative policies associated with it. Type: string Example: gp3 Default value: Not specified
	startupProbe	Configure a startup probe to check if the AMQ Broker application within the broker container has started. For information about startup probe properties, see properties in the OpenShift Container Platform documentation.
	livenessProbe	Configures a periodic health check on a running broker container to check that the broker is running. For information about liveness probe properties, see properties in the OpenShift Container Platform documentation.

Entry	Sub-entry	Description and usage
	readinessProbe	Configures a periodic health check on a running broker container to check that the broker is accepting network traffic. For information about readiness probe properties, see properties in the OpenShift Container Platform documentation.
	extraMounts	Mounts a secret or configMAP, that contains configuration information, as a file on a broker Pod. For example, you can mount a secret that contains customized logging configuration for AMQ Broker. Type: object Example See Section 4.18, “Configuring logging for brokers” Default value: Not specified
	labels	Assign labels to a broker pod. Type: string Example: location: "production" Default value: Not specified
	podSecurityContext	Defines the security options used to run the broker pods. The following default security values allow the broker pods to run on a OpenShift Container Platform restricted security context constraint (SCC): runAsNonRoot: true

Entry	Sub-entry	seccompProfile: Description and usage type: RuntimeDefault
		If you want the broker to run on a custom SCC, you can configure the following podSecurityContext options in the CR. If you configure any podSecurityContext option in the CR, none of the defaults apply, so you must configure all the options that are required to run under the custom SCC. • fsGroup • fsGroupChangePolicy • runAsGroup • runAsUser • runAsNonRoot • seLinuxOptions • seccompProfile • supplementalGroups • sysctls • windowsOptions For information on the podSecurityContext options, see properties in the OpenShift Container Platform documentation.
	containerSecurityContext	Defines the security options used to run the broker containers in the pods. With the following default values, the containers run on a OpenShift Container Platform restricted security context constraint (SCC):

Entry	Sub-entry	● allowPrivilegeEscalation Description and usage
		● capabilities: drop:ALL ● runAsNonRoot: true ● seccompProfile: type:RuntimeDefault If you want the broker to run on a custom SCC, you can configure the following containerSecurityContext options in the CR. If you configure any containerSecurityContext option in the CR, none of the defaults apply, so you must configure all the options that are required to run under the custom SCC. ● allowPrivilegeEscalation ● capabilities ● privileged ● procMount ● readOnlyRootFilesystem ● runAsGroup ● runAsNonRoot ● runAsUser ● seLinuxOptions ● seccompProfile ● windowsOptions For information on the containerSecurityContext options, see link:https://docs.redhat.com/en/documentation/openshift-container-platform/latest/en_US/installing_and_upgrading/containers/containers.html

Entry	Sub-entry	Description and usage
		OpenShift Container Platform/OperatorSub_a pis/clusterserviceversion-operators-coreos-com-v1alpha1#spec-install-spec-deployments-spec-template-spec-containers-securitycontext in the OpenShift Container Platform documentation.
	podSecurity.serviceAccountName	Specify a service account name for the broker pod. Type: string Example: amq-broker-controller-manager Default value: default
console		Configuration of broker management console.
	expose	Specify whether to expose the management console to clients outside OpenShift Container Platform. Type: Boolean Example: true Default value: false
	exposeMode	Specify whether to expose the management console by using a route or an ingress. By default, the management console is exposed by using a route only. Type: String Example: ingress Default value: route If you expose the console by using an ingress, you must specify an ingressHost or an ingressDomain value in the CR.

Entry	Sub-entry	Description and usage
	ingressHost	Specify a custom host value for routes and ingresses exposed for the management console. You can include any the following variables in the host value: * \$(CR_NAME) - The value of the metadata.name attribute in the CR. * \$(CR_NAMESPACE) - The namespace of the custom resource. * \$(BROKER_ORDINAL) - The ordinal number assigned to the broker pod by the StatefulSet. * \$(ITEM_NAME) - The name of the console. The default name is wconsj * \$(RES_TYPE) - The resource type. A route has a resource type of rte. An ingress has a resource type of ing. * \$(INGRESS_DOMAIN) - The value of the spec.ingressDomain attribute if it is configured in the CR. Type: string Example: console-\$(CR_NAME)-\$(ITEM_NAME)-\$(BROKER_ORDINAL).mydomain.com
	sslEnabled	Specify whether to use SSL on the management console port. Type: Boolean Example: true Default value: false

Entry	Sub-entry	Description and usage
	sslSecret	Secret where broker key store, trust store, and their corresponding passwords (all Base64-encoded) are stored. If you do not specify a value for sslSecret, the console uses a default secret name. The default secret name is in the form of <custom_resource_name>-console-secret. This property applies only when the sslEnabled property is set to true. Type: string Example: my-broker-deployment-console-secret Default value: Not specified
	useClientAuth	Specify whether the management console requires that the client presents a certificate to authenticate. Type: Boolean Example: true Default value: false
acceptors.acceptor		A single acceptor configuration instance.
	name*	Name of acceptor. Type: string Example: my-acceptor Default value: Not applicable

Entry	Sub-entry	Description and usage
	port	Port number to use for the acceptor instance. Type: int Example: 5672 Default value: 61626 for the first acceptor that you define. The default value then increments by 10 for every subsequent acceptor that you define.
	protocols	Messaging protocols to be enabled on the acceptor instance. Type: string Example: amqp,core Default value: all
	sslEnabled	Specify whether SSL is enabled on the acceptor port. If set to true, look in the secret name specified in sslSecret for the credentials required by TLS/SSL. Type: Boolean Example: true Default value: false

Entry	Sub-entry	Description and usage
	sslSecret	Secret where broker key store, trust store, and their corresponding passwords (all Base64-encoded) are stored. If you do not specify a custom secret name for sslSecret, the acceptor assumes a default secret name. The default secret name has a format of <custom_resource_name><acceptor_name>-secret. You must always create this secret yourself, even when the acceptor assumes a default name. Type: string Example: my-broker-deployment-my-acceptor-secret Default value: <custom_resource_name>-<acceptor_name>-secret

Entry	Sub-entry	Description and usage
	enabledCipherSuites	Comma-separated list of cipher suites to use for TLS communication. Specify the most secure cipher suite(s) supported by your client application. If you specify a comma-separated list of cipher suites that are common to both the broker and the client, or you do not specify any cipher suites, the broker and client mutually negotiate a cipher suite to use. If you do not know which cipher suites to specify, you can first establish a broker-client connection with your client running in debug mode to verify the cipher suites that are common to both the broker and the client. Then, configure enabledCipherSuites on the broker. The cipher suites available depend on the TLS protocol versions used by the broker and clients. If the default TLS protocol version changes after you upgrade the broker, you might need to select an earlier TLS protocol version to ensure that the broker and the clients can use a common cipher suite. For more information, see enabledProtocols. Type: string Default value: Not specified

Entry	Sub-entry	Description and usage
	enabledProtocols	Comma-separated list of protocols to use for TLS communication. Type: string Example: TLSv1,TLSv1.1,TLSv1.2 Default value: Not specified If you don't specify a TLS protocol version, the broker uses the JVM's default version. If the broker uses the JVM's default TLS protocol version and that version changes after you upgrade the broker, the TLS protocol versions used by the broker and clients might be incompatible. While it is recommended that you use the later TLS protocol version, you can specify an earlier version in enabledProtocols to interoperate with clients that do not support a newer TLS protocol version.
	keyStoreProvider	The name of the provider of the keystore that the broker uses. Type: string Example: SunJCE Default value: Not specified

Entry	Sub-entry	Description and usage
	trustStoreProvider	The name of the provider of the truststore that the broker uses. Type: string Example: SunJCE Default value: Not specified
	trustStoreType	The type of truststore that the broker uses. Type: string Example: JCEKS Default value: JKS
	needClientAuth	Specify whether the broker informs clients that two-way TLS is required on the acceptor. This property overrides wantClientAuth. Type: Boolean Example: true Default value: Not specified
	wantClientAuth	Specify whether the broker informs clients that two-way TLS is <i>requested</i> on the acceptor, but not required. This property is overridden by needClientAuth. Type: Boolean Example: true Default value: Not specified

Entry	Sub-entry	Description and usage
	verifyHost	Specify whether to compare the Common Name (CN) of a client's certificate to its host name, to verify that they match. This option applies only when two-way TLS is used. Type: Boolean Example: true Default value: Not specified
	sslProvider	Specify whether the SSL provider is JDK or OPENSSL. Type: string Example: OPENSSL Default value: JDK
	sniHost	Regular expression to match against the server_name extension on incoming connections. If the names don't match, connection to the acceptor is rejected. Type: string Example: some_regular_expression Default value: Not specified
	expose	Specify whether to expose the acceptor to clients outside OpenShift Container Platform. Type: Boolean Example: true Default value: false

Entry	Sub-entry	Description and usage
	exposeMode	Specify whether to expose the acceptor by using a route or an ingress. By default, an acceptor is exposed using a route only. Type: String Example: ingress Default value: route If you expose a connector by using an ingress, you must include the ingressHost or the ingressDomain attribute in the CR.

Entry	Sub-entry	Description and usage
	ingressHost	Specify a custom host value for routes and ingress exposed for the acceptor. You can include any of the following variables for the host: * \$(CR_NAME) - The value of the metadata.name attribute in the CR. * \$(CR_NAMESPACE) - The namespace of the custom resource. * \$(BROKER_ORDINAL) - The ordinal number assigned to the broker pod by the StatefulSet. * \$(ITEM_NAME) - The name of the acceptor. * \$(RES_TYPE) - The resource type. A route has a resource type of rte. An ingress has a resource type of ing. * \$(INGRESS_DOMAIN) - The value of the spec.ingressDomain attribute if it is configured in the CR. Type: string Example: my-acceptor-\$(CR_NAME)-\$(ITEM_NAME)-\$(BROKER_ORDINAL).mydomain.com

Entry	Sub-entry	Description and usage
	anycastPrefix	Prefix used by a client to specify that the anycast routing type should be used. Type: string Example: jms.queue Default value: Not specified
	multicastPrefix	Prefix used by a client to specify that the multicast routing type should be used. Type: string Example: /topic/ Default value: Not specified
	connectionsAllowed	Number of connections allowed on the acceptor. When this limit is reached, a DEBUG message is issued to the log, and the connection is refused. The type of client in use determines what happens when the connection is refused. Type: integer Example: 2 Default value: 0 (unlimited connections)

Entry	Sub-entry	Description and usage
	amqpMinLargeMessageSize	Minimum message size, in bytes, required for the broker to handle an AMQP message as a large message. If the size of an AMQP message is equal or greater to this value, the broker stores the message in a large messages directory (/opt/<custom_resource_name>/data/large-messages, by default) on the persistent volume (PV) used by the broker for message storage. Setting the value to -1 disables large message handling for AMQP messages. Type: integer Example: 204800 Default value: 102400 (100 KB)
	BindToAllInterfaces	If set to true, configures the broker acceptors with a 0.0.0.0 IP address instead of the internal IP address of the pod. When the broker acceptors have a 0.0.0.0 IP address, they bind to all interfaces configured for the pod and clients can direct traffic to the broker by using OpenShift Container Platform port-forwarding. Normally, you use this configuration to debug a service. For more information about port-forwarding, see Using port-forwarding in the OpenShift Container Platform documentation.

Entry	Sub-entry	Description and usage NOTE
		 If port-forwarding is used incorrectly, it can create a security risk for your environment. Where possible, do not use port-forwarding in a production environment. Type: Boolean Example: true Default value: false
connectors.connector		A single connector configuration instance.
	name*	Name of connector. Type: string Example: my-connector Default value: Not applicable
	type	The type of connector to create; tcp or vm. Type: string Example: vm Default value: tcp

Entry	Sub-entry	Description and usage
	host*	Host name or IP address to connect to. Type: string Example: 192.168.0.58 Default value: Not specified
	port*	Port number to be used for the connector instance. Type: int Example: 22222 Default value: Not specified
	sslEnabled	Specify whether SSL is enabled on the connector port. If set to true, look in the secret name specified in sslSecret for the credentials required by TLS/SSL. Type: Boolean Example: true Default value: false

Entry	Sub-entry	Description and usage
	sslSecret	Secret where broker key store, trust store, and their corresponding passwords (all Base64-encoded) are stored. If you do not specify a custom secret name for sslSecret, the connector assumes a default secret name. The default secret name has a format of <custom_resource_name><connector_name>-secret. You must always create this secret yourself, even when the connector assumes a default name. Type: string Example: my-broker-deployment-my-connector-secret Default value: <custom_resource_name>-<connector_name>-secret
	enabledCipherSuites	Comma-separated list of cipher suites to use for TLS communication. Type: string NOTE: For a connector, it is recommended that you do not specify a list of cipher suites. Default value: Not specified

Entry	Sub-entry	Description and usage
	keyStoreProvider	The name of the provider of the keystore that the broker uses. Type: string Example: SunJCE Default value: Not specified
	trustStoreProvider	The name of the provider of the truststore that the broker uses. Type: string Example: SunJCE Default value: Not specified
	trustStoreType	The type of truststore that the broker uses. Type: string Example: JCEKS Default value: JKS
	enabledProtocols	Comma-separated list of protocols to use for TLS communication. Type: string Example: TLSv1,TLSv1.1,TLSv1.2 Default value: Not specified

Entry	Sub-entry	Description and usage
	needClientAuth	Specify whether the broker informs clients that two-way TLS is required on the connector. This property overrides wantClientAuth. Type: Boolean Example: true Default value: Not specified
	wantClientAuth	Specify whether the broker informs clients that two-way TLS is <i>requested</i> on the connector, but not required. This property is overridden by needClientAuth. Type: Boolean Example: true Default value: Not specified
	verifyHost	Specify whether to compare the Common Name (CN) of client's certificate to its host name, to verify that they match. This option applies only when two-way TLS is used. Type: Boolean Example: true Default value: Not specified

Entry	Sub-entry	Description and usage
	sslProvider	Specify whether the SSL provider is JDK or OPENSSL. Type: string Example: OPENSSL Default value: JDK
	sniHost	Regular expression to match against the server_name extension on outgoing connections. If the names don't match, the connector connection is rejected. Type: string Example: some_regular_expression Default value: Not specified
	expose	Specify whether to expose the connector to clients outside OpenShift Container Platform. Type: Boolean Example: true Default value: false

Entry	Sub-entry	Description and usage
	exposeMode	Specify whether to expose the connector by using a route or an ingress. By default, a connector is exposed using a route only. Type: string Example: ingress Default value: route If you expose a connector by using an ingress, you must include the ingressHost or the ingressDomain attribute in the CR.

Entry	Sub-entry	Description and usage
	ingressHost	Specify a custom host value for routes and ingresses exposed for the connector. You can include any the following variables in the host value: * <code>\$(CR_NAME)</code> - The value of the metadata.name attribute in the CR. * <code>\$(CR_NAMESPACE)</code> - The namespace of the custom resource. * <code>\$(BROKER_ORDINAL)</code> - The ordinal number assigned to the broker pod by the StatefulSet. * <code>\$(ITEM_NAME)</code> - The name of the connector. * <code>\$(RES_TYPE)</code> - The resource type. A route has a resource type of rte. An ingress has a resource type of ing. * <code>\$(INGRESS_DOMAIN)</code> - The value of the spec.ingressDomain attribute if it is configured in the CR. Type: string Example: my-connector- \$(CR_NAME)- \$(ITEM_NAME)- \$(BROKER_ORDINAL).\$(I NGRESS_DOMAIN).mydo main.com
addressSettings.applyRule		Specifies how the Operator applies the configuration that you add to the CR for each matching address or set of addresses. The values that you can specify are:

Entry	Sub-entry	merge_all Description and usage For address settings
		specified in both the CR and the default configuration that match the same address or set of addresses: ● Replace any property values specified in the default configuration with those specified in the CR. ● Keep any property values that are specified uniquely in the CR or the default configuration. Include each of these in the final, merged configuration. For address settings specified in either the CR or the default configuration that uniquely match a particular address or set of addresses, include these in the final, merged configuration. merge_replace For address settings specified in both the CR and the default configuration that match the same address or set of addresses, include the settings specified in the CR in the final, merged configuration. Do not include any properties specified in the default configuration, even if these are not specified in the CR. + For address settings specified in either the CR or the default configuration that uniquely match a

Entry	Sub-entry	Description and usage
		particular address or set of addresses, include these in the final, merged configuration. replace_all Replace all address settings specified in the default configuration with those specified in the CR. The final, merged configuration corresponds exactly to that specified in the CR. Type: string Example: replace_all Default value: merge_all
addressSettings.addressSetting		Address settings for a matching address or set of addresses.

Entry	Sub-entry	Description and usage
	addressFullPolicy	Specify what happens when an address configured with maxSizeBytes becomes full. The available policies are: PAGE Messages sent to a full address are paged to disk. DROP Messages sent to a full address are silently dropped. FAIL Messages sent to a full address are dropped and the message producers receive an exception. BLOCK Message producers will block when they try to send any further messages. The BLOCK policy works only for AMQP, OpenWire, and Core Protocol, because those protocols support flow control. Type: string Example: DROP Default value: PAGE

Entry	Sub-entry	Description and usage
	autoCreateAddresses	Specify whether the broker automatically creates an address when a client sends a message to, or attempts to consume a message from, a queue that is bound to an address that does not exist. Type: Boolean Example: false Default value: true
	autoCreateDeadLetterResources	Specify whether the broker automatically creates a dead letter address and queue to receive undelivered messages. If the parameter is set to true, the broker automatically creates a dead letter address and an associated dead letter queue. The name of the automatically-created address matches the value that you specify for deadLetterAddress. Type: Boolean Example: true Default value: false

Entry	Sub-entry	Description and usage
	autoCreateExpiryResources	Specify whether the broker automatically creates an address and queue to receive expired messages. If the parameter is set to true, the broker automatically creates an expiry address and an associated expiry queue. The name of the automatically-created address matches the value that you specify for expiryAddress. Type: Boolean Example: true Default value: false
	autoCreateJmsQueues	This property is deprecated. Use autoCreateQueues instead.
	autoCreateJmsTopics	This property is deprecated. Use autoCreateQueues instead.
	autoCreateQueues	Specify whether the broker automatically creates a queue when a client sends a message to, or attempts to consume a message from, a queue that does not yet exist. Type: Boolean Example: false Default value: true

Entry	Sub-entry	Description and usage
	autoDeleteAddresses	Specify whether the broker automatically deletes automatically-created addresses when the broker no longer has any queues. Type: Boolean Example: false Default value: true
	autoDeleteAddressDelay	Time, in milliseconds, that the broker waits before automatically deleting an automatically-created address when the address has no queues. Type: integer Example: 100 Default value: 0
	autoDeleteJmsQueues	This property is deprecated. Use autoDeleteQueues instead.
	autoDeleteJmsTopics	This property is deprecated. Use autoDeleteQueues instead.
	autoDeleteQueues	Specify whether the broker automatically deletes an automatically-created queue when the queue has no consumers and no messages. Type: Boolean Example: false Default value: true

Entry	Sub-entry	Description and usage
	autoDeleteCreatedQueues	Specify whether the broker automatically deletes a manually-created queue when the queue has no consumers and no messages. Type: Boolean Example: true Default value: false
	autoDeleteQueuesDelay	Time, in milliseconds, that the broker waits before automatically deleting an automatically-created queue when the queue has no consumers. Type: integer Example: 10 Default value: 0
	autoDeleteQueuesMessageCount	Maximum number of messages that can be in a queue before the broker evaluates whether the queue can be automatically deleted. Type: integer Example: 5 Default value: 0

Entry	Sub-entry	Description and usage
	configDeleteAddresses	When the configuration file is reloaded, this parameter specifies how to handle an address (and its queues) that has been deleted from the configuration file. You can specify the following values: OFF The broker does not delete the address when the configuration file is reloaded. FORCE The broker deletes the address and its queues when the configuration file is reloaded. If there are any messages in the queues, they are removed also. Type: string Example: FORCE Default value: OFF

Entry	Sub-entry	Description and usage
	configDeleteQueues	When the configuration file is reloaded, this setting specifies how the broker handles queues that have been deleted from the configuration file. You can specify the following values: OFF The broker does not delete the queue when the configuration file is reloaded. FORCE The broker deletes the queue when the configuration file is reloaded. If there are any messages in the queue, they are removed also. Type: string Example: FORCE Default value: OFF
	deadLetterAddress	The address to which the broker sends dead (that is, <i>undelivered</i>) messages. Type: string Example: DLA Default value: None
	deadLetterQueuePrefix	Prefix that the broker applies to the name of an automatically-created dead letter queue. Type: string Example: myDLQ. Default value: DLQ.

Entry	Sub-entry	Description and usage
	deadLetterQueueSuffix	Suffix that the broker applies to an automatically-created dead letter queue. Type: string Example: .DLQ Default value: None
	defaultAddressRoutingType	Routing type used on automatically-created addresses. Type: string Example: ANYCAST Default value: MULTICAST
	defaultConsumersBeforeDispatch	Number of consumers needed before message dispatch can begin for queues on an address. Type: integer Example: 5 Default value: 0
	defaultConsumerWindowSize	Default window size, in bytes, for a consumer. Type: integer Example: 300000 Default value: 1048576 (1024*1024)

Entry	Sub-entry	Description and usage
	defaultDelayBeforeDispatch	Default time, in milliseconds, that the broker waits before dispatching messages if the value specified for defaultConsumersBeforeDispatch has not been reached. Type: integer Example: 5 Default value: -1 (no delay)
	defaultExclusiveQueue	Specifies whether all queues on an address are exclusive queues by default. Type: Boolean Example: true Default value: false
	defaultGroupBuckets	Number of buckets to use for message grouping. Type: integer Example: 0 (message grouping disabled) Default value: -1 (no limit)
	defaultGroupFirstKey	Key used to indicate to a consumer which message in a group is first. Type: string Example: firstMessageKey Default value: None

Entry	Sub-entry	Description and usage
	defaultGroupRebalance	Specifies whether to rebalance groups when a new consumer connects to the broker. Type: Boolean Example: true Default value: false
	defaultGroupRebalancePauseDispatch	Specifies whether to pause message dispatch while the broker is rebalancing groups. Type: Boolean Example: true Default value: false
	defaultLastValueQueue	Specifies whether all queues on an address are last value queues by default. Type: Boolean Example: true Default value: false
	defaultLastValueKey	Default key to use for a last value queue. Type: string Example: stock_ticker Default value: None
	defaultMaxConsumers	Maximum number of consumers allowed on a queue at any time. Type: integer Example: 100 Default value: -1 (no limit)

Entry	Sub-entry	Description and usage
	defaultNonDestructive	Specifies whether all queues on an address are non-destructive by default. Type: Boolean Example: true Default value: false
	defaultPurgeOnNoConsumers	Specifies whether the broker purges the contents of a queue once there are no consumers. Type: Boolean Example: true Default value: false
	defaultQueueRoutingType	Routing type used on automatically-created queues. The default value is MULTICAST. Type: string Example: ANYCAST Default value: MULTICAST
	defaultRingSize	Default ring size for a matching queue that does not have a ring size explicitly set. Type: integer Example: 3 Default value: -1 (no size limit)

Entry	Sub-entry	Description and usage
	enableMetrics	Specifies whether a configured metrics plugin such as the Prometheus plugin collects metrics for a matching address or set of addresses. Type: Boolean Example: false Default value: true
	expiryAddress	Address that receives expired messages. Type: string Example: myExpiryAddress Default value: None
	expiryDelay	Expiration time, in milliseconds, applied to messages that are using the default expiration time. Type: integer Example: 100 Default value: -1 (no expiration time applied)
	expiryQueuePrefix	Prefix that the broker applies to the name of an automatically-created expiry queue. Type: string Example: myExp. Default value: EXP.

Entry	Sub-entry	Description and usage
	expiryQueueSuffix	Suffix that the broker applies to the name of an automatically-created expiry queue. Type: string Example: .EXP Default value: None
	lastValueQueue	Specify whether a queue uses only last values or not. Type: Boolean Example: true Default value: false
	managementBrowsePageSize	Specify how many messages a management resource can browse. Type: integer Example: 100 Default value: 200
	match*	String that matches address settings to addresses configured on the broker. You can specify an exact address name or use a wildcard expression to match the address settings to a set of addresses. If you use a wildcard expression as a value for the match property, you must enclose the value in single quotation marks, for example, 'myAddresses*'. Type: string Example: 'myAddresses*' Default value: None

Entry	Sub-entry	Description and usage
	maxDeliveryAttempts	Specifies how many times the broker attempts to deliver a message before sending the message to the configured dead letter address. Type: integer Example: 20 Default value: 10
	maxExpiryDelay	Expiration time, in milliseconds, applied to messages that are using an expiration time greater than this value. Type: integer Example: 20 Default value: -1 (no maximum expiration time applied)
	maxRedeliveryDelay	Maximum value, in milliseconds, between message redelivery attempts made by the broker. Type: integer Example: 100 Default value: The default value is ten times the value of redeliveryDelay, which has a default value of 0.

Entry	Sub-entry	Description and usage
	maxSizeBytes	Maximum memory size, in bytes, for an address. Used when addressFullPolicy is set to PAGING, BLOCK, or FAIL. Also supports byte notation such as "K", "Mb", and "GB". Type: string Example: 10Mb Default value: -1 (no limit)
	maxSizeBytesRejectThreshold	Maximum size, in bytes, that an address can reach before the broker begins to reject messages. Used when the address-full-policy is set to BLOCK. Works in combination with maxSizeBytes for the AMQP protocol only. Type: integer Example: 500 Default value: -1 (no maximum size)
	messageCounterHistoryDayLimit	Number of days for which a broker keeps a message counter history for an address. Type: integer Example: 5 Default value: 0

Entry	Sub-entry	Description and usage
	minExpiryDelay	Expiration time, in milliseconds, applied to messages that are using an expiration time lower than this value. Type: integer Example: 20 Default value: -1 (no minimum expiration time applied)
	pageMaxCacheSize	Number of page files to keep in memory to optimize I/O during paging navigation. Type: integer Example: 10 Default value: 5
	pageSizeBytes	Paging size in bytes. Also supports byte notation such as K, Mb, and GB. Type: string Example: 20971520 Default value: 10485760 (approximately 10.5 MB)
	redeliveryDelay	Time, in milliseconds, that the broker waits before redelivering a cancelled message. Type: integer Example: 100 Default value: 0

Entry	Sub-entry	Description and usage
	redistributionDelay	Time, in milliseconds, that the broker waits after the last consumer is closed on a queue before redistributing any remaining messages. Type: integer Example: 100 Default value: -1 (not set)
	retroactiveMessageCount	Number of messages to keep for future queues created on an address. Type: integer Example: 100 Default value: 0
	sendToDlaOnNoRoute	Specify whether a message will be sent to the configured dead letter address if it cannot be routed to any queues. Type: Boolean Example: true Default value: false
	slowConsumerCheckPeriod	How often, in seconds, that the broker checks for slow consumers. Type: integer Example: 15 Default value: 5

Entry	Sub-entry	Description and usage
	slowConsumerPolicy	Specifies what happens when a slow consumer is identified. Valid options are KILL or NOTIFY. KILL kills the consumer's connection, which impacts any client threads using that same connection. NOTIFY sends a CONSUMER_SLOW management notification to the client. Type: string Example: KILL Default value: NOTIFY
	slowConsumerThreshold	Minimum rate of message consumption, in messages per second, before a consumer is considered slow. Type: integer Example: 100 Default value: -1 (not set)
env	<variable name>=<value>	Set environment variables for the broker. Type: array Example: name: TZ value: Europe/Vienna Default value: Not applicable
brokerProperties		Configure broker properties that are not exposed in the broker's custom resource definitions (CRDs) and are, otherwise, not configurable in a custom resource(CR).

Entry	Sub-entry	Description and usage
	<property name>=<value>	A list of property names and values to configure for the broker. Type: string Example: globalMaxSize=512m Default value: Not applicable
version		Specify the version of the AMQ Broker container images that you want the Operator to deploy. For example, if you change the value of version from 7.12.4 to 7.13.0, the Operator upgrades the broker images to 7.13.0. You can omit the micro and minor digits from the version number to automatically upgrade to the broker images that are available for the latest micro or minor release. For example, if you specify a version of 7.12, the Operator upgrades to the images for the latest 7.12.x release. Or, if you specify a version of 7, the Operator upgrades to the images for the latest 7.x.x release. Type: string Example: 7.14.0 Default value: Current version of AMQ Broker

8.1.2. Address Custom Resource configuration reference

In a CR instance based on the **ActiveMQArtemisAddresss** CRD, you can define addresses and queues for the brokers in your deployment.

The following table details the items that you can configure in a CR instance based on the **ActiveMQArtemisAddresss** CRD.



IMPORTANT

Configuration items marked with an asterisk (*) are required in any corresponding Custom Resource (CR) that you deploy. If you do not explicitly specify a value for a non-required item, the configuration uses the default value.

Table 8.2. Address CR reference

Entry	Description and usage
addressName*	Address name to be created on broker. Type: string Example: address0 Default value: Not specified
queueName	Queue name to be created on broker. If queueName is not specified, the CR creates only the address. Type: string Example: queue0 Default value: Not specified
removeFromBrokerOnDelete*	Specify whether the Operator removes existing addresses for all brokers in a deployment when you remove the address CR instance for that deployment. The default value is false, which means the Operator does not delete existing addresses when you remove the CR. Type: Boolean Example: true Default value: false
routingType*	Routing type to be used; anycast or multicast. Type: string Example: anycast Default value: multicast

8.1.3. Security Custom Resource configuration reference

In a CR instance based on the **ActiveMQArtemisSecurity** CRD, you can define the security configuration for the brokers in your deployment.

Security information includes:

- users and roles
- login modules, including **propertiesLoginModule**, **guestLoginModule** and **keycloakLoginModule**
- role based access control
- console access control



NOTE

Many of the options require you understand the broker security concepts described in [Securing brokers](#)

The following table details the items that you can configure.



IMPORTANT

Configuration items marked with an asterisk (*) are required in any corresponding custom resource (CR) that you deploy. If you do not explicitly specify a value for a non-required item, the configuration uses the default value.

Table 8.3. Security CR reference

Entry	Sub-entry	Description and usage
loginModules		One or more login module configurations. A login module can be one of the following types: • propertiesLoginModule - allows you define broker users directly. • guestLoginModule - for a user who does not have login credentials, or whose credentials fail authentication, you can grant limited access to the broker using a guest account. • keycloakLoginModule. - allows you secure brokers using Red Hat Single Sign-On.
propertiesLoginModule	name*	Name of login module. Type: string Example: my-login Default value: Not applicable

Entry	Sub-entry	Description and usage
	users.name*	Name of user. Type: string Example: jdoe Default value: Not applicable
	users.password*	password of user. Type: string Example: password Default value: Not applicable
	users.roles	Names of roles. Type: string Example: viewer Default value: Not applicable
guestLoginModule	name*	Name of guest login module. Type: string Example: guest-login Default value: Not applicable
	guestUser	Name of guest user. Type: string Example: myguest Default value: Not applicable
	guestRole	Name of role for guest user. Type: string Example: guest Default value: Not applicable

Entry	Sub-entry	Description and usage
keycloakLoginModule	name	Name for KeycloakLoginModule Type: string Example: sso Default value: Not applicable
	moduleType	Type of KeycloakLoginModule (directAccess or bearerToken) Type: string Example: bearerToken Default value: Not applicable
	configuration	The following configuration items are related to Red Hat Single Sign-On and detailed information is available from the OpenID Connect documentation.
	configuration.realm*	Realm for KeycloakLoginModule Type: string Example: myrealm Default value: Not applicable
	configuration.realmPublicKey	Public key for the realm Type: string Default value: Not applicable
	configuration.authServerUrl*	URL of the keycloak authentication server Type: string Default value: Not applicable
	configuration.sslRequired	Specify whether SSL is required Type: string Valid values are 'all', 'external' and 'none'.

Entry	Sub-entry	Description and usage
	configuration.resource*	Resource Name The client-id of the application. Each application has a client-id that is used to identify the application.
	configuration.publicClient	Specify whether it is public client. Type: Boolean Default value: false Example: false
	configuration.credentials.key	Specify the credentials key. Type: string Default value: Not applicable Type: string Default value: Not applicable
	configuration.credentials.value	Specify the credentials value Type: string Default value: Not applicable
	configuration.useResourceRoleMappings	Specify whether to use resource role mappings Type: Boolean Example: false
	configuration.enableCors	Specify whether to enable Cross-Origin Resource Sharing (CORS) It will handle CORS preflight requests. It will also look into the access token to determine valid origins. Type: Boolean Default value: false
	configuration.corsMaxAge	CORS max age If CORS is enabled, this sets the value of the Access-Control-Max-Age header.

Entry	Sub-entry	Description and usage
	configuration.corsAllowedMethods	CORS allowed methods If CORS is enabled, this sets the value of the Access-Control-Allow-Methods header. This should be a comma-separated string.
	configuration.corsAllowedHeaders	CORS allowed headers If CORS is enabled, this sets the value of the Access-Control-Allow-Headers header. This should be a comma-separated string.
	configuration.corsExposedHeaders	CORS exposed headers If CORS is enabled, this sets the value of the Access-Control-Expose-Headers header. This should be a comma-separated string.
	configuration.exposeToken	Specify whether to expose access token Type: Boolean Default value: false
	configuration.bearerOnly	Specify whether to verify bearer token Type: Boolean Default value: false
	configuration.autoDetectBearerOnly	Specify whether to only auto-detect bearer token Type: Boolean Default value: false
	configuration.connectionPoolSize	Size of the connection pool Type: Integer Default value: 20
	configuration.allowAnyHostName	Specify whether to allow any host name Type: Boolean Default value: false

Entry	Sub-entry	Description and usage
	configuration.disableTrustManager	Specify whether to disable trust manager Type: Boolean Default value: false
	configuration.trustStore*	Path of a trust store This is REQUIRED unless ssl-required is none or disable-trust-manager is true.
	configuration.trustStorePassword*	Truststore password This is REQUIRED if truststore is set and the truststore requires a password.
	configuration.clientKeyStore	Path of a client keystore Type: string Default value: Not applicable
	configuration.clientKeyStorePassword	Client keystore password Type: string Default value: Not applicable
	configuration.clientKeyPassword	Client key password Type: string Default value: Not applicable
	configuration.alwaysRefreshToken	Specify whether to always refresh token Type: Boolean Example: false
	configuration.registerNodeAtStartup	Specify whether to register node at startup Type: Boolean Example: false

Entry	Sub-entry	Description and usage
	configuration.registerNodePeriod	Period for re-registering node Type: string Default value: Not applicable
	configuration.tokenStore	Type of token store (session or cookie) Type: string Default value: Not applicable
	configuration.tokenCookiePath	Cookie path for a cookie store Type: string Default value: Not applicable
	configuration.principalAttribute	OpenID Connect ID Token attribute to populate the UserPrincipal name with OpenID Connect ID Token attribute to populate the UserPrincipal name with. If token attribute is null, defaults to sub. Possible values are sub, preferred_username, email, name, nickname, given_name, family_name.
	configuration.proxyUrl	The proxy URL
	configuration.turnOffChangeSessionIdOnLogin	Specify whether to change session id on a successful login Type: Boolean Example: false
	configuration.tokenMinimumTimeToLive	Minimum time to refresh an active access token Type: Integer Default value: 0
	configuration.minTimeBetweenJwksRequests	Minimum interval between two requests to Keycloak to retrieve new public keys Type: Integer Default value: 10

Entry	Sub-entry	Description and usage
	configuration.publicKeyCacheTtl	Maximum interval between two requests to Keycloak to retrieve new public keys Type: Integer Default value: 86400
	configuration.ignoreOAuthQueryParameter	Whether to turn off processing of the access_token query parameter for bearer token processing Type: Boolean Example: false
	configuration.verifyTokenAudience	Verify whether the token contains this client name (resource) as an audience Type: Boolean Example: false
	configuration.enableBasicAuth	Whether to support basic authentication Type: Boolean Default value: false
	configuration.confidentialPort	The confidential port used by the Keycloak server for secure connections over SSL/TLS Type: Integer Example: 8443
	configuration.redirectRewriteRules.key	The regular expression used to match the Redirect URI. Type: string Default value: Not applicable
	configuration.redirectRewriteRules.value	The replacement String Type: string Default value: Not applicable

Entry	Sub-entry	Description and usage
	configuration.scope	The OAuth2 scope parameter for DirectAccessGrantsLoginModule Type: string Default value: Not applicable
securityDomains		Broker security domains
	brokerDomain.name	Broker domain name Type: string Example: activemq Default value: Not applicable
	brokerDomain.loginModules	One or more login modules. Each entry must be previously defined in the loginModules section above.
	brokerDomain.loginModules.name	Name of login module Type: string Example: prop-module Default value: Not applicable
	brokerDomain.loginModules.flag	Same as propertiesLoginModule, required, requisite, sufficient and optional are valid values. Type: string Example: sufficient Default value: Not applicable
	brokerDomain.loginModules.debug	Debug
	brokerDomain.loginModules.reload	Reload

Entry	Sub-entry	Description and usage
	consoleDomain.name	Broker domain name Type: string Example: activemq Default value: Not applicable
	consoleDomain.loginModules	A single login module configuration.
	consoleDomain.loginModules.name	Name of login module Type: string Example: prop-module Default value: Not applicable
	consoleDomain.loginModules.flag	Same as propertiesLoginModule, required , requisite , sufficient and optional are valid values. Type: string Example: sufficient Default value: Not applicable
	consoleDomain.loginModules.debug	Debug Type: Boolean Example: false
	consoleDomain.loginModules.reload	Reload Type: Boolean Example: true Default: false
securitySettings		Additional security settings to add to broker.xml or management.xml
	broker.match	The address match pattern for a security setting section. See AMQ Broker wildcard syntax for details about the match pattern syntax.

Entry	Sub-entry	Description and usage
	broker.permissions.operation Type	The operation type of a security setting, as described in Setting permissions. Type: string Example: createAddress Default value: Not applicable
	broker.permissions.roles	The security settings are applied to these roles, as described in Setting permissions. Type: string Example: root Default value: Not applicable
securitySettings.management		Options to configure management.xml .
	hawtioRoles	The roles allowed to log into the Broker console. Type: string Example: root Default value: Not applicable
	connector.host	The connector host for connecting to the management API. Type: string Example: myhost Default value: localhost
	connector.port	The connector port for connecting to the management API. Type: integer Example: 1099 Default value: 1099

Entry	Sub-entry	Description and usage
	connector.jmxRealm	The JMX realm of the management API. Type: string Example: activemq Default value: activemq
	connector.objectName	The JMX object name of the management API. Type: String Example: connector:name=rmi Default: connector:name=rmi
	connector.authenticatorType	The management API authentication type. Type: String Example: password Default: password
	connector.secured	Whether the management API connection is secured. Type: Boolean Example: true Default value: false
	connector.keyStoreProvider	The keystore provider for the management connector. Required if you have set connector.secured="true". The default value is JKS.
	connector.keyStorePath	Location of the keystore. Required if you have set connector.secured="true".
	connector.keyStorePassword	The keystore password for the management connector. Required if you have set connector.secured="true".

Entry	Sub-entry	Description and usage
	connector.trustStoreProvider	The truststore provider for the management connector. Required if you have set <code>connector.secured="true"</code>. Type: String Example: JKS Default: JKS
	connector.trustStorePath	Location of the truststore for the management connector. Required if you have set <code>connector.secured="true"</code>. Type: string Default value: Not applicable
	connector.trustStorePassword	The truststore password for the management connector. Required if you have set <code>connector.secured="true"</code>. Type: string Default value: Not applicable
	connector.passwordCodec	The password codec for management connector. The fully qualified class name of the password codec to use as described in Encrypting a password in a configuration file.
	authorisation.allowedList.domain	The domain of allowedList Type: string Default value: Not applicable
	authorisation.allowedList.key	The key of allowedList Type: string Default value: Not applicable
	authorisation.defaultAccess.method	The method of defaultAccess List Type: string Default value: Not applicable

Entry	Sub-entry	Description and usage
	authorisation.defaultAccess.roles	The roles of defaultAccess List Type: string Default value: Not applicable
	authorisation.roleAccess.domain	The domain of roleAccess List Type: string Default value: Not applicable
	authorisation.roleAccess.key	The key of roleAccess List Type: string Default value: Not applicable
	authorisation.roleAccess.accessList.method	The method of roleAccess List Type: string Default value: Not applicable
	authorisation.roleAccess.accessList.roles	The roles of roleAccess List Type: string Default value: Not applicable
	applyToCrNames	Apply this security config to the brokers defined by the named CRs in the current namespace. A value of * or empty string means applying to all brokers. Type: string Example: my-broker Default value: All brokers defined by CRs in the current namespace.

8.2. EXAMPLE JAAS LOGIN MODULE CONFIGURATIONS

Review an example JAAS login module configuration that has both a properties login module and an LDAP login module configured.

In the example, the properties login module references the default login module that contains the credentials used by the Operator to authenticate with the broker.

```
activemq {
```

```

org.apache.activemq.artemis.spi.core.security.jaas.LDAPLoginModule required
  debug=true
  initialContextFactory=com.sun.jndi.LdapCtxFactory
  connectionURL="LDAP://localhost:389"
  connectionUsername="CN=Administrator,CN=Users,OU=System,DC=example,DC=com"
  connectionPassword=redhat.123
  connectionProtocol=s
  connectionTimeout="5000"
  authentication=simple
  userBase="dc=example,dc=com"
  userSearchMatching="(CN={0})"
  userSearchSubtree=true
  readTimeout="5000"
  roleBase="dc=example,dc=com"
  roleName=cn
  roleSearchMatching="(member={0})"
  roleSearchSubtree=true;

org.apache.activemq.artemis.spi.core.security.jaas.PropertiesLoginModule
  reload=true
  org.apache.activemq.jaas.properties.user="artemis-users.properties"
  org.apache.activemq.jaas.properties.role="artemis-roles.properties"
  baseDir="/home/jboss/amq-broker/etc";
};

```

The following example shows a JAAS login module configuration that has two properties login modules in separate realms.

- The default properties login module is in a realm named **console** and has the properties files that are used by the Operator and AMQ Management Console to authenticate with the broker.
- The login module in the **activemq** realm has new properties files, which, for example, could contain the credentials to authenticate users for messaging.

You might want to create separate realms to, for example, apply specific security controls to the realm that contains the login module used by the Operator to authenticate with the broker.

```

activemq {
  org.apache.activemq.artemis.spi.core.security.jaas.PropertiesLoginModule
  reload=true
  org.apache.activemq.jaas.properties.user="new-users.properties"
  org.apache.activemq.jaas.properties.role="new-roles.properties"
};

console {
  org.apache.activemq.artemis.spi.core.security.jaas.PropertiesLoginModule
  reload=true
  org.apache.activemq.jaas.properties.user="artemis-users.properties"
  org.apache.activemq.jaas.properties.role="artemis-roles.properties"
  baseDir="/home/jboss/amq-broker/etc";
};

```




NOTE

By default, AMQ Management Console uses the default properties login module in the **activemq** realm for authentication. If the default properties login module is configured in another realm, as in the example, you must set an environment variable in the broker CR to configure AMQ Management Console to use that realm. For example:

```
spec:
  ...
  env:
  - name: JAVA_ARGS_APPEND
    value: --Hawtio.realm=console
  ...
```

For more information about setting environment variables in a CR, see [Section 4.9, “Setting environment variables for the broker containers”](#).

8.3. EXAMPLE: CONFIGURING AMQ BROKER TO USE RED HAT SINGLE SIGN-ON

Review an example of how to configure AMQ Broker to use Red Hat Single Sign-On for authentication and authorization by using JAAS login modules.

Prerequisites

- A Red Hat Single Sign-On instance integrated with an LDAP directory.
 - The LDAP directory is populated with users and role information for AMQ Broker.
 - Red Hat Single Sign-On is configured to federate users from the LDAP server.
 - Red Hat Single Sign-On is configured to use the role-ldap-mapper to map role information from LDAP to Red Hat Single Sign-On.
- A Red Hat Single Sign-On realm that has:
 - A client configured with the following settings for applications, such as AMQ Management Console, that can use the oAuth protocol to obtain a token:
Authentication flow: Standard flow

Valid Redirect URIs: An OpenShift Container Platform route for AMQ Management Console. For example, http://artemis-wconsj-0-svc-rte-kc-ldap-tests-0eae49.apps.redhat-412t.broker.app-services-dev.net/console/*
 - A separate client configured with the following settings if you have messaging client applications that cannot use the oAuth protocol to obtain a token:
Authentication flow: Direct Access Grants

Valid Redirect URIs: *



NOTE

Each realm in Red Hat Single Sign-On includes a client named **Broker**. This client is not related to AMQ Broker.

Procedure

1. Create a text file named **login.config** and add the JAAS login module configuration to connect AMQ Broker with Red Hat Single Sign-On. For example:

```
console {
    // ensure the operator can connect to the broker by referencing the existing properties
    config
    org.apache.activemq.artemis.spi.core.security.jaas.PropertiesLoginModule sufficient
    org.apache.activemq.jaas.properties.user="artemis-users.properties"
    org.apache.activemq.jaas.properties.role="artemis-roles.properties"
    baseDir="/home/jboss/amq-broker/etc";

    org.keycloak.adapters.jaas.BearerTokenLoginModule sufficient
    keycloak-config-file="/amq/extra/secrets/sso-jaas-config/_keycloak-bearer-token.json"
    role-principal-class=org.apache.activemq.artemis.spi.core.security.jaas.RolePrincipal;
};
activemq {
    org.keycloak.adapters.jaas.BearerTokenLoginModule sufficient
    keycloak-config-file="/amq/extra/secrets/sso-jaas-config/_keycloak-bearer-token.json"
    role-principal-class=org.apache.activemq.artemis.spi.core.security.jaas.RolePrincipal;

    org.keycloak.adapters.jaas.DirectAccessGrantsLoginModule sufficient
    keycloak-config-file="/amq/extra/secrets/sso-jaas-config/_keycloak-direct-access.json"
    role-principal-class=org.apache.activemq.artemis.spi.core.security.jaas.RolePrincipal;

    org.apache.activemq.artemis.spi.core.security.jaas.PrincipalConversionLoginModule
    required
    principalClassList=org.keycloak.KeycloakPrincipal;
};
```

NOTE

- The path to the **.json** configuration files must be in the format **/amq/extra/secrets/*name*-jaas-config**. For *name*, specify a string value. You must use the same string value and a **-jaas-config** suffix to name the secret that you create later in this procedure.
- In the example **login.config** file, a realm named **console** is used to authenticate AMQ Management Console users and a realm named **activemq** to authenticate messaging clients.

The following login modules are configured in the example **login.config** file.

Table 8.4. Login modules

Login module	Description and usage
--------------	-----------------------

Login module	Description and usage
org.apache.activemq.artemis.spi.core.security.jaas.PropertiesLoginModule	This is the default login module and contains the artemis-users.properties file, which contains a default user that is required by the Operator to authenticate with the broker.
org.keycloak.adapters.jaas.BearerTokenLoginModule	This login module is for applications, for example, AMQ Management Console, that can use the OAuth protocol to obtain a token. When a user opens AMQ Management Console in a browser window, they are redirected to the Red Hat Single Sign-On console to log in to obtain a bearer token.
org.keycloak.adapters.jaas.DirectAccessGrantsLoginModule	This login module is required for non-HTTP applications, such as messaging clients, which cannot use the OAuth protocol. Using this login module, the broker first authenticates the client using a secret that is configured in Red Hat Single Sign-On and then obtains a token on behalf of the client.
org.apache.activemq.artemis.spi.core.security.jaas.PrincipalConversionLoginModule	This login module is required to convert the Keycloak principal received into a JAAS principal that can be used by AMQ Broker.



NOTE

In the **login.config** file example, each **.json** properties file name has an underscore prefix. The Operator ignores files prefixed with an underscore when it reports the status of the **JaasPropertiesApplied** condition. If the file names do not have an underscore prefix, the status of the **JaasPropertiesApplied** condition shows **OutOfSync** permanently because the broker does not recognize properties files used by third party login modules. For more information about status reporting, see [Section 4.3.2.1, “Setting up user authentication by using the default JAAS login module”](#).

1. Create text files for each of the **.json** properties files that are referenced in the login modules and configure the details required to connect AMQ Broker to Red Hat Single Sign-On. For example:

_keycloak-bearer-token.json

```
{
```

```

    "realm": "amq-broker-ldap",
    "resource": "amq-console",
    "auth-server-url": "https://keycloak-svc-rte-kc-ldap-tests-
0eae49.apps.412t.broker.app-services-dev.net",
    "principal-attribute": "preferred_username",
    "use-resource-role-mappings": false,
    "ssl-required": "external",
    "confidential-port": 0
  }

```

`_keycloak-direct-access.json`

```

{
  "realm": "amq-broker-ldap",
  "resource": "amq-broker",
  "auth-server-url": "https://keycloak-svc-rte-kc-ldap-tests-
0eae49.apps.412t.broker.app-services-dev.net",
  "principal-attribute": "preferred_username",
  "use-resource-role-mappings": false,
  "ssl-required": "external",
  "credentials": {
    "secret": "Lfk6g1ZKIGzNT6eRkz0d1scM4M29Ohmn"
  }
}

```

realm

The realm configured to authenticate the AMQ Broker applications and services in Red Hat Single Sign-On.

resource

The client ID of a client that is configured in Red Hat Single Sign-On.

auth-server-url

The base URL of the Red Hat Single Sign-On server.

principal-attribute

The token attribute with which to populate the UserPrincipal name.

use-resource-role-mappings

If set to true, Red Hat Single Sign-On looks inside the token for application level role mappings for the user. If false, it looks at the realm level for user role mappings. The default value is false.

ssl-required

Ensures that all communication to and from the Red Hat Single Sign-On server is over HTTPS. The default value is **external**, which means that HTTPS is required by default for external requests.

credentials

A secret configured in Red Hat Single Sign-On which the broker uses to log in to Red Hat Single Sign-On and obtain a token on behalf of the client.

2. Create a text file named **`_keycloak-js-client.json`** and add the configuration required for AMQ Management Console to redirect users to the URL of the Red Hat Single Sign-On Admin Console, where they enter their credentials. For example:

```
{
  "realm": "amq-broker-ldap",
  "clientId": "amq-console",
  "url": "https://keycloak-svc-rte-kc-ldap-tests-0eae49.apps.412t.broker.app-services-dev.net"
}
```

3. Use the **oc create secret** command to create a secret that contains the files that are referenced in the login module configuration. For example:

```
oc create secret generic sso-jaas-config --from-file=login.config --from-file=artemis-
users.properties --from-file=artemis-roles.properties --from-file=_keycloak-bearer-
token.json --from-file=_keycloak-direct-access.json --from-file=_keycloak-js-client.json
```



NOTE

- The secret name must have a suffix of **-jaas-config** so the Operator can recognize that the secret contains login module configuration and propagate any updates to each broker Pod.
- The secret name must match the last directory name in the path to the **.json** configuration files, which you specified in the **login.config** file. For example, if the path to the configuration files is **/amq/extra/secrets/sso-jaas-config**, you must specify a secret name of **sso-jaas-config**.

For more information about how to create secrets, see [Secrets](#) in the Kubernetes documentation.

4. Add the secret you created to the ActiveMQArtemis Custom Resource (CR) instance for your broker deployment.
 - a. Using the OpenShift command-line interface:
 - i. Log in to OpenShift as a user that has privileges to deploy CRs in the project for the broker deployment.
 - ii. Edit the CR for your deployment.

```
oc edit ActiveMQArtemis <CR instance name> -n <namespace>
```

- b. Using the OpenShift Container Platform web console:
 - i. Log in to the console as a user that has privileges to deploy CRs in the project for the broker deployment.
 - ii. In the left pane, click **Ecosystem > Installed Operators**
 - iii. Click the **Red Hat Integration - AMQ Broker for RHEL 9 (Multiarch)** operator.
 - iv. Click the **AMQ Broker** tab.
 - v. Click the name of the ActiveMQArtemis instance name.
 - vi. Click the **YAML** tab.
Within the console, a YAML editor opens, enabling you to configure a CR instance.

5. Create an **extraMounts** attribute and a **secrets** attribute and add the name of the secret. The following example adds a secret named **custom-jaas-config** to the CR.

```
deploymentPlan:
...
extraMounts:
  secrets:
    - "sso-jaas-config"
...
```

6. In the **ActiveMQArtemis** CR, create an environment variable that contains the hawtio settings required by AMQ Management Console to use Red Hat Single Sign-On for authentication. The contents of the environment variable are passed as arguments to the Java application launcher when the JVM that hosts a broker is started. For example:

```
env:
- name: JAVA_ARGS_APPEND
  value: -
    Dhawtio.rolePrincipalClasses=org.apache.activemq.artemis.spi.core.security.jaas.RolePrincipal
    -Dhawtio.keycloakEnabled=true -
    Dhawtio.keycloakClientConfig=/amq/extra/secrets/sso-jaas-config/_keycloak-js-client.json
    -Dhawtio.authenticationEnabled=true -Dhawtio.realm=console
```

For more information on hawtio settings, see the [hawtio documentation](#).

7. In the **spec** section of the **ActiveMQArtemis** CR, add a **brokerProperties** attribute and add permissions for the roles configured in the LDAP directory. You can grant a role permissions to a single address. Or, you can specify a wildcard match using the **#** sign to grant a role permissions to all addresses. For example:

```
spec:
...
brokerProperties:
- securityRoles.#.producers.send=true
- securityRoles.#.consumers.consume=true
...
```

8. Save the CR.
The Operator mounts the files in the secret in a **/amq/extra/secrets/secret name** directory on each Pod and configures the broker JVM to read the mounted **login.config** file, which contains the SSO configuration, instead of the default **login.config** file.

8.4. LOGGING

In addition to viewing the OpenShift logs, you can troubleshoot a running AMQ Broker on OpenShift image by viewing the AMQ logs that are output to the container's console.

Procedure

1. At the command line, run the following command:

```
$ oc logs -f <pass:quotes[<pod-name>]> <pass:quotes[<container-name>]>
```

■

Revised on 2026-07-14 17:54:46 UTC